



GMAT Official Guide Data Insights Review 2025–2026

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Scaled section scores and a total score that aligns to the actual test



Detailed score performance report, including time management

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275+ **practice questions** with detailed answer explanations



Digital flashcards, games, a **customizable question bank**, and more



Exclusive **tips and tricks** for exam success from the exam creators

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Table of Contents

[Cover](#)

[Title Page](#)

[Copyright](#)

[Table of Contents](#)

[Letter from the President and CEO, GMAC™](#)

[GMAT™ Official Guide Data Insights Review 2025–2026](#)

[1.0 What Is the GMAT™ Exam?](#)

[1.1 What Is the GMAT™ Exam?](#)

[1.2 Why Take the GMAT™ Exam?](#)

[1.3 GMAT™ Exam Format](#)

[1.4 What Is the Testing Experience Like?](#)

[1.5 What Is the Exam Content Like?](#)

[1.6 Data Insights Section](#)

[1.7 How Are Scores Calculated?](#)

[2.0 How to Prepare](#)

[2.1 How Should I Prepare for the GMAT™ Exam?](#)

[2.2 Getting Ready for Exam Day](#)

[2.3 How to Use the GMAT™ Official Guide Data Insights Review 2025–2026](#)

[2.4 How to Use Other GMAT™ Official Prep Products](#)

[2.5 Tips for Taking the Exam](#)

[2.6 Data Insights Section Strategies](#)

[3.0 Math Review](#)

[3.0 Math Review](#)

[3.1 Value, Order, and Factors](#)

[3.2 Algebra, Equalities, and Inequalities](#)

[3.3 Rates, Ratios, and Percents](#)

[3.4 Statistics, Sets, Counting, Probability, Estimation, and Series](#)

[3.5 Reference Sheets](#)

[4.0 Verbal Review](#)

[4.0 Verbal Review](#)

[4.1 Analyzing Passages](#)

[4.2 Inductive Reasoning](#)

[4.3 Deductive Reasoning](#)

[5.0 Data Insights Review](#)

[5.0 Data Insights Review](#)

[5.1 Data Sets and Types](#)

[5.2 Data Displays](#)

[5.3 Data Patterns](#)

[6.0 Data Insights](#)

[6.0 Data Insights](#)

[6.1 What Is Measured](#)

[6.2 Question Types and Test-Taking Strategies](#)

[6.3 Online Question Bank Instructions](#)

[6.4 Practice Questions: Data Sufficiency](#)

[6.5 Answer Key: Data Sufficiency](#)

[6.6 Answer Explanations: Data Sufficiency](#)

[6.7 Practice Questions: Two-Part Analysis](#)

[6.8 Answer Key: Two-Part Analysis](#)

[6.9 Answer Explanations: Two-Part Analysis](#)

[7.0 GMAT™ Official Guide Data Insights Review Question Index](#)

[7.0 GMAT™ Official Guide Data Insights Review Question Index](#)

[Appendix A Answer Sheets](#)

[Data Sufficiency Answer Sheet](#)

[Two-Part Analysis Answer Sheet](#)

[End User License Agreement](#)

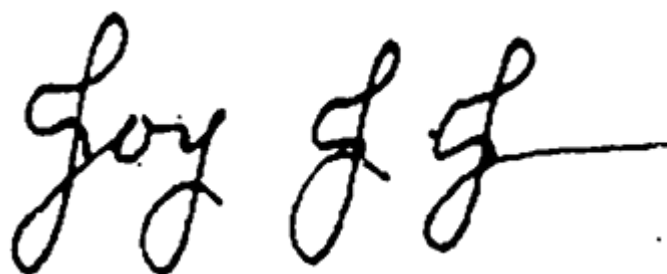
Dear GMAT™ Test Taker, Thank you for taking this important step toward your graduate management education. The GMAT exam stands as the global standard in business school admissions, used by more than 7,700 graduate programs worldwide to identify excellence in candidates for MBA, business master's, and other graduate-level management degrees. Seven out of ten business school applicants choose to differentiate themselves with a GMAT exam score.*

By choosing the GMAT™ Official Guide, you're preparing with the most authoritative resource available—the only guide featuring real GMAT questions published by the Graduate Management Admission Council (GMAC™), the makers of the GMAT exam. Combined with our prep tools at www.mba.com, this guide will build your confidence and help you achieve your personal best on exam day.

For more than 70 years, the GMAT exam has empowered candidates like you to demonstrate their readiness for academic success and signal their commitment to business education. Schools use and trust the GMAT because it consistently predicts classroom performance and indicates your potential to excel in your chosen program.

At GMAC, our purpose is to ensure no talent goes undiscovered. We're committed to providing the tools and insights you need throughout your business school journey, continuously enhancing the GMAT exam, and helping you connect with programs that match your aspirations. We applaud your commitment to advancing your education and wish you every success in your future educational and professional endeavors.

Sincerely,

A handwritten signature in black ink, appearing to read "Joy G. G." followed by a horizontal line.

Joy J. Jones
CEO, Graduate Management Admission Council

* Top 100 *Financial Times* full-time MBA programs

GMAT™ Official Guide Data Insights Review 2025–2026

1.0 What Is the GMAT™ Exam?

1.1 What Is the GMAT™ Exam?

The Graduate Management Admission Test™ (GMAT™) is used in admissions decisions by more than 7,700 graduate management programs at over 2,400 business schools worldwide. Unlike undergraduate grades and courses, whose meanings vary across regions and institutions, your GMAT scores are a standardized, statistically valid, and reliable measure for both you and these schools to predict your future performance and success in core courses of graduate-level management programs.

Hundreds of studies across hundreds of schools have demonstrated the validity of GMAT scores as being an accurate indicator of business school success. Together, these studies have shown that performance on the GMAT predicts success in business school even better than undergraduate grades.

The exam tests you on skills expected by management faculty and admission professionals for incoming graduate students. These skills include problem-solving, data analysis, and critical thinking, which all require complex judgments and are tested in the three sections of the GMAT exam: Quantitative Reasoning, Data Insights, and Verbal Reasoning. These three sections feature content relevant to today's business challenges and opportunities, ensuring you are prepared for graduate business school and beyond.

Your GMAT Official Score is meant to be an objective, numeric measure of your ability and potential for success. Business schools will use it as part of their holistic admissions processes, which may also consider recommendation letters, essays, interviews, work experiences, and other signs of social and emotional intelligence, as well as leadership. Even if your program does not require a GMAT score, you can stand out from the crowd by doing well on the exam to show you are serious about business school and have the skills to succeed.

The exam is always delivered in English on a computer, either online (such as at home) or at a test center. The exam tests your ability to apply foundational knowledge in the following areas: algebra and arithmetic,

analyzing and interpreting data, reading and comprehending written material, and reasoning and evaluating arguments.

Myth -vs- FACT

M – My GMAT score does not predict my success in business school.

F – The GMAT exam measures your critical thinking skills, which you will need in business school and your career.

1.2 Why Take the GMAT™ Exam?

Taking the exam helps you stand out as an applicant and shows you're ready for and committed to a graduate management education. Schools use GMAT scores in choosing the most qualified applicants. They know an applicant who has taken the exam is serious about earning a graduate business degree, and they know the exam scores reliably predict how well applicants can do in graduate business programs.

No matter how you do on the exam, you should contact schools that interest you to learn more about them, and to ask how they use GMAT scores and other criteria in admissions decisions. School admissions offices, websites, and publications are key sources of information when you are researching business schools. Note that schools' published GMAT scores are *averages* of the scores of their admitted students, not minimum scores needed for admission.

While you might aim to get a high or perfect score, such a score is not required to get into top business school programs around the world. You should try your best to achieve a competitive score that aligns with the ranges provided by the schools of your choice. Admissions officers will use GMAT scores as one factor in admissions decisions along with undergraduate records, application essays, interviews, letters of recommendation, and other information.

To learn more about the exam, test preparation materials, registration, and how to use your GMAT Official Score in applying to business schools, please visit www.mba.com/gmat.

Myth -vs- FACT

M – If I don’t get a high GMAT score, I won’t get into my top-choice schools.

F – Schools use your GMAT score as a part of their holistic evaluation process.

1.3 GMAT™ Exam Format

The GMAT exam has three separately timed sections (see the table on the following page). The Quantitative Reasoning section and the Verbal Reasoning section consist of only multiple-choice questions. The Data Insights section includes multiple-choice questions along with other kinds of graphical and data analysis questions. Before you start the exam, you can choose any order in which you will take the three sections. For example, you can choose to start with Verbal Reasoning, then do Quantitative Reasoning, and end with Data Insights. Or you can choose to do Data Insights first, followed by Verbal Reasoning, and then Quantitative Reasoning. You can take one optional ten-minute break after either the first or second section.

All three GMAT sections are computer adaptive. This means the test chooses from a large bank of questions to adjust itself to your ability level, so you will not get many questions that are too hard or too easy for you. The first question will be of medium difficulty. As you answer each question, the computer uses your answer, along with your responses to earlier questions, to choose the next question with the right level of difficulty. Because the computer uses your answers to choose your next question, you cannot skip questions.

Computer adaptive tests get harder as you answer more questions correctly. But getting a question that seems easier than the last one doesn’t always

mean your last answer was wrong. At the end of each section, you can review any question(s) and edit up to three answers within the allotted section time.

Though each test taker gets different questions, the mix of question types is always consistent. Your score depends on the difficulty and statistical traits of the questions you answer, as well as on which of your answers are correct. If you don't know how to answer a question, try to rule out as many wrong answer choices as possible. Then pick the answer choice you think is best. By adapting to each test taker, the exam can accurately and efficiently gauge a full range of skill levels, from very high to very low. Many factors may make the questions easier or harder, so don't waste time worrying if some questions seem easy.

To make sure every test taker gets equivalent content, the test gives specific numbers of questions of each type. While the test covers the same kinds of questions for everyone, some questions may seem harder or easier for you because you may be stronger in some questions than in others.

At the end of the exam, you will see your unofficial score displayed on the screen. A few days after your exam, you will receive your Official Score Report, which includes detailed performance insights. Once you receive your report, you can select to send your Official Score to schools of your choice.

Format of the GMAT™ Exam		
	Questions	Timing
Quantitative Reasoning Problem-Solving	21	45 min.
Data Insights Data Sufficiency Multi-Source Reasoning Table Analysis Graphics Interpretation Two-Part Analysis	20	45 min.
Verbal Reasoning Reading Comprehension Critical Reasoning	23	45 min.
Total Time		135 min.

A ten-minute optional break can be taken after the first or the second section.

Each section of the GMAT exam contains the following features:

- **Bookmarking:** Mark any questions you are unsure about so you can easily get back to them after you complete the section. Bookmarking can make the Question Review & Edit process more efficient.
 - **Question Review & Edit:** Review as many questions as you would like (whether or not they're bookmarked) and change or edit up to three answers per section, within the section's allotted time.
-

Myth -vs- FACT

M – Getting an easier question means I answered the previous one wrong.

F – Many factors may make the questions easier or harder, so don't waste time worrying if some questions seem easy.

1.4 What Is the Testing Experience Like?

You can take the exam either online (such as at home) or at a test center—whichever you prefer. You may feel more comfortable at home with the online delivery format. Or you may prefer the uninterrupted, structured environment of a test center. It is your choice. Both options have the same content, structure, features, optional ten-minute break, scores, and score scales.

At the Test Center: Over 700 test centers worldwide administer the GMAT exam under standardized conditions. Each test center has proctored testing rooms with individual computer workstations that allow you to take the exam in a peaceful, quiet setting, with some privacy. To learn more about exam day, visit www.mba.com/gmat.

Online: In available regions, the GMAT exam is delivered online and is proctored remotely, so you can take it in the comfort of your home or office. You will need a quiet workspace with a desktop or laptop computer that meets minimum system requirements, a webcam, microphone, and a reliable internet connection. For more information about taking the exam online, visit www.mba.com/gmat.

Whether you're taking the GMAT exam online or at a test center, there are several accommodations available. To learn more about available accommodations for the exam, visit www.mba.com/accommodations.

1.5 What Is the Exam Content Like?

The GMAT exam measures several types of analytical reasoning skills. The Quantitative Reasoning section gives you basic arithmetic and algebra problems. The questions present you with a mix of word or pure math problems. The Data Insights section asks you to use diverse reasoning skills to solve real-world problems involving data. It also asks you to interpret and combine data from different sources and in different formats to reach conclusions. The Verbal Reasoning section tests your ability to read and comprehend written material and to reason through and evaluate arguments.

The test questions are contextualized in various subject areas, but each question provides you everything you need to know to answer it correctly. In other words, you do not need detailed outside knowledge of the subject areas.

1.6 Data Insights Section

The GMAT Data Insights section tests the skills that today's business managers need to analyze intricate data sources and solve complex real-world problems. It tests how well you can assess multiple sources and types of information—graphic, numeric, and verbal—as they relate to one another. It also tests how well you can analyze a practical math and non-math problem to determine if enough data is given to solve it.

The Data Insights section has five types of questions:

- **Data Sufficiency:** Measures your ability to analyze a quantitative or logical problem, recognize which data is relevant, and determine at what point there is enough data to solve the problem. Questions can be presented in mathematical or non-mathematical real-world contexts.
- **Multi-Source Reasoning:** Measures your ability to examine data from several sources including text passages, tables, or graphics, and analyze each source of data carefully to answer multiple questions. Some questions will require you to recognize discrepancies among different sources of data, while others will ask you to draw inferences, or require you to determine whether data is relevant.

- **Table Analysis:** Measures your ability to sort and analyze a table of data, similar to a spreadsheet, in order to determine what information is relevant or meets certain conditions.
- **Graphics Interpretation:** Measures your ability to interpret the information presented in a graph or other graphical image (scatter plot, x/y graph, bar chart, pie chart, or statistical curve distribution) to discern relationships and make inferences.
- **Two-Part Analysis:** Measures your ability to solve complex problems. They could be quantitative, verbal, or a combination of both. The format is intentionally versatile to cover a wide range of content. These questions measure your ability to evaluate trade-offs, solve simultaneous equations, and discern relationships between two entities.

Data Insights questions may require math, data analysis, verbal reasoning, or all three. You will have to interpret graphs and sort data tables to answer some questions, but you won't need advanced statistics or spreadsheet skills. For both online and test center exam delivery, you will have access to an on-screen calculator with basic functions. It should be noted that this calculator is provided for the Data Insights section **only**, and **not** for the Quantitative Reasoning section.

[Chapter 5](#) of this book, “Data Insights Review,” reviews the basic data analysis skills you need to answer questions in the Data Insights section. [Chapter 6](#), “Data Insights,” explains the Data Insights question types, provides practice questions and answer explanations both in this book and in the Online Question Bank, and offers test-taking tips.

Because the Data Insights questions need to be rendered online, certain question types, such as Multi-Source Reasoning, Table Analysis, and Graphics Interpretation, are only available in the **Online Question Bank** that is included as a part of this guide.

You can access the **Online Question Bank** by going to www.mba.com/my-account and using your unique Access Code on the inside front cover of this book.

1.7 How Are Scores Calculated?

The Quantitative Reasoning, Data Insights, and Verbal Reasoning sections are each scored on a scale from 60 to 90, in 1-point increments. You will get four scores: a Section Score each for Quantitative Reasoning, Data Insights, and Verbal Reasoning, along with a Total Score based on your three section scores. The Total Score ranges from 205 to 805. Your scores depend on:

- Which questions you answered correctly.
- How many questions you answered.
- Each question's difficulty and other statistical characteristics.

There is a penalty for not completing each section of the exam. If you do not finish in the allotted time, your score will be penalized, reflecting the number of unanswered questions. Your GMAT exam score will be the best reflection of your performance when all questions are answered within the time limit.

Immediately after completing the exam, your unofficial scores and percentile for the Quantitative Reasoning, Data Insights, and Verbal Reasoning, as well as your Total Score, are displayed on-screen. You are **not** allowed to record, save, screenshot, or print your unofficial score. You will receive an email notification when your Official Score Report is available in your www.mba.com account.

The following table summarizes the different types of scores and their scale properties.

Score Type	Scale	Increment
Quantitative Reasoning	60–90	1
Data Insights	60–90	1
Verbal Reasoning	60–90	1
Total	205–805	10

Your GMAT Official Scores are valid for five years from your exam date. Your Total GMAT Score includes a percentile ranking, which shows the percentage of tests taken with scores lower than your score.

In addition to reviewing your Total and Section Scores, it's important to pay attention to your percentile ranking. Percentile rankings indicate what percentage of test takers you performed better than. For example, a percentile ranking of 75% means that you performed better than 75% of other test takers, and 25% of test takers performed better than you. Percentile ranks are calculated using scores from the most recent five years. Visit www.mba.com/scores to view the most recent predicted percentile rankings tables.

To better understand the exam experience and view score reports before exam day, we recommend taking at least one GMAT official practice exam to simulate the test-taking experience and gauge your potential score. The more practice exams you take, the better prepared you will be on your actual testing day. Visit www.mba.com/examprep to learn more about the practice exams offered by GMAC.

To register for the GMAT™ exam, go to www.mba.com/register

2.0 How to Prepare

2.1 How Should I Prepare for the GMAT™ Exam?

The GMAT™ exam has several unique question formats. We recommend that you familiarize yourself with the test format and the different question types before you take the test. The key to prepping for any exam is setting a pace that works for you and your lifestyle. That might be easier said than done, but the **GMAT™ Official 6-Week Study Planner** does the planning for you! Our step-by-step planner will help you stick to a schedule, inform your activities, and track your progress. Go to www.mba.com/examprep to download the planner.

Here are our recommended steps to starting your prep journey with your best foot forward.

1. **Study the structure.** Use the study plan in our **free GMAT™ Official Starter Kit** to become familiar with the exam format and structure. The study plan will guide you through each question type and give you sample questions. This will boost your confidence come test day when you know what to expect.
2. **Understand the question types.** Beyond knowing how to answer questions correctly, learn what each type of question is asking of you. GMAT questions rely on logic and analytical skills, not underlying subject matter mastery, as detailed in the **GMAT™ Official Guide 2025–2026 and Online Question Bank**.
3. **Establish your baseline.** Take the **GMAT™ Official Practice Exam 1 (FREE)** to establish your baseline. It uses the same format and scoring algorithm as the real test, so you can use the Official Score Report to accurately assess your strengths and growth areas.
4. **Study the answer explanations.** Take advantage of each question you get wrong by studying the correct answers, so you know how to get it right the next time. **GMAT™ Official Practice Questions** provide detailed answer explanations for hundreds of real GMAT questions. This will help you understand why you got a question right or wrong.

5. **Simulate the test-taking experience.** Take the **GMAT™ Official Practice Exams**. All GMAT™ Official Practice Exams use the same algorithm, scoring, and timing as the real exam, so take them with test-day-like conditions (e.g., quiet space, use the tools allowed on test day) for the truest prep experience.

Remember, the exam is timed, so learning to pace yourself and understanding the question formats and the skills you need can be a stepping stone to achieving your desired score. The timed practice in the **Online Question Bank** can help you prepare for this. The time management performance chart provided in practice exam score reports can also help you practice your pacing.

Because the exam assesses reasoning rather than knowledge, memorizing facts probably won't help you. You don't need to study advanced math, but you should know basic arithmetic and algebra. Likewise, you don't need to study advanced vocabulary words, but you should know English well enough to understand writing at an undergraduate level.

2.2 Getting Ready for Exam Day

Whether you take the exam online or in a test center, knowing what to expect will help you feel confident and succeed. To understand which exam delivery is right for you, visit www.mba.com/plan-for-exam-day.

Our Top Exam Day Strategies:

1. Get a good night's sleep the night before.
2. Pacing is key. Consult your on-screen timer periodically to avoid having to rush through sections.
3. Read each question carefully to fully understand what is being asked.
4. Don't waste time trying to solve a problem you recognize as too difficult or time-consuming. Instead, eliminate answers you know are wrong and select the best from the remaining choices.
5. Leverage the bookmarking tool to make your question review and edit process more efficient.

“Don’t take a ‘brute force’ approach to GMAT questions—think strategically instead.”

—A test instructor from GMAT Genius

“Compile summarized notes that can be reviewed on the morning of the test. Awareness of the key points and types of mistakes will boost your scores.”

—A test instructor from LEADERSMBA

2.3 How to Use the *GMAT™ Official Guide Data Insights Review 2025–2026*

The GMAT™ Official Guide series is the largest official source of actual GMAT questions. You can use this series of books and the included Online Question Bank to practice answering the different types of questions. The *GMAT™ Official Guide Data Insights Review* is designed for those who have completed the Data Insights questions in the *GMAT™ Official Guide 2025–2026* and are looking for additional practice questions, as well as those who are interested in practicing only Data Insights questions. Questions of each type are organized by difficulty level of easy, medium, and hard. Your rate of accuracy in each category might differ from what you expect. You might be able to answer the “hard” questions easily, while the “easy” ones are challenging. This is common and is not an indicator of exam performance. The questions in this book are not adaptive based on your performance but are meant to serve as exposure to the range of question types and formats you might encounter on the exam. Also, the proportions of questions about different content areas in this book don’t reflect the proportions in the actual exam. To find questions of a specific type and difficulty level (for example, easy arithmetic questions), use the index of questions in Chapter 9.

We recommend the steps below for how to best use this book:

1. Start with the review chapters to gain an overview of the required concepts.

“Building a strong foundation is crucial to achieve a high score. If you struggle with fundamental questions, your progress on more advanced questions will be hindered.”

—A test instructor from XY Education

2. Go through the practice questions in this book. Once you’ve familiarized yourself with the concepts and question types, use the Online Question Bank to further customize your practice by choosing your preferred level of difficulty, category of concepts, or question types.

“The Official Guide offers in-depth answer explanations. After completing a question, reviewing it alongside the explanation helps you gain a deeper understanding of the question’s key concepts and solution strategies. On the result interface of the online question bank, it serves a dual purpose: on one hand, it encourages us to review incorrect answers; on the other hand, it provides detailed insights into the time spent on each question, aiding in optimizing our pace during exercises.”

—A test instructor from JiangjiangGMAT

3. Use the **Online Question Bank** to continue practicing based on your progress. To better customize and enhance your practice, use the Online Question Bank to:
 - a. Review and retry practice questions to improve performance by using the untimed or timed features along with a study mode or an exam mode.
 - b. Analyze key performance metrics to help assess focus area and track improvement.
 - c. Use flashcards to master key concepts.

“The biggest mistake students make is completing too many new problems. Completing problems doesn’t move your score! Learning from problems does. Keep a list of questions you want to go back to and redo. Redo at least three of these questions every time you study. This book is one of the most important resources you can use to create the future you want. Make sure you understand every question you complete extremely well. You should be able to explain every problem you complete to someone who is new to the exam. Don’t focus on simply ‘getting through’ the book. That mindset will work against you and your dreams.”

—A test instructor from The GMAT Strategy



TIP

Since the exam is given on a computer, we suggest you practice the questions in this book using the **Online Question Bank** accessed via www.mba.com/my-account. It includes all the questions in this book, and it lets you create practice sets—both timed and untimed—and track your progress more easily. The Online Question Bank is also available on your mobile device through the GMAT™ Official Practice mobile app. To access the Online Question Bank on your mobile device, first create an account at www.mba.com, and then sign into your account on the mobile app.



TIP

Remember: Some of the Data Insights section questions are only available in the **Online Question Bank**.

2.4 How to Use Other GMAT™ Official Prep Products

We recommend using our other GMAT™ Official Prep products along with this guidebook.

- **For a realistic simulation of the exam:** GMAT™ Official Practice Exams 1–6 are the only practice exams that use real exam questions along with the scoring algorithm and user interface from the actual exam. The first two practice exams are free to all test takers at www.mba.com/gmatprep.
- **For more practice questions:** *GMAT™ Official Guide Verbal Review 2025–2026*, and *GMAT™ Official Guide Quantitative Review 2025–2026* offer over 400 additional practice questions not included in this book.
- **For focused practice:** GMAT™ Official Practice Questions for Quantitative, Data Insights, and Verbal products offer 100+ questions that are not included in the Official Guide series.

“Build teaching-level depth; don’t just finish content mindlessly. Even if you solve thousands of questions on a shaky foundation, you will remain stuck on a really low accuracy.”

—A test instructor from Top One Percent

2.5 Tips for Taking the Exam

Tips for answering questions of the different types are given later in this book. Here are some general tips to help you do your best on the test.

1. **Before the actual exam, decide in what order to take the sections.**

The exam lets you choose in which order you'll take the sections. Use the GMAT™ Official Practice Exams to practice and find your preferred order. No order is “wrong.” Some test takers prefer to complete the section that challenges them the most first, while others prefer to ease into the exam by starting with a section that they're stronger in. Practice each order and see which one works best for you.

2. **Try the practice questions and practice exams.**

Timing yourself as you answer the practice questions and taking the practice exams can give you a sense of how long you will have for each question on the actual test, and whether you are answering them fast enough to finish in time.



TIP

After you've learned about all the question types, use the practice questions in this book and practice them online at www.mba.com/my-account to prepare for the actual test. Reminder: Most types of Data Insights practice questions are only available online.

3. **Review all test directions ahead of time.**

The directions explain exactly what you need to do to answer questions of each type. You can review the directions in the GMAT™ Official Practice Exams ahead of time so that you don't miss anything you need to know to answer properly. To review directions during the test, you can click on the Help icon. But note that your time spent reviewing directions counts against your available time for that section of the test.

4. Study each question carefully.

Before you answer a question, understand exactly what it is asking, then pick the best answer choice. Never skim a question. Skimming may make you miss important details or nuances.

5. Use your time wisely.

Although the exam stresses accuracy over speed, you should use your time wisely. On average, you have just about 2 minutes and 9 seconds per Quantitative Reasoning question; 2 minutes and 15 seconds per Data Insights question; and under 2 minutes per Verbal Reasoning question. Once you start the test, an on-screen clock shows how much time you have left. You can hide this display if you want, but by checking the clock periodically, you can make sure to finish in time.

6. Do not spend too much time on any one question.

If finding the right answer is taking too long, try to rule out answer choices you know are wrong. Then pick the best of the remaining choices and move on to the next question.

Not finishing sections or randomly guessing answers can lower your score significantly. As long as you've worked on each section, you will get a score even if you didn't finish one or more sections in time. You don't earn points for questions you never get to see.

Pacing is important. If a question stumps you, pick the answer choice that seems best and move on. If you guess wrong, the computer will likely give you an easier question, which you're more likely to answer correctly. Soon the computer will return to giving you questions matched to your ability. You can bookmark questions you get stuck on, then return to change up to three of your answers if you still have time left at the end of the section. But if you don't finish the section, your score will be reduced.

7. Confirm your answers ONLY when you are ready to move on.

In the GMAT Quantitative Reasoning, Data Insights, and Verbal Reasoning sections, once you choose your answer to a question, you are asked to confirm it. As soon as you confirm your response, the next question appears. You can't skip questions. In the Data Insights and

Verbal Reasoning sections, several questions based on the same prompt may appear at once. When more than one question is on a single screen, you can change your answers to any questions on that screen before moving on to the next screen. But until you've reached the end of the section, you can't navigate back to a previous screen to change any answers.

Myth -vs- FACT

M – Avoiding wrong answers is more important than finishing the test.

F – Not finishing can lower your score a lot.

Myth -vs- FACT

M – The first ten questions are critical, so you should spend the most time on them.

F – All questions impact your score.

2.6 Data Insights Section Strategies

Utilize the strategies below to better prepare for the exam. Creating a solid study plan and selecting the right prep materials are two key elements of getting accepted into your top business schools. But knowing how to strategically approach the exam is another crucial factor that can increase your confidence going into test day and help you perform your best.

“Don’t just read explanations. Review your notes and learn from your mistakes.”

—A test instructor from Admit Master

Data Sufficiency Questions

- Decide whether the problem allows only one value or a range of values. Remember: You are only determining whether you have enough data to solve the problem.
- Avoid making unwarranted assumptions based on figures. Figures are not necessarily drawn to scale.

Multi-Source Reasoning Questions

- Don’t expect to be completely familiar with the material. All of the information you need to answer the questions is provided.
- Analyze each source of data carefully, as the questions require detailed understanding of the data presented. Text passages often build ideas in sequences, so be mindful of how each statement adds to the main idea of the passage. Graphic elements come in various forms, such as tables, graphs, diagrams, or charts.
- Make sure you understand what is being asked for each question. Some questions will require recognizing discrepancies among different sources of data. Others will ask you to draw inferences. Still others may require you to determine which one of the data sources is relevant.

- Select the answer choices that have the most support based on the data provided. Don't let your knowledge of the subject matter influence your answer choice. Answer the questions using only the data provided to you.

Table Analysis Questions

- Examine the table and accompanying text to determine the type of information provided.
- Read the question carefully to determine the data analysis required and know the choices you have to make by reviewing the answers.
- Judge each answer statement carefully based on the condition specified (i.e., yes or no, true or false). Focus your attention on whether the given condition has been met.

Graphics Interpretation Questions

- Familiarize yourself with the data presented in the graphic. Make note of the scales on the axis, marked values, and labels. Pay attention to any discrepancies between the units in the graph and the units discussed in the text.
- Read any accompanying text carefully. The text might present data that isn't contained in the graphic but that you need to answer the question.
- Make sure you understand what the problem is asking you to do. You will interpret and integrate data, discern relationships, and make inferences from a set of data.
- Read all the choices in the drop-down menu. By checking the menu options, you will get additional information about your assigned task.
- Choose the option that best completes the statement. More than one option in the drop-down menu may seem plausible. You will need to choose the one that makes the statement most accurate or logical.

Two-Part Analysis Questions

- Read the information carefully. It may cover a wide range of content, including quantitative, verbal, or some combination of both. All the material presented is designed to be challenging. Don't let any familiarity with the subject matter influence your response. Only use the data presented in the question.
- Determine exactly what the question is asking. Pay close attention to how the question describes the tasks. Sometimes the response column headings lack the details that could help you better understand what you are supposed to do.
- When making your answer choices, determine whether your tasks are dependent or independent. Some questions will pose two tasks that can be carried out individually. Others pose one task with two dependent parts.
- Keep in mind that the same answer choice might be the correct response for both columns. It is possible that one answer option satisfies the conditions of both response columns.

Important Note for Data Insights Practice

Multi-Source Reasoning, Table Analysis, and Graphics Interpretation questions are only available in the Online Question Bank. To access the Online Question Bank, go to www.mba.com/my-account and use your unique Access Code, which can be found on the inside front cover of this book.

To register for the GMAT™ exam, go to www.mba.com/register

3.0 Math Review

3.0 Math Review

This chapter reviews the math you need to answer GMAT™ Quantitative Reasoning questions and Data Insights questions. This is only a brief overview. If you find unfamiliar terms, consult other resources to learn more.

Unlike some math problems you may have solved in school, GMAT math questions ask you to **apply** your math knowledge. For example, rather than asking you to list a number's prime factors to show you understand prime factorization, a GMAT question may ask you to **use** prime factorization and exponents to simplify an algebraic expression with a radical.

To prepare for the GMAT Quantitative Reasoning section and the Data Insights section, first review basic math to make sure you know enough to answer the questions. Then practice with GMAT questions from past exams.

[Section 3.1](#), “Value, Order, and Factors,” includes:

1. Numbers and the Number Line
2. Factors, Multiples, and Remainders
3. Exponents
4. Decimals and Place Value
5. Arithmetic Shortcuts and Properties of Operations

[Section 3.2](#), “Algebra, Equalities, and Inequalities,” includes:

1. Algebraic Expressions and Equations
2. Linear Equations
3. Factoring and Quadratic Equations
4. Inequalities
5. Functions
6. Graphing
7. Formulas and Measurement Conversion

[Section 3.3](#), “Rates, Ratios, and Percents,” includes:

1. Ratio and Proportion
2. Fractions
3. Percents
4. Converting Decimals, Fractions, and Percents
5. Working with Decimals, Fractions, and Percents
6. Rate, Work, and Mixture Problems

[Section 3.4](#), “Statistics, Sets, Counting, Probability, Estimation, and Series,” includes:

1. Statistics
2. Sets
3. Counting Methods
4. Probability
5. Estimation
6. Sequences and Series

[Section 3.5](#), “Reference Sheets”

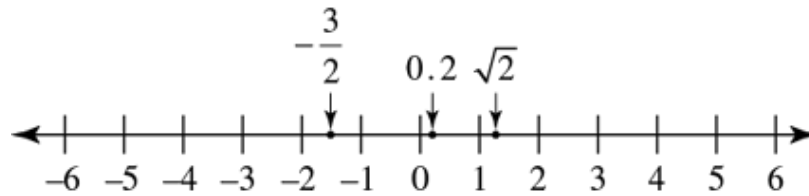
3.1 Value, Order, and Factors

1. Numbers and the Number Line

- A. All ***real numbers*** match points on ***the number line***, and all points on the number line represent real numbers.

The figure below shows the number line with labeled points standing for the real numbers $-\frac{3}{2}$, 0.2, and $\sqrt{2}$.

The Number Line



- B. On the number line, points to the left of zero stand for **negative** numbers, and points to the right of zero stand for **positive** numbers. Every real number except zero is either positive or negative.
- C. On the number line, each number is less than any number to its right. So, as the figure above shows, $-4 < -3 < -\frac{3}{2} < -1$, and $1 < \sqrt{2} < 2$.
- D. If a number n is between 1 and 4 on the number line, then $n > 1$ and $n < 4$; that is, $1 < n < 4$. If n is “between 1 and 4, inclusive,” then $1 \leq n \leq 4$.
- E. The **absolute value** of a real number x , written as $|x|$, is x if $x \geq 0$ and $-x$ if $x < 0$. A number’s absolute value is the distance between that number and zero on the number line. Thus, -3 and 3 have the same absolute value since each is three units from zero on the number line. The absolute value of any nonzero number is positive.

Examples:

$$|-5| = |5| = 5, |0| = 0, \text{ and}$$

$$\left| -\frac{7}{2} \right| = \frac{7}{2}.$$

For any real numbers x and y , $|x + y| \leq |x| + |y|$, and $|xy| = |x||y|$.

Examples:

If $x = 10$ and $y = 2$, then $|x + y| = |12| = 12 = |x| + |y|$.

And $|xy| = |20| = 20 = |x||y|$.

If $x = 10$ and $y = -2$, then $|x + y| = |8| = 8 < 12 = |x| + |y|$.

And $|xy| = |-20| = 20 = |x||y|$.

2. Factors, Multiples, and Remainders

- A. An **integer** is any number in the set $\{\dots -3, -2, -1, 0, 1, 2, 3, \dots\}$. For any integer n , the numbers in the set $\{n, n + 1, n + 2, n + 3, \dots\}$ are **consecutive integers**.
- B. For integers x and y , if $x \neq 0$, x is a **divisor** or **factor** of y if $y = xn$ for some integer n . Then y is **divisible** by x and is a **multiple** of x .

Example:

Since $28 = (7)(4)$, both 4 and 7 are divisors or factors of 28.

But 8 isn't a divisor or factor of 28 since n isn't an integer if $28 = 8n$.

- C. Dividing a positive integer y by a positive integer x and then rounding down to the nearest nonnegative integer gives the **quotient** of the division.

To find the **remainder** of the division, multiply x by the quotient, then subtract the result from y . The quotient and the remainder are the unique positive integers q and r , respectively, such that $y = xq + r$ and $0 \leq r < x$.

Example:

When 28 is divided by 8, the quotient is 3 and the remainder is 4 because $28 = (8)(3) + 4$.

The remainder r is 0 if and only if y is **divisible** by x . Then x is a divisor or factor of y , and y is a multiple of x .

Example:

Since 32 divided by 8 has a remainder of 0, 32 is divisible by 8. So, 8 is a divisor of 32, and 32 is a multiple of 8.

When a smaller integer is divided by a larger integer, the quotient is 0 and the remainder is the smaller integer.

Example:

When 5 is divided by 7, the quotient is 0 and the remainder is 5 since $5 = (7)(0) + 5$.

- D. For any positive integers x , y , and z , suppose that r is the remainder when x is divided by z , and that s is the remainder when y is divided by z . Then the remainder when $x + y$ is divided by z equals the remainder when $r + s$ is divided by z . Similarly, the remainder when xy is divided by z equals the remainder when rs is divided by z .

Examples:

When 142 is divided by 13, the remainder is 12 because $132 = (13 \times 10) + 12$. And when 29 is divided by 13, the remainder is 3 because $29 = (13 \times 2) + 3$.

So the remainder when $142 + 29$ is divided by 13 equals the remainder when $12 + 3 = 15$ is divided by 13; that is, it's 2.

And the remainder when 142×29 is divided by 13 equals the remainder when $12 \times 3 = 36$ is divided by 13; that is, it's 10.

- E. Any integer divisible by 2 is **even**. The set of even integers is $\{\dots -4, -2, 0, 2, 4, 6, 8, \dots\}$. Integers not divisible by 2 are **odd**, so $\{\dots -3, -1, 1, 3, 5, \dots\}$ is the set of odd integers. For any integer n , the numbers in the set $\{2n, 2n + 2, 2n + 4, \dots\}$ are **consecutive even integers**, and the numbers in $\{2n + 1, 2n + 3, 2n + 5, \dots\}$ are **consecutive odd integers**.

If a product of integers has at least one even factor, the product is even; otherwise, it's odd. If two integers are both even or both odd, their sum and

their difference are even. Otherwise, their sum and their difference are odd.

- F. A **prime** number is a positive integer with exactly two positive divisors: 1 and itself. That is, a prime number is divisible by no integer but itself and 1.

Examples:

The first six prime numbers are 2, 3, 5, 7, 11, and 13.

But 15 is not a prime number because it has four positive divisors: 1, 3, 5, and 15.

Nor is 1 a prime number because it has only one positive divisor: itself.

3. Exponents

- A. An expression of the form k^n means the n^{th} **power** of k , or k raised to the n^{th} power, where n is the **exponent** and k is the **base**.
- B. A positive integer exponent shows how many instances of the base are multiplied together. That is, when n is a positive integer, k^n is the product of n instances of k .

Examples:

x^5 is $(x)(x)(x)(x)(x)$; that is, the product in which x is a factor 5 times with no other factors. We can also say x^5 is the 5^{th} power of x , or x raised to the 5^{th} power.

The second power of 2, also called 2 **squared**, is $2^2 = 2 \times 2 = 4$. The third power of 2, also called 2 **cubed**, is $2^3 = 2 \times 2 \times 2 = 8$.

Squaring a number greater than 1, or raising it to any power greater than 1, gives a larger number.

Squaring a number between 0 and 1 gives a smaller number.

Examples:

$$3^2 = 9, \text{ and } 9 > 3.$$

$$(0.1)^2 = 0.01, \text{ and } 0.01 < 0.1.$$

- C. A **composite number** is an integer greater than 1 that's not prime. Any composite number is a product of a unique set of prime factors, each raised to some positive integer power. The **prime factorization** of a composite number shows it as such a product.

Examples:

Some prime factorizations of composite numbers are $14 = (2)(7)$, $81 = 3^4$, and $484 = (2^2)(11^2)$.

- D. For any two positive integers x and y , y is divisible by x if and only if the prime factorization of y includes each prime number in the prime factorization of x , raised to at least as great a power as it is in the prime factorization of x .

Examples:

$1080 = (2^3)(3^3)(5)$ is divisible by $24 = (2^3)(3)$, because the prime factorization of 1080 includes both 2 and 3, which are the only prime factors in the prime factorization of 24, and each is raised to at least as great a power as it is in the prime factorization of 24.

But 1080 is not divisible by $16 = 2^4$, because the prime factor 2 in the prime factorization of 16 is raised to a higher power than it is in the prime factorization of 1080.

Nor is 1080 divisible by $21 = (3)(7)$, because its prime factor 7 isn't in the prime factorization of 1080.

- E. A **square root** of a number n is a number x such that $x^2 = n$. Every positive number has two real square roots, one positive and the other

negative. The positive square root of n is written as $\sqrt{n}$ or $n^{\frac{1}{2}}$.

Example:

The two square roots of 9 are $\sqrt{9} = 3$ and $-\sqrt{9} = -3$.

For any x , the nonnegative square root of x^2 equals the absolute value of x ; that is, $\sqrt{x^2} = |x|$.

The square root of a negative number is not a real number. Only real numbers appear on the GMAT.

- F. Every real number r has exactly one real **cube root**, the number s such that $s^3 = r$. The real cube root of r is written as $\sqrt[3]{r}$ or $r^{\frac{1}{3}}$.

Examples:

Since $2^3 = 8$, $\sqrt[3]{8} = 2$.

Likewise, $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$.

4. Decimals and Place Value

- A. A **decimal** is a real number written as a series of digits, often with a period called a **decimal point**. The decimal point's position sets the **place values** of the digits.

Example:

The digits in the decimal 7,654.321 have these place values:

Thousands			Hundreds	Tens	Ones or units		Tenths	Hundredths	Thousandths
7	,	6	5	4	.	3	2	1	

- B. Some decimal numbers extend rightward by infinitely many digits. In a **repeating decimal**, a finite sequence of digits that are not all 0s repeats forever. A repeating decimal may be written showing the finite sequence of digits to the right of the decimal point only once, with a bar over them. There are never digits to the right of the repeating sequence.

Example:

The repeating decimal $12.34\overline{56}$ has the digits 12.3456565656..., with the digit sequence 56 repeating forever.

- C. In **scientific notation**, a decimal is written with only one nonzero digit to the decimal point's left, multiplied by an integer power of 10. To convert a number from scientific notation to regular decimal notation, move the decimal point by the number of places equal to the absolute value of the exponent on the 10. Move the decimal point to the right if the exponent is positive or to the left if the exponent is negative.

Examples:

In scientific notation, 231 is written as 2.31×10^2 , and 0.0231 is written as 2.31×10^{-2} .

To convert 2.013×10^4 to regular decimal notation, move the decimal point 4 places to the right, giving 20,130.

Likewise, to convert 1.91×10^{-4} to regular decimal notation, move the decimal point 4 places to the left, giving 0.000191.

- D. To add or subtract decimals with many finite digits, line up their decimal points. If one decimal has fewer digits to the right of its decimal point than another, insert zeros to the right of its last digit.

Examples:

To add 17.6512 and 653.27, insert zeros to the right of the last digit in 653.27 to line up the decimal points when the numbers are in a column:

$$\begin{array}{r} 17.6512 \\ + 653.2700 \\ \hline 670.9212 \end{array}$$

Likewise for 653.27 minus 17.6512:

$$\begin{array}{r} 653.2700 \\ - 17.6512 \\ \hline 635.6188 \end{array}$$

- E. Multiply decimals with many finite digits as if they were integers. Then insert the decimal point in the product so that the number of digits to the right of the decimal point is the sum of the numbers of digits to the right of the decimal points in the numbers being multiplied.

Example:

To multiply 2.09 by 1.3, first multiply the integers 209 and 13 to get 2,717. Since $2 + 1 = 3$ digits are to the right of the decimal points in 2.09 and 1.3, put 3 digits in 2,717 to the right of the decimal point to find the product:

$$\begin{array}{r} 2.09 \quad (2 \text{ digits to the right}) \\ \times 1.3 \quad (1 \text{ digit to the right}) \\ \hline 627 \\ \\ 2090 \\ \hline 2.717 \quad (2 + 1 = 3 \text{ digits to the right}) \end{array}$$

- F. To divide a number (the **dividend**) by a decimal (the **divisor**), when both have many finite digits, move the decimal points of the dividend and

divisor the same number of digits to the right until the divisor is an integer. Then divide as you would integers. The decimal point in the quotient goes directly above the decimal point in the new dividend.

Example:

To divide 698.12 by 12.4, first move the decimal points in both the divisor 12.4 and the dividend 698.12 one place to the right to make the divisor an integer. That is, replace $698.12/12.4$ with $6981.2/124$. Then do the long division normally:

$$\begin{array}{r} 56.3 \\ 124 \overline{)6981.2} \\ \underline{620} \\ 781 \\ \underline{744} \\ 372 \\ \underline{372} \\ 0 \end{array}$$

5. Arithmetic Shortcuts and Properties of Operations

- A. For any two nonnegative integers x and y , the final digit of $x + y$ is the final digit of the sum of x 's final digit and y 's final digit. And for any two integers x and y , the final digit of xy is the final digit of the product of x 's final digit and y 's final digit.

Examples:

The final digit of $345 + 789$ is the final digit of $5 + 9 = 14$, which is 4.

The final digit of 345×789 is the final digit of $5 \times 9 = 45$, which is 5.

- B. Here are some shortcuts to help tell whether one integer is divisible by another:

An integer is divisible by 2 if and only if its final digit is even.

An integer is divisible by 3 if and only if the sum of its digits is divisible by 3.

An integer with 3 or more digits is divisible by 4 if and only if its final 2 digits make a number divisible by 4.

An integer is divisible by 5 if and only if its final digit is 5 or 0.

An integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

An integer is divisible by 10 if and only if its final digit is 0.

Examples:

The integer 180 is divisible by 2, 5, and 10, because its final digit is 0. It's also divisible by 3 and 9, because its digits sum to $1 + 8 + 0 = 9$, which is divisible by both 3 and 9. And it's divisible by 4, because its final 2 digits, 80, make a number divisible by 4.

But the integer 121 is not divisible by 2, because its final digit is odd. Nor is it divisible by 3 or 9, because its digits sum to $1 + 2 + 1 = 4$, which is divisible by neither 3 nor 9. And it's not divisible by 4, because its final two digits, 21, make a number not divisible by 4. Finally, because its final digit is neither 5 nor 0, it's not divisible by 5 or 10.

C. Here are some basic properties of arithmetical operations for any real numbers x , y , and z :

- Addition and Subtraction

$$x + 0 = x = x - 0$$

$$x - x = 0$$

$$x + y = y + x$$

$$x - y = -(y - x) = x + (-y)$$

$$(x + y) + z = x + (y + z)$$

If x and y are both positive, then $x + y$ is positive.

If x and y are both negative, then $x + y$ is negative.

- Multiplication and Division

$$x \times 1 = x = \frac{x}{1}$$

$$x \times 0 = 0$$

$$\text{If } x \neq 0, \text{ then } \frac{x}{x} = 1.$$

$$\frac{x}{0} \text{ is undefined.}$$

$$xy = yx$$

$$\text{If } x \neq 0 \text{ and } y \neq 0, \text{ then } \frac{x}{y} = \frac{1}{\left(\frac{y}{x}\right)}.$$

$$(xy)z = x(yz)$$

$$xy + xz = x(y + z)$$

$$\text{If } y \neq 0, \text{ then } \left(\frac{x}{y}\right) + \left(\frac{z}{y}\right) = \frac{(x + z)}{y}.$$

If x and y are both positive, then xy is positive.

If x and y are both negative, then xy is positive.

If x is positive and y is negative, then xy is negative.

If $xy = 0$, then $x = 0$ or $y = 0$, or both.

- Exponentiation

$$x^1 = x$$

$$\text{If } x \neq 0, \text{ then } x^0 = 1.$$

$$\text{If } x \neq 0, \text{ then } x^{-1} = \frac{1}{x}.$$

$$(x^y)^z = x^{yz} = (x^z)^y$$

$$x^{y+z} = x^y x^z$$

$$\text{If } x \neq 0, \text{ then } x^{y-z} = \frac{x^y}{x^z}.$$

$$(xz)^y = x^y z^y$$

$$\text{If } z \neq 0, \text{ then } \left(\frac{x}{z}\right)^y = \frac{x^y}{z^y}.$$

$$\text{If } z \neq 0, \text{ then } x^{\frac{y}{z}} = (x^y)^{\frac{1}{z}} = \left(x^{\frac{1}{z}}\right)^y.$$

3.2 Algebra, Equalities, and Inequalities

1. Algebraic Expressions and Equations

A. Algebra is based on arithmetic and on the concept of an **unknown quantity**. Letters like ***x*** or ***n*** are **variables** that stand for unknown quantities. Other numerical expressions called **constants** stand for known quantities. A combination of variables, constants, and arithmetical operations is an **algebraic expression**.

Solving word problems often requires translating words into algebraic expressions. The table below shows how some words and phrases can be translated as math operations in algebraic expressions:

3.2 Translating Words into Math Operations				
$x + y$	$x - y$	xy	$\frac{x}{y}$	x^y
x added to y x increased by y x more than y x plus y the sum of x and y the total of x and y	x decreased by y difference of x and y y fewer than x y less than x x minus y x reduced by y y subtracted from x	x multiplied by y the product of x and y x times y	x divided by y x over y the quotient of x and y the ratio of x to y	x to the power of y x to the y^{th} power
		If $y = 2$: double x twice x	If $y = 2$: half of x x halved	If $y = 2$: x squared
		If $y = 3$: triple x		If $y = 3$: x cubed

- B. In an algebraic expression, a **term** is a constant, a variable, or a product of terms that are each a constant or a variable. A variable in a term may be raised to an exponent. A term with no variables is a **constant term**. A constant in a term with one or more variables is a **coefficient**.

Example:

Suppose Pam has 5 more pencils than Fred. If F is the number of pencils Fred has, then Pam has $F + 5$ pencils. The algebraic expression $F + 5$ has two terms: the variable F and the constant 5.

- C. A **polynomial** is an algebraic expression that's a sum of terms and has exactly one variable. Each term in a polynomial is a variable raised to some power and multiplied by some coefficient. If the highest power a variable is raised to is 1, the expression is a **first degree** (or **linear**) **polynomial** in that

variable. If the highest power a variable is raised to is 2, the expression is a **second degree** (or **quadratic**) **polynomial** in that variable.

Examples:

$F + 5$ is a linear polynomial in F , since the highest power of F is 1.

$19x^2 - 6x + 3$ is a quadratic polynomial in x , since the highest power of x is 2.

$\frac{3x^2}{(2x - 5)}$ is not a polynomial because it's not a sum of powers of x multiplied by coefficients.

- D. You can simplify many algebraic expressions by factoring or combining **like** terms.

Examples:

In the expression $6x + 5x$, x is a factor common to both terms. So, $6x + 5x$ is equivalent to $(6 + 5)x$, or $11x$.

In the expression $9x - 3y$, 3 is a factor common to both terms:
 $9x - 3y = 3(3x - y)$.

The expression $5x^2 + 6y$ has no like terms and no common factors.

- E. In a fraction $\frac{n}{d}$, n is the **numerator** and d is the **denominator**. In a numerator and denominator, you can divide out any common factors not equal to zero.

Example:

$$\text{If } x \neq 3, \text{ then } \frac{(x-3)}{(x-3)} = 1.$$

$$\text{So, } \frac{(3xy-9y)}{(x-3)} = \frac{3y(x-3)}{(x-3)} = 3y(1) = 3y.$$

- F. To multiply two algebraic expressions, multiply each term of one expression by each term of the other.

Example:

$$\begin{aligned}(3x-4)(9y+x) &= 3x(9y+x) - 4(9y+x) \\ &= 3x(9y) + 3x(x) - 4(9y) - 4(x) \\ &= 27xy + 3x^2 - 36y - 4x\end{aligned}$$

- G. To **evaluate** an algebraic expression, replace its variables with constants.

Example:

If $x = 3$ and $y = -2$, we can evaluate $3xy - x^2 + y$ as

$$3(3)(-2) - (3)^2 + (-2) = -18 - 9 - 2 = -29.$$

- H. An **algebraic equation** is an equation with only algebraic expression. An algebraic equation's **solutions** are the sets of assignments of constant values to its variables that make it true, or "satisfy the equation." An equation may have no solution, one solution, or more than one solution. For equations solved together, the solutions must satisfy all the equations at once. An equation's solutions are also called its **roots**. To confirm the roots are correct, you can substitute them into the equation.
- I. Two equations with the same solution or solutions are **equivalent**.

Examples:

The equations $2 + x = 3$ and $4 + 2x = 6$ are equivalent, because each has the unique solution $x = 1$. Notice the second equation is the first equation multiplied by 2.

Likewise, the equations $3x - y = 6$ and $6x - 2y = 12$ are equivalent, although each has infinitely many solutions. For any value of x , giving the value $3x - 6$ to y satisfies both these equations. For example, $x = 2$ with $y = 0$ is a solution to both equations, and so is $x = 5$ with $y = 9$.

2. Linear Equations

- A. A **linear equation** has a linear polynomial on one side of the equals sign and either a linear polynomial or a constant on the other side—or can be converted to that form. A linear equation with only one variable is a **linear equation with one unknown**. A linear equation with two variables is a **linear equation with two unknowns**.

Examples:

$5x - 2 = 9 - x$ is a linear equation with one unknown.

$3x + 1 = y - 2$ is a linear equation with two unknowns.

- B. To solve a linear equation with one unknown (that is, to find what value of the unknown satisfies the equation), isolate the unknown on one side of the equation by doing the same operations on both sides. Adding or subtracting the same number on both sides of the equation doesn't change the equality. Likewise, multiplying or dividing both sides by the same nonzero number doesn't change the equality.

Example:

To solve the equation $\frac{5x - 6}{3} = 4$, isolate the variable x like this:

$$5x - 6 = 12 \quad \text{multiply both sides by 3}$$

$$5x = 18 \quad \text{add 6 to both sides}$$

$$x = \frac{18}{5} \quad \text{divide both sides by 5}$$

To check the answer $\frac{18}{5}$, substitute it for x in the original equation to confirm it satisfies that equation:

$$\frac{\left(5\left(\frac{18}{5}\right) - 6\right)}{3} = \frac{(18 - 6)}{3} = \frac{12}{3} = 4$$

So, $x = \frac{18}{5}$ is the solution.

- C. Two equivalent linear equations with the same two unknowns have infinitely many solutions, as in the example of the equivalent equations $3x - y = 6$ and $6x - 2y = 12$ in Section 3.2.1.I above. But if two linear equations with the same two unknowns aren't equivalent, they have at most one solution.

When solving two linear equations with two unknowns, if you reach a trivial equation like $0 = 0$, the equations are equivalent and have infinitely many solutions. But if you reach a contradiction, the equations have no solution.

Example:

Consider the two equations $3x + 4y = 17$ and $6x + 8y = 35$. Note that $3x + 4y = 17$ implies $6x + 8y = 34$, contradicting the second equation. So, no values of x and y can satisfy both equations at once. The two equations have no solution.

If neither a trivial equation nor a contradiction is reached, a unique solution can be found.

- D. To solve two linear equations with two unknowns, you can use one of the equations to express one unknown in terms of the other unknown. Then substitute this result into the second equation to make a new equation with only one unknown. Next, solve this new equation. Substitute the value of its unknown into either original equation to solve for the remaining unknown.

Example:

Let's solve these two equations for x and y :

$$(1) \quad 3x + 2y = 11$$

$$(2) \quad x - y = 2$$

In equation (2), $x = 2 + y$. So, in equation (1), substitute $2 + y$ for x :

$$3(2 + y) + 2y = 11$$

$$6 + 3y + 2y = 11$$

$$6 + 5y = 11$$

$$5y = 5$$

$$y = 1$$

Since $y = 1$, we find $x - 1 = 2$, so $x = 2 + 1 = 3$.

- E. Another way to remove one unknown and solve for x and y is to make the coefficients of one unknown the same in both equations (ignoring the sign). Then either add the equations or subtract one from the other.

Example:

Let's solve the equations:

$$(1) \ 6x + 5y = 29 \text{ and}$$

$$(2) \ 4x - 3y = -6$$

Multiply equation (1) by 3 and equation (2) by 5 to get

$$18x + 15y = 87 \text{ and}$$

$$20x - 15y = -30$$

Add the two equations to remove y . This gives us $38x = 57$, or

$$x = \frac{3}{2}.$$

Substituting $\frac{3}{2}$ for x in either original equation gives $y = 4$. To check these answers, substitute both values into both the original equations.

3. Factoring and Quadratic Equations

- A. Some equations can be solved by **factoring**. To do this, first add or subtract to bring all the expressions to one side of the equation, with 0 on the other side. Then try to express the nonzero side as a product of factors that are algebraic expressions. When that's possible, setting any of these factors equal to 0 makes a simpler equation. The solutions of the simpler equations made this way are also solutions of the factored equation.

Example:

Factor to find the solutions of the equation

$$x^3 - 2x^2 + x = -5(x - 1)^2:$$

$$x^3 - 2x^2 + x + 5(x - 1)^2 = 0$$

$$x(x^2 - 2x + 1) + 5(x - 1)^2 = 0$$

$$x(x - 1)^2 + 5(x - 1)^2 = 0$$

$$(x + 5)(x - 1)^2 = 0$$

$$x + 5 = 0 \text{ or } x - 1 = 0$$

$$x = -5 \text{ or } x = 1$$

So, $x = -5$ or $x = 1$.

- B. When factoring to solve equations with algebraic fractions, note that a fraction equals 0 if and only if its numerator equals 0 and its denominator doesn't.

Example:

Find the solutions of the equation $\frac{x(x - 3)(x^2 + 5)}{x - 4} = 0$.

The denominator must not equal 0, so $x - 4 \neq 0$. Thus, $x \neq 4$.

But the numerator must equal 0: $x(x - 3)(x^2 + 5) = 0$.

Thus, $x = 0$, or $x - 3 = 0$, or $x^2 + 5 = 0$. So, $x = 0$, or $x = 3$, or $x^2 + 5 = 0$.

But $x^2 + 5 = 0$ has no real solution because $x^2 + 5 \geq 0$ for every real number x . So, the original equation's solutions are 0 and 3.

- C. Here are some common algebraic equations used for factoring. They hold for any real numbers x , y , and z :

Examples:

$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$$

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x - y)^2 = x^2 - 2xy + y^2$$

$$(x + y)(x - y) = x^2 - y^2$$

$$x^2 + y^2 = (x + y)^2 - 2xy = (x - y)^2 + 2xy$$

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$$

D. A **quadratic equation** has the standard form $ax^2 + bx + c = 0$, where a , b , and c are real numbers and $a \neq 0$.

Examples:

$$x^2 + 6x + 5 = 0,$$

$$3x^2 - 2x = 0, \text{ and}$$

$$x^2 + 4 = 0.$$

E. Some quadratic equations are easily solved by factoring.

Example (1):

$$x^2 + 6x + 5 = 0$$

$$(x + 5)(x + 1) = 0$$

$$x + 5 = 0 \text{ or } \\ x + 1 = 0$$

$$x = -5 \text{ or } x = -1$$

Example (2):

$$3x^2 - 3 = 8x$$

$$3x^2 - 8x - 3 = 0$$

$$(3x + 1)(x - 3) = 0$$

$$3x + 1 = 0 \text{ or } \\ x - 3 = 0$$

$$x = -\frac{1}{3} \text{ or } x = 3$$

F. A quadratic equation has at most two real roots but may have just one or even no real root.

Examples:

The equation $x^2 - 6x + 9 = 0$ can be written as $(x - 3)^2 = 0$ or $(x - 3)(x - 3) = 0$. So, its only root is 3.

The equation $x^2 + 4 = 0$ has no real root. Since any real number squared is greater than or equal to zero, $x^2 + 4$ must be greater than or equal to 4 if x is a real number.

G. An expression of the form $a^2 - b^2$ can be factored as $(a - b)(a + b)$.

Example:

We can solve the quadratic equation $9x^2 - 25 = 0$ like this:

$$\begin{aligned}(3x - 5)(3x + 5) &= 0 \\ 3x - 5 = 0 \text{ or } 3x + 5 &= 0 \\ x = \frac{5}{3} \text{ or } x &= -\frac{5}{3}\end{aligned}$$

H. If a quadratic expression isn't easily factored, we can still find its roots with the **quadratic formula**: If $ax^2 + bx + c = 0$ and $a \neq 0$, the roots are

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

These roots are two distinct real numbers unless $b^2 - 4ac \leq 0$.

If $b^2 - 4ac = 0$, the two root expressions both equal $-\frac{b}{2a}$, so the equation has only one root.

If $b^2 - 4ac < 0$, then $\sqrt{b^2 - 4ac}$ is not a real number, so the equation has no real root.

4. Inequalities

A. An **inequality** is a statement with one of these symbols:

$\neq$ is not equal to

$>$ is greater than

$\geq$ is greater than or equal to

$<$ is less than

$\leq$ is less than or equal to

Examples:

$$5x - 3 < 9 \text{ and } 6x \geq y$$

- B. Solve a linear inequality with one unknown the same way you solve a linear equation: isolate the unknown on one side. As in an equation, the same number can be added to or subtracted from both sides of the inequality. And you can multiply or divide both sides by a positive number without changing the order of the inequality. However, multiplying or dividing an inequality by a negative number reverses the order of the inequality. Thus, $6 > 2$, but $(-1)(6) < (-1)(2)$.

Example (1):

To solve the inequality
 $3x - 2 > 5$ for x , isolate x :

$$3x - 2 > 5$$

$$3x > 7 \quad (\text{add 2 to both sides})$$

$$x > \frac{7}{3} \quad (\text{divide both sides by 3})$$

Example (2):

To solve the inequality
 $\frac{5x - 1}{-2} < 3$ for x , isolate x :

$$\frac{5x - 1}{-2} < 3$$

$$5x - 1 > -6 \quad (\text{multiply both sides by } -2)$$

$$5x > -5 \quad (\text{add 1 to both sides})$$

$$x > -1 \quad (\text{divide both sides by 5})$$

5. Functions

- A. An algebraic expression in one variable can define a **function** of that variable. A function is written as a letter like f or g along with the variable

in the expression. Function notation is a short way to express that a value is being substituted for a variable.

Examples:

i. The expression $x^3 - 5x^2 + 2$ can define a function f written as $f(x) = x^3 - 5x^2 + 2$.

ii. The expression $\frac{2z + 7}{\sqrt{z + 1}}$ can define a function g written as $g(z) = \frac{2z + 7}{\sqrt{z + 1}}$.

In these examples, the symbols “ $f(x)$ ” and “ $g(z)$ ” don’t stand for products. Each is just a symbol for a function. Pronounce “ $f(x)$ ” as “ f of x ” and “ $g(z)$ ” as “ g of z .”

The substitution of 1 for x in the first expression can be written as $f(1) = -2$. Then $f(1)$ is called the “value of f at $x = 1$.”

Likewise, in the second expression the value of g at $z = 0$ is $g(0) = 7$.

B. Once a function $f(x)$ is defined, think of x as an input and $f(x)$ as the output. In any function, any one input gives at most one output. But different inputs can give the same output.

Example:

If $h(x) = |x + 3|$, then $h(-4) = 1 = h(-2)$.

C. The set of all allowed inputs for a function is the function’s **domain**. In the examples in Section 3.2.5.A above, the domain of f is the set of all real numbers, and the domain of g is the set of all numbers greater than -1 .

Any function’s definition can restrict the function’s domain. For example, the definition “ $a(x) = 9x - 5$ for $0 \leq x \leq 10$ ” restricts the domain of a to real numbers greater than or equal to 0 but less than or equal to 10. If the

definition has no restrictions, the domain is the set of all values of x that each give a real output when input into the function.

D. The set of a function's outputs is the function's **range**.

Examples:

- i. For the function $h(x) = |x + 3|$ in the example in Section 3.2.5.B above, the range is the set of all numbers greater than or equal to 0.
- ii. For the function $a(x) = 9x - 5$ for $0 \leq x \leq 10$ defined in Section 3.2.5.C above, the range is the set of all real numbers y such that $-5 \leq y \leq 85$.

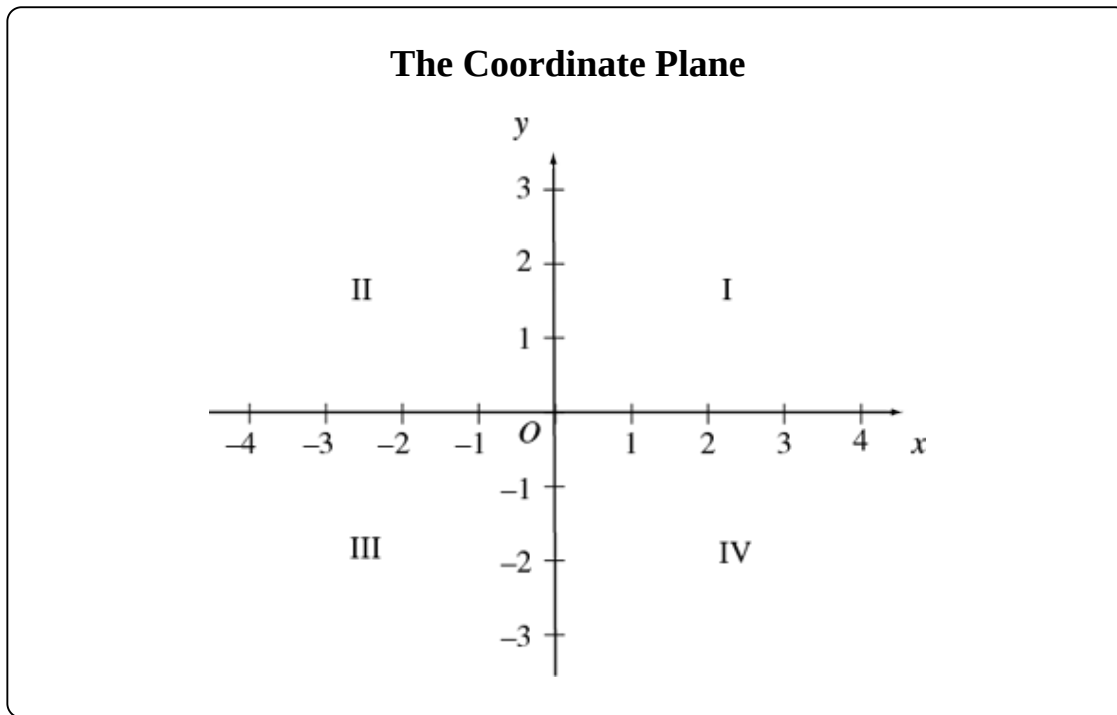
E. A **quadratic function** has the form $f(x) = ax^2 + bx + c$ where a , b , and c are constants and $a \neq 0$. If $a > 0$, then $f(x)$ has a unique minimum value but no maximum value. Otherwise, it has a unique maximum value but no minimum value. In either case, the unique minimum or maximum value of $f(x)$ is $c - \frac{b^2}{4a}$. This value occurs only when $x = -\frac{b}{2a}$.

Examples:

- i. The quadratic function $f(x) = 2x^2 - 3x + 4$ has a unique minimum value of $4 - \frac{(-3)^2}{4 \times 2} = 4 - \frac{9}{8} = \frac{25}{8}$. This minimum value occurs only when $x = -\left(\frac{-3}{2 \times 2}\right) = \frac{3}{4}$.
- ii. The quadratic function $g(x) = -3x^2 + 2x + 1$ has a unique maximum value of $1 - \frac{2^2}{4 \times -3} = 1 - \frac{-1}{3} = \frac{4}{3}$. This maximum value occurs only when $x = -\left(\frac{2}{2 \times -3}\right) = \frac{1}{3}$.

6. Graphing

- A. The figure below shows the rectangular **coordinate plane**. The horizontal line is the **x -axis** and the vertical line is the **y -axis**. These two axes intersect at the **origin**, called **O** . The axes divide the plane into four quadrants: I, II, III, and IV, as shown.

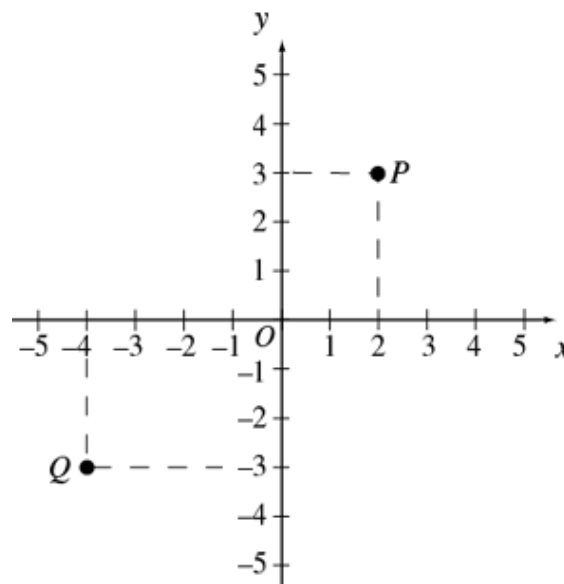


- B. Any ordered pair (x, y) of real numbers defines a point in the coordinate plane. The point's **x -coordinate** is the first number in this pair. It shows how far the point is to the right or left of the y -axis. If the x -coordinate is positive, the point is to the right of the y -axis. If it's negative, the point is to the left of the y -axis. If it's 0, the point is on the y -axis. The point's **y -coordinate** is the second number in the ordered pair. It shows how far the point is above or below the x -axis. If the y -coordinate is positive, the point is above the x -axis. If it's negative, the point is below the x -axis. If it's 0, the point is on the x -axis.

Example:

In the graph below, the (x, y) coordinates of point P are $(2, 3)$. P is 2 units to the right of the y -axis, so $x = 2$. Since P is 3 units above the x -axis $y = 3$.

Likewise, the (x, y) coordinates of point Q are $(-4, -3)$. The origin O has coordinates $(0, 0)$.



- C. The coordinates of the points on a straight line in the coordinate plane satisfy a linear equation of the form $y = mx + b$ (or the form $x = a$ if the line is vertical).

In the equation $y = mx + b$, the coefficient m is the line's **slope**, and the constant b is the line's ***y*-intercept**.

The ***y*-intercept** is the y -coordinate of the point where the line intersects the y -axis. Likewise, the ***x*-intercept** is the x -coordinate of the point where the line intersects the x -axis.

For any two points on the line, the line's slope is the ratio of the difference in their y -coordinates to the difference in their x -coordinates. To find the slope, subtract one point's y -coordinate from the other's. Then subtract the former point's x -coordinate from the latter's—not the other way around! Then divide the first difference by the second.

If a line's slope is negative, the line slants down from left to right.

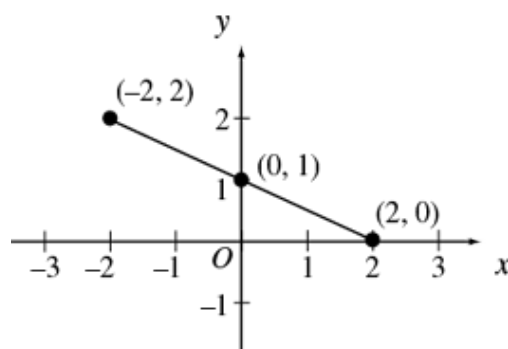
If the slope is positive, the line slants up.

If the slope is 0, the line is horizontal. A horizontal line's equation has the form $y = b$, since $m = 0$.

A vertical line's slope is undefined.

Example:

In the graph below, each point on the line satisfies the equation $y = -\frac{1}{2}x + 1$. To check this for the points $(-2, 2)$, $(2, 0)$, and $(0, 1)$, substitute each point's coordinates for x and y in the equation.



You can use the points $(-2, 2)$ and $(2, 0)$ to find the line's slope:

$$\frac{\text{the difference in the } y\text{-coordinates}}{\text{the difference in the } x\text{-coordinates}} = \frac{0 - 2}{2 - (-2)} = \frac{-2}{4} = -\frac{1}{2}.$$

The y -intercept is 1. That's y 's value when x is 0 in $y = -\frac{1}{2}x + 1$.

To find the x -intercept, set y to 0 in the same equation:

$$\begin{aligned} -\frac{1}{2}x + 1 &= 0 \\ -\frac{1}{2}x &= -1 \\ x &= 2 \end{aligned}$$

Thus, the x -intercept is 2.

D. You can use the definition of slope to find an equation of the line through two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$. The slope is

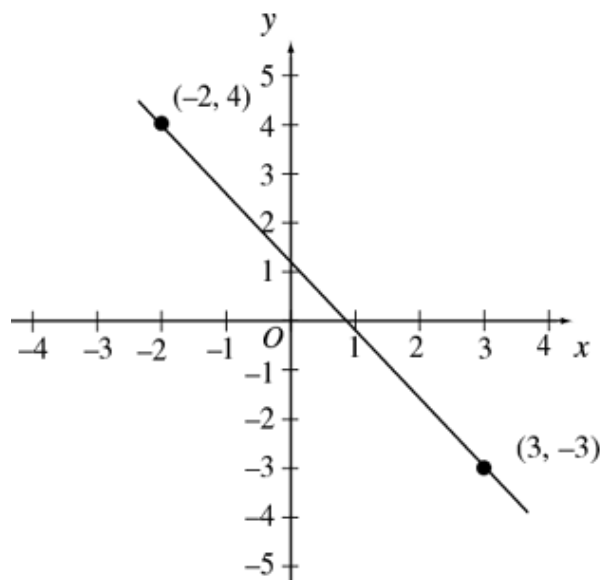
$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Given the known point (x_1, y_1) and the slope m , any

other point (x, y) on the line satisfies the equation $m = \frac{(y - y_1)}{(x - x_1)}$, or
equivalently $(y - y_1) = m(x - x_1)$. Using (x_2, y_2) instead of (x_1, y_1)
as the known point gives an equivalent equation.

Example:

The graph below shows points $(-2, 4)$ and $(3, -3)$.



The line's slope is $\frac{(-3 - 4)}{(3 - (-2))} = -\frac{7}{5}$. To find an equation of this line, let's use the point $(3, -3)$:

$$y - (-3) = \left(-\frac{7}{5}\right)(x - 3)$$

$$y + 3 = \left(-\frac{7}{5}\right)x + \frac{21}{5}$$

$$y = \left(-\frac{7}{5}\right)x + \frac{6}{5}$$

So, the y -intercept is $\frac{6}{5}$.

Find the x -intercept like this:

$$\begin{aligned} 0 &= -\frac{7}{5}x + \frac{6}{5} \\ \frac{7}{5}x &= \frac{6}{5} \\ x &= \frac{6}{7} \end{aligned}$$

The graph shows both these intercepts.

- E. If two linear equations with unknowns x and y have a unique shared solution, their graphs are two lines intersecting at the shared point that is the solution.

If two linear equations are equivalent, they both stand for the same line and have infinitely many shared solutions.

Two linear equations with no shared solution stand for two parallel lines.

- F. Graph any function $f(x)$ in the coordinate plane by equating y with the function's value: $y = f(x)$. For any x in the function's domain, the point $(x, f(x))$ is on the function's graph. For every point on the graph, the y -coordinate is the function's value at the x -coordinate.

Example:

Consider the function $f(x) = -\frac{7}{5}x + \frac{6}{5}$.

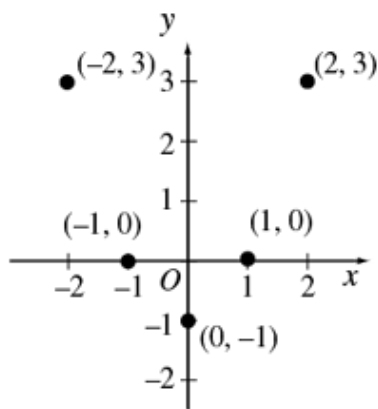
If $f(x)$ is equated with the variable y , the function's graph is the graph of $y = -\frac{7}{5}x + \frac{6}{5}$ in the example above.

- G. For any function f , the x -intercepts are the solutions of the equation $f(x) = 0$. The y -intercept is the value $f(0)$.

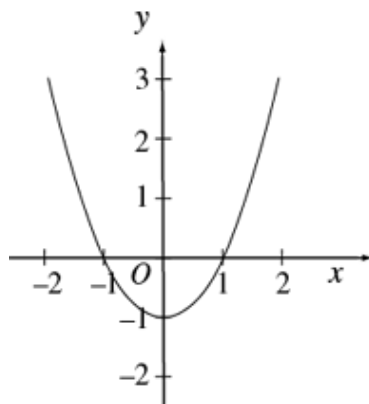
Example:

To see how the quadratic function $f(x) = x^2 - 1$ relates to its graph, let's plot some points $(x, f(x))$ in the coordinate plane:

x	$f(x)$
-2	3
-1	0
0	-1
1	0
2	3



The graph below shows all the points for $-2 \leq x \leq 2$:



The roots of this equation $f(x) = x^2 - 1 = 0$ are $x = 1$ and $x = -1$. They match the x -intercepts since x -intercepts are found by setting $y = 0$ and solving for x .

The y -intercept is $f(0) = -1$ because that's the value of y for $x = 0$.

7. Formulas and Measurement Conversion

- A. A **formula** is an algebraic equation with variables that have specific meanings. To use a formula, assign quantities to its variables to match these meanings.

Example:

In the physics formula $F = ma$, the variable F stands for force, the variable m stands for mass, and the variable a stands for acceleration. The standard metric unit of force, the newton, is just enough force to accelerate a mass of 1 kilogram by 1 meter/second².

If we know a rock with a mass of 2 kilograms is accelerating at 5 meters/second², we can use the formula $F = ma$ by setting the variable m to 2 kilograms, and the variable a to 5 meters/second². Then we find that 10 newtons of force F are pushing the rock.

Note: You don't need to learn physics formulas or terms like this for the GMAT, but some specific GMAT questions may give you the formulas and terms you need to solve them.

- B. Any quantitative relationship between units of measure can be written as a formula.

Examples:

- i. Since 1 kilometer is 1,000 meters, the formula $m = 1000k$ can stand for the relationship between kilometers (k) and meters (m).
- ii. The formula $C = \frac{5}{9}(F - 32)$ can stand for the relationship between temperature measurements in degrees Celsius (C) and degrees Fahrenheit (F).

- C. Except for units of time, a GMAT question that requires converting one unit of measure to another will give the relationship between those units.

Example:

A train travels at a constant 25 meters per second. How many kilometers does it travel in 5 minutes? (1 kilometer = 1,000 meters)

Solution: In 1 minute the train travels $(25)(60) = 1,500$ meters, so in 5 minutes it travels 7,500 meters. Since 1 kilometer = 1,000 meters, we find 7,500 meters = 7.5 kilometers.

3.3 Rates, Ratios, and Percents

1. Ratio and Proportion

- A. The **ratio** of a number x to a nonzero number y may be written as $x:y$, or $\frac{x}{y}$, or x to y . In each of these expressions, x is the numerator and y is the denominator. The order of a ratio's terms is important. Unless the absolute values of x and y are equal, $\frac{x}{y} \neq \frac{y}{x}$.

Examples:

The ratio of 2 to 3 may be written as 2:3, or $\frac{2}{3}$, or 2 to 3.

The ratio of the number of months with exactly 30 days to the number of months with exactly 31 days is 4:7, not 7:4.

- B. A **proportion** is an equation between two ratios. Remember, the denominator on either side of the proportion must not equal zero.

Example:

2:3 = 8:12 is a proportion.

- C. One way to solve for an unknown in a proportion is to cross multiply, then solve the resulting equation.

Example:

To solve for n in the proportion $\frac{2}{3} = \frac{n}{12}$, cross multiply to get $3n = 24$, then divide both sides by 3 to find $n = 8$.

- D. Some word problems can be solved using ratios.

Example:

If 5 shirts cost a total of \$44, then what is the total cost of 8 shirts at the same cost per shirt?

Solution: If c is the cost of the 8 shirts, then $\frac{5}{44} = \frac{8}{c}$. Cross multiplying gives $5c = 8 \times 44 = 352$, so $c = \frac{352}{5} = 70.4$. Thus, the 8 shirts cost a total of \$70.40.

2. Fractions

- A. In a fraction $\frac{n}{d}$, n is the **numerator** and d is the **denominator**. A fraction's denominator can never be 0 because division by 0 is undefined.
- B. **Equivalent** fractions stand for the same number. To check whether two fractions with numerators and denominators that are integers are equivalent, divide each fraction's numerator and denominator by the largest factor common to that numerator and that denominator, their **greatest common divisor** (GCD). This is called **reducing each fraction to its lowest terms**. Two fractions are equivalent if and only if reducing each to its lowest terms makes them identical.

Example:

To check whether $\frac{8}{36}$ and $\frac{14}{63}$ are equivalent, first reduce each to its lowest terms. In the first fraction, 4 is the GCD of the numerator 8 and the denominator 36. Dividing both the numerator and the denominator of $\frac{8}{36}$ by 4 gives $\frac{2}{9}$. In the second fraction, 7 is the GCD of the numerator 14 and the denominator 63. Dividing both the numerator and the denominator of $\frac{14}{63}$ by 7 also gives $\frac{2}{9}$. Since reducing $\frac{8}{36}$ and $\frac{14}{63}$ to their lowest terms makes them identical, they're equivalent.

- C. To add or subtract two fractions with the same denominator, just add or subtract the numerators, leaving the denominators the same.

Examples:

$$\frac{3}{5} + \frac{4}{5} = \frac{3+4}{5} = \frac{7}{5} \text{ and}$$

$$\frac{5}{7} - \frac{2}{7} = \frac{5-2}{7} = \frac{3}{7}$$

- D. To add or subtract two fractions with different denominators, first express them as fractions with the same denominator.

Example:

To add $\frac{3}{5}$ and $\frac{4}{7}$, multiply the numerator and denominator of $\frac{3}{5}$ by 7 to get $\frac{21}{35}$. Then multiply the numerator and denominator of $\frac{4}{7}$ by 5 to get $\frac{20}{35}$. Since both fractions now have the same denominator 35, you can easily add them: $\frac{3}{5} + \frac{4}{7} = \frac{21}{35} + \frac{20}{35} = \frac{41}{35}$

- E. To multiply two fractions, multiply their numerators, and also multiply their denominators.

Example:

$$\frac{2}{3} \times \frac{4}{7} = \frac{2 \times 4}{3 \times 7} = \frac{8}{21}$$

- F. The **reciprocal** of a fraction $\frac{n}{d}$ is $\frac{d}{n}$ if neither n nor d is 0.

Example:

The reciprocal of $\frac{4}{7}$ is $\frac{7}{4}$.

G. To divide by a fraction, multiply by its reciprocal.

Example:

$$\frac{2}{3} \div \frac{4}{7} = \frac{2}{3} \times \frac{7}{4} = \frac{14}{12} = \frac{7}{6}$$

H. A **mixed number** is written as an integer next to a fraction. It equals the integer plus the fraction.

Example:

The mixed number $7\frac{2}{3} = 7 + \frac{2}{3}$

I. To write a mixed number as a fraction, multiply the integer part of the mixed number by the denominator of the fractional part. Add this product to the numerator. Then put this sum over the denominator.

Example:

$$7\frac{2}{3} = \frac{(7 \times 3) + 2}{3} = \frac{23}{3}$$

3. Percents

A. The word **percent** means **per hundred** or **number out of 100**.

Example:

Saying that 37 percent, or 37%, of the houses in a city are painted blue means that 37 houses per 100 in the city are painted blue.

B. A percent may be greater than 100.

Example:

Saying that the number of blue houses in a city is 150% of the number of red houses means the city has 150 blue houses for every 100 red houses. Since $150:100 = 3:2$, this means the city has 3 blue houses for every 2 red houses.

C. A percent need not be an integer.

Example:

Saying that the number of pink houses in a city is 0.5% of the number of blue houses means the city has 0.5 of a pink house for every 100 blue houses. Since $0.5:100 = 1:200$, this means the city has 1 pink house for every 200 blue houses.

Likewise, saying that the number of orange houses is 12.5% of the number of blue houses means the ratio of orange houses to blue houses is $12.5:100 = 1:8$. Therefore, there is 1 orange house for every 8 blue houses.

4. Converting Decimals, Fractions, and Percents

A. Decimals with many finite digits can be rewritten as fractions or sums of fractions.

Examples:

$$0.321 = \frac{3}{10} + \frac{2}{100} + \frac{1}{1,000} = \frac{321}{1,000}$$

$$0.0321 = \frac{0}{10} + \frac{3}{100} + \frac{2}{1,000} + \frac{1}{10,000} = \frac{321}{10,000}$$

$$1.56 = 1 + \frac{5}{10} + \frac{6}{100} = \frac{156}{100}$$

- B. Repeating decimals can also be rewritten as fractions. To do this, set a variable x equal to the repeating decimal. Next multiply both sides by a power of 10 just large enough to make the repeating digits start right after the decimal if they don't already. Call this Equation A. Then make another equation by multiplying both sides of Equation A by a power of 10 just large enough to move one copy of the repeating digits from the right to the left of the decimal point. From this, subtract Equation A, and solve for x .

Examples:

i. To rewrite the repeating decimal $3.\overline{6}$ as a fraction, first let $x = 3.\overline{6}$. This equation already has the repeating digit 6 starting right after the decimal, so it's our Equation A. Next we multiply both sides by 10 to move one repeating 6 from the right to the left of the decimal point: $10x = 36.\overline{6}$. From this, subtract Equation A to get $9x = 36.\overline{6} - 3.\overline{6} = 33$. Finally divide both sides by 9 to solve for x : $x = \frac{33}{9} = 3\frac{2}{3}$.

ii. To rewrite the repeating decimal $1.2\overline{79}$ as a fraction, first let $x = 1.2\overline{79}$. Next multiply both sides by 10 to make the repeating digits start right after the decimal: $10x = 12.\overline{79}$. This is our Equation A. Then multiply both sides by 100 to move one copy of the repeating digits 79 from the right to the left of the decimal point: $1000x = 1279.\overline{79}$. From this, subtract Equation A to get $990x = 1279.\overline{79} - 12.\overline{79} = 1267$. Finally divide both sides by 990 to solve for x : $x = \frac{1267}{990} = 1\frac{277}{990}$.

C. To rewrite a percent as a fraction, write the percent number as the numerator over a denominator of 100. To rewrite a percent as a decimal, move the decimal point in the percent two places to the left and drop the percent sign. To rewrite a decimal as a percent, move the decimal point two places to the right, then add a percent sign.

Examples:

$$37\% = \frac{37}{100} = 0.37$$

$$300\% = \frac{300}{100} = 3$$

$$0.5\% = \frac{0.5}{100} = 0.005$$

- D. To find a percent of a number, multiply the number by the percent written as a fraction or decimal.

Examples:

$$20\% \text{ of } 90 = 90 \left(\frac{20}{100} \right) = 90 \left(\frac{1}{5} \right) = \frac{90}{5} = 18$$

$$20\% \text{ of } 90 = 90(0.2) = 18$$

$$250\% \text{ of } 80 = 80 \left(\frac{250}{100} \right) = 80(2.5) = 200$$

$$0.5\% \text{ of } 12 = 12 \left(\frac{0.5}{100} \right) = 12(0.005) = 0.06$$

5. Working with Decimals, Fractions, and Percents

- A. To find the percent increase or decrease from one positive quantity to another, first find the amount of increase or decrease. Then divide this amount by the original quantity. Write this quotient as a percent.

Examples:

Suppose a price increases from \$24 to \$30. To find the percent increase, first find the amount of increase: $\$30 - \$24 = \$6$. Divide this \$6 by the original price of \$24 to find the percent increase:

$$\frac{6}{24} = 0.25 = 25\%.$$

Now suppose a price falls from \$30 to \$24. The amount of decrease is $\$30 - \$24 = \$6$. So, the percent decrease is $\frac{6}{30} = 0.20 = 20\%$.

Notice the percent **increase** from 24 to 30 (25%) doesn't equal the percent **decrease** from 30 to 24 (20%).

A percent increase may be greater than 100%, but a percent decrease cannot be.

Example:

Suppose a house's price in 2018 was 300% of its price in 2003, which was above 0. By what percent did the price increase?

Solution: If n is the price in 2003, the percent increase is

$$\left| \frac{(3n - n)}{n} \right| = \left| \frac{2n}{n} \right| = 2, \quad \frac{(3n - n)}{n} = \frac{2n}{n} = 2, \text{ or } 200\%.$$

B. A price discounted by n percent is $(100 - n)$ percent of the original price.

Example:

A customer paid \$24 for a dress. If the customer got a 25% discount off the original price of the dress, what was the original price before the discount?

Solution: The discounted price is $100\% - 25\% = 75\%$ of the original price. So, if p is the original price, $0.75p = \$24$ is the discounted price. Thus, $p = (\$24/0.75) = \32 , the original price before the discount.

Two discounts can be combined to make a larger discount.

Example:

A price is discounted 20%. Then this reduced price is discounted another 30%. These two discounts together make an overall discount of what percent?

Solution: If p is the original price, then $0.8p$ is the price after the first discount. The price after the second discount is $(0.7)(0.8)p = 0.56p$. The overall discount is $100\% - 56\% = 44\%$.

C. **Profit** equals revenues minus expenses, or selling price minus total costs.

Example:

A certain appliance costs a merchant a total of \$30 to acquire and sell. At what price should the merchant sell the appliance to make a profit of 50% of the appliance's total cost to the merchant?

Solution: The merchant should sell the appliance for a price $\$s$ such that $s - 30 = (0.5)(30)$. So, $\$s = \$30 + \$15 = \45 .

D. **Simple annual interest** on a loan or investment is based only on the original loan or investment amount (the **principal**). It equals (principal) $\times$ (interest rate) $\times$ (time).

Example:

If \$8,000 is invested at 6% simple annual interest, how much interest is earned in 3 months?

Solution: Since the annual interest rate is 6%, the interest for 1 year is $(0.06)(\$8,000) = \480 . A year has 12 months, so the interest earned in 3 months is $\left(\frac{3}{12}\right)(\$480) = \$120$.

E. **Compound interest** is based on the principal plus any interest already earned.

Compound interest over
 n periods = (principal) $\times$ $(1 + \text{interest per period})^n$ - principal
.

Example:

If \$10,000 is invested at 10% annual interest, compounded every 6 months, what is the investment's value after 1 year?

Solution: Since the interest is compounded every 6 months, or twice a year, the interest rate for each 6-month period is 5%, half the 10% annual rate. So, the investment's value after the first 6 months is $10,000 + (10,000)(0.05) = \$10,500$.

For the second 6 months, the interest is based on the \$10,500 balance after the first 6 months. So, the investment's value after 1 year is $10,500 + (10,500)(0.05) = \$11,025$.

The investment's value after 1 year can also be written as

$$10,000 \times \left(1 + \frac{0.10}{2}\right)^2 \text{ dollars.}$$

- F. To solve some word problems with percents and fractions, you can organize the information in a table.

Example:

In a production lot, 40% of the toys are red, and the rest are green. Half of the toys are small, and half are large. If 10% of the toys are red and small, and 40 toys are green and large, how many of the toys are red and large?

Solution: First make a table to organize the information:

	Red	Green	Total
Small	10%		50%
Large			50%
Total	40%	60%	100%

Then fill in the missing percents so that the “Red” and “Green” percents in each row add up to that row’s total, and the “Small” and “Large” percents in each column add up to that column’s total:

	Red	Green	Total
Small	10%	40%	50%
Large	30%	20%	50%
Total	40%	60%	100%

The number of large green toys, 40, is 20% of the total number of toys (n), so $0.20n = 40$. Thus, the total number of toys $n = 200$. So, 30% of the 200 toys are red and large. Since $(0.3)(200) = 60$, we find that 60 of the toys are red and large.

6. Rate, Work, and Mixture Problems

- A. The distance an object travels is its average speed multiplied by the time it takes to travel that distance. That is, ***distance = rate × time***.

Example:

How many kilometers did a car travel in 4 hours at an average speed of 70 kilometers per hour?

Solution: Since distance = rate × time, multiply 70 kilometers/hour × 4 hours to find that the car went 280 kilometers.

- B. To find an object's average travel speed, divide the total travel distance by the total travel time.

Example:

On a 600-kilometer trip, a car went half the distance at an average speed of 60 kilometers per hour (kph), and the other half at an average speed of 100 kph. The car didn't stop between the two halves of the trip. What was the car's average speed over the whole trip?

Solution: First find the total travel time. For the first 300 kilometers, the car went at 60 kph, taking $\frac{300 \text{ kilometers}}{60 \text{ kph}} = 5$ hours. For the

second 300 kilometers, the car went at 100 kph, taking $\frac{300 \text{ kilometers}}{100 \text{ kph}} = 3$ hours. So, the total travel time was $5 + 3 = 8$

hours. The car's average speed was $\frac{600 \text{ kilometers}}{8 \text{ hours}} = 75$ kph.

Notice the average speed was not $\frac{(60 + 100) \text{ kilometers}}{2 \text{ hours}} = 80$ kph.

- C. A ***work problem*** usually says how fast certain individuals work alone, then asks you to find how fast they work together, or vice versa.

The basic formula for work problems is $\frac{1}{r} + \frac{1}{s} = \frac{1}{h}$, where r is how long an amount of work takes a certain individual, s is how long that much work takes a different individual, and h is how long that much work takes both individuals working at the same time.

Example:

Suppose one machine takes 4 hours to make 1,000 bolts, and a second machine takes 5 hours to make 1,000 bolts. How many hours do both machines working at the same time take to make 1,000 bolts?

Solution:

$$\begin{aligned}\frac{1}{4} + \frac{1}{5} &= \frac{1}{h} \\ \frac{5}{20} + \frac{4}{20} &= \frac{1}{h} \\ \frac{9}{20} &= \frac{1}{h} \\ 9h &= 20 \\ h &= \frac{20}{9} = 2\frac{2}{9}\end{aligned}$$

Working together, the two machines can make 1,000 bolts in $2\frac{2}{9}$ hours.

If a work problem says how long it takes two individuals to do an amount of work together, and how long it takes one of them to do that much work alone, you can use the same formula to find how long it takes the other individual to do that much work alone.

Example:

Suppose Art and Rita both working at the same time take 4 hours to do an amount of work, and Art alone takes 6 hours to do that much work. Then how many hours does Rita alone take to do that much work?

Solution:

$$\begin{aligned}\frac{1}{6} + \frac{1}{R} &= \frac{1}{4} \\ \frac{1}{R} &= \frac{1}{4} - \frac{1}{6} = \frac{1}{12} \\ R &= 12\end{aligned}$$

Rita alone takes 12 hours to do that much work.

- D. In ***mixture problems***, substances with different properties are mixed, and you must find the mixture's properties.

Example:

If 6 kilograms of nuts that cost \$1.20 per kilogram are mixed with 2 kilograms of nuts that cost \$1.60 per kilogram, how much does the mixture cost per kilogram?

Solution: The 8 kilograms of nuts cost a total of $6(\$1.20) + 2(\$1.60) = \$10.40$. So, the cost per kilogram of the mixture is $\frac{\$10.40}{8} = \1.30 .

Some mixture problems use percents.

Example:

How many liters of a solution that is 15% salt must be added to 5 liters of a solution that is 8% salt to make a solution that is 10% salt?

Solution: Let n be the needed number of liters of the 15% solution. The amount of salt in n liters of 15% solution is $0.15n$. The amount of salt in the 5 liters of 8% solution is $(0.08)(5)$. These amounts add up to the amount of salt in the 10% mixture, which is $0.10(n + 5)$. So,

$$0.15n + 0.08(5) = 0.10(n + 5)$$

$$15n + 40 = 10n + 50$$

$$5n = 10$$

$$n = 2 \text{ liters}$$

So, 2 liters of the 15% salt solution must be added to the 8% solution to make the 10% solution.

3.4 Statistics, Sets, Counting, Probability, Estimation, and Series

1. Statistics

- A. A common statistical measure is the **average** or **(arithmetic) mean**, a type of center for the numerical values in a data set. The average or mean of n values is the sum of the n values divided by n . If a number appears more than once in the data set, each instance of that number is a separate value and is added separately to find the sum. To learn more about values in data sets, see [Chapter 5](#), “Data Insights Review.”

Example:

The average of the 5 values 6, 4, 7, 10, and 4 is

$$\frac{(6 + 4 + 7 + 10 + 4)}{5} = \frac{31}{5} = 6.2.$$
 Notice that 4 appears twice in the data set of 5 values, so it is added twice when finding the sum.

- B. The **median** is another type of center for ordered values in a data set. As above, if a number appears more than once in the data set, each instance of that number is a separate value. To find the median, arrange the values from least to greatest. If the number of entries in the data set is odd, the median is the middle entry in the data set. But if the number of entries in the data set is even, the median is the average of the two middle entries. The median may be less than, equal to, or greater than the mean.

Example:

To find the median of the 5 values 6, 4, 7, 10, and 4, arrange them from least to greatest: 4, 4, 6, 7, 10. The median is 6, the middle entry in this data set.

The median of the 6 values 4, 6, 6, 8, 9, and 12 is $\frac{(6 + 8)}{2} = 7$. But the mean of these 6 values is

$$\frac{(4 + 6 + 6 + 8 + 9 + 12)}{6} = \frac{45}{6} = 7.5.$$

Often about half the values in a data set are less than the median, and about half are greater than the median. But not always.

Example:

For the 15 values 3, 5, 7, 7, 7, 7, 7, 7, 8, 9, 9, 9, 9, 10, and 10, the median is 7. Only $\frac{2}{15}$ of the values are less than the median.

C. The **mode** of a data set is the value appearing most often in the data set.

Example:

The mode of the list of the values 1, 3, 6, 4, 3, and 5 is 3, since 3 is the only value appearing more than once in the data set.

A data set may have more than one mode.

Example:

The data set 1, 2, 3, 3, 3, 5, 7, 10, 10, 10, and 20 has two modes, 3 and 10.

D. The **dispersion** of numerical data is how spread out the data is. The simplest measure of dispersion is the **range**, which is the greatest value in the data minus the least value.

Example:

The range of the values 11, 10, 5, 13, and 21 is $21 - 5 = 16$. Notice the range depends on only 2 of the values.

E. Another common measure of dispersion is the **standard deviation**. Generally, the farther the values are from the mean, the greater the standard deviation. To find the standard deviation of n values:

1. Find their mean
2. Find the differences between the mean and each of the n values
3. Square each difference
4. Find the average of the squared differences, and
5. Take the nonnegative square root of this average

Examples:

Let's use the table below to find the standard deviation of the 5 values 0, 7, 8, 10, and 10, which have the mean 7.

x	$x - 7$	$(x - 7)^2$
0	-7	49
7	0	0
8	1	1
10	3	9
10	3	9
Total		68

The standard deviation is $\sqrt{\frac{68}{5}} \approx 3.7$

The standard deviation depends on every value in the data set, but more on those farther from the mean. This is why the standard deviation is smaller for a data set grouped closer around its mean.

As a second example, consider the values 6, 6, 6.5, 7.5, and 9, which also have the mean 7. These values are grouped closer around the mean 7 than are those in the first example. That makes the standard deviation in this second example only about 1.1, far below the standard deviation of 3.7 in the first example.

- F. How many times a value occurs in a data set is its **frequency** in the set. When different values have different frequencies, a **frequency distribution** can help show how the values are distributed.

Example:

Consider this data set of 20 numbers:

-4 0 0 -3 -2 -1 -1 0 -1 -4
-1 -5 0 -2 0 -5 -2 0 0 -1

We can show its frequency distribution in a table showing each value x and x 's frequency f :

Value x	Frequency f
-5	2
-4	2
-3	1
-2	3
-1	5
0	7
Total	20

This frequency distribution table makes finding statistical measures easier:

Mean:

$$= \frac{(-5)(2) + (-4)(2) + (-3)(1) + (-2)(3) + (-1)(5) + (0)(7)}{20} = -1.6$$

Median: -1 (the average of the 10th and 11th values)

Mode: 0 (the value that occurs most often)

Range: $0 - (-5) = 5$

Standard deviation:

$$\sqrt{\frac{(-5 + 1.6)^2 (2) + (-4 + 1.6)^2 (2) + \dots + (0 + 1.6)^2 (7)}{20}} \approx 1.7$$

2. Sets

- A. In math, a **set** is a collection of numbers or other things. The things in the set are its **elements**. A list of a set's elements between braces stands for the set. The list's order doesn't matter.

Example:

$\{-5, 0, 1\}$ is the same set as $\{0, 1, -5\}$. That is, $\{-5, 0, 1\} = \{0, 1, -5\}$.

- B. The number of elements in a finite set S is written as $|S|$.

Example:

$S = \{-5, 0, 1\}$ is a set with $|S| = 3$.

- C. If all the elements in a set S are also in a set T , then S is a **subset** of T . This is written as $S \subseteq T$ or $T \supseteq S$.

Example:

$\{-5, 0, 1\}$ is a subset of $\{-5, 0, 1, 4, 10\}$. That is, $\{-5, 0, 1\} \subseteq \{-5, 0, 1, 4, 10\}$.

- D. The **union** of two sets A and B is the set of all elements that are each in A or in B or both. The union is written as $A \cup B$.

Example:

$$\{3, 4\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$$

- E. The **intersection** of two sets A and B is the set of all elements that are each in **both** A and B . The intersection is written as $A \cap B$.

Example:

$$\{3, 4\} \cap \{4, 5, 6\} = \{4\}$$

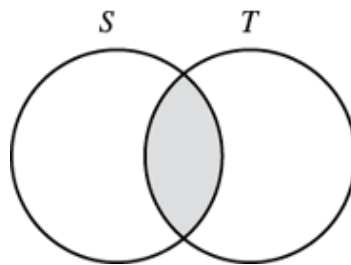
- F. Two sets sharing no elements are **disjoint** or **mutually exclusive**.

Example:

$\{-5, 0, 1\}$ and $\{4, 10\}$ are disjoint.

- G. The *empty set* is the set with no elements. It's written as $\emptyset$ or $\{ \}$.
- H. A **Venn diagram** shows the unions and intersections of two or more sets. Suppose sets S and T aren't disjoint, and neither is a subset of the other. The Venn diagram below shows their intersection $S \cap T$ as a shaded area.

A Venn Diagram of Two Intersecting Sets



A Venn diagram of sets S and T , with their intersection $S \cap T$ shaded.

- I. The number of elements in the union of two finite sets S and T is the number of elements in S , plus the number of elements in T , minus the number of elements in the intersection of S and T . That is,

$|S \cup T| = |S| + |T| - |S \cap T|$. This is the **general addition rule for two sets**.

Example:

$$|\{3, 4\} \cup \{4, 5, 6\}| = |\{3, 4\}| + |\{4, 5, 6\}| - |\{3, 4\} \cap \{4, 5, 6\}| = |\{3, 4\}| + |\{4, 5, 6\}| - 1|\{4\}| = 2 + 3 - 1 = 4.$$

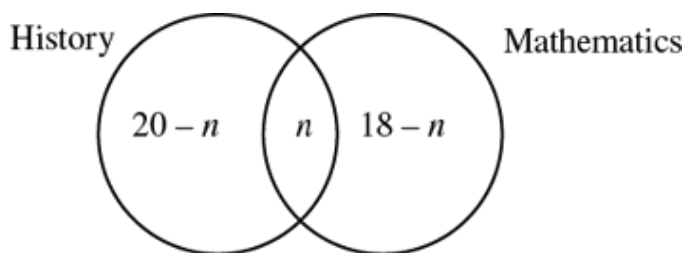
If S and T are disjoint, then $|S \cup T| = |S| + |T|$, since $|S \cap T| = 0$.

- J. You can often solve word problems involving sets by using Venn diagrams and the general addition rule.

Example:

Each of 25 students is taking history, mathematics, or both. If 20 of them are taking history and 18 of them are taking mathematics, how many of them are taking both history and mathematics?

Solution: Separate the 25 students into three disjoint sets: the students taking history only, those taking mathematics only, and those taking both history and mathematics. This gives us the Venn diagram below, where n is the number of students taking both courses, $20 - n$ is the number taking history only, and $18 - n$ is the number taking mathematics only.



Since there are 25 students total, $(20 - n) + n + (18 - n) = 25$, so $n = 13$. So, 13 students are taking both history and mathematics. Notice that $20 + 18 - 13 = 25$ uses the general addition rule for two sets.

3. Counting Methods

- A. To count elements in sets without listing them, you can sometimes use this ***multiplication principle***:

If an object will be chosen from a set of m different objects, and another object will be chosen from a disjoint set of n different objects, then mn different choices are possible.

Example:

Suppose a meal at a restaurant must include exactly 1 entree and 1 dessert. The entree can be any 1 of 5 options, and the dessert can be any 1 of 3 options. Then $5 \times 3 = 15$ different meals are available.

- B. Here's a more general version of the multiplication principle: The number of possible choices of 1 object apiece out of any number of sets is the product of the numbers of objects in those sets. For example, when choosing 1 object apiece out of 3 sets with x , y , and z elements, respectively, xyz different choices are possible. The general multiplication principle also means that when choosing 1 object apiece out of n sets of exactly m objects apiece, m^n different choices are possible.

Example:

Each time a coin is flipped, the 2 possible results are heads and tails. In a sequence of 8 consecutive coin flips, think of each flip as a set of those 2 possible results. The 8 flips give us a sequence of 8 of these 2-element sets. So, the sequence of 8 flips has 2^8 possible results.

- C. A concept often used with the multiplication principle is the ***factorial***. For any integer $n > 1$, n factorial is written as $n!$ and is the product of all the integers from 1 through n . Also, by definition, $0! = 1! = 1$.

Examples:

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24, \text{ etc.}$$

A useful equation with factorials is $n! = (n - 1)!(n)$.

- D. Any sequential ordering of a set's elements is a **permutation** of the set. A permutation is a way to choose elements one by one in a certain order.

The factorial is useful for finding how many permutations a set has. If a set of n objects is being ordered from 1st to n^{th} , there are n choices for the 1st object, $n - 1$ choices left for the 2nd object, $n - 2$ choices left for the 3rd object, and so on, until only 1 choice is left for the n^{th} object. So, by the multiplication principle, a set of n objects has $n(n - 1)(n - 2) \dots (3)(2)(1) = n!$ permutations.

Example:

The set of letters A, B, and C has $3! = 6$ permutations: ABC, ACB, BAC, BCA, CAB, and CBA.

- E. When $0 \leq k \leq n$, each possible choice of k objects out of n objects is a **combination** of n objects taken k at a time. The number of these

combinations is written as $\binom{n}{k}$. This is also the number of k -element subsets of a set with n elements, since the combinations simply are these subsets. It can be calculated as $\binom{n}{k} = \frac{n!}{k!(n - k)!}$. Note that

$$\binom{n}{k} = \binom{n}{n - k}.$$

Example:

The 2-element subsets of $S = \{A, B, C, D, E\}$ are the combinations of the 5 letters in S taken 2 at a time. There are

$\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{(2)(6)} = 10$ of these subsets: $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, $\{A, E\}$, $\{B, C\}$, $\{B, D\}$, $\{B, E\}$, $\{C, D\}$, $\{C, E\}$, and $\{D, E\}$.

For each of its 2-element subsets, a 5-element set also has exactly one 3-element subset containing the elements not in that 2-element subset. For example, in S the 3-element subset $\{C, D, E\}$ contains the elements not in the 2-element subset $\{A, B\}$, the 3-element subset $\{B, D, E\}$ contains the elements not in the 2-element subset $\{A, C\}$, and so on. This shows that a 5-element set like S has exactly as many 2-element subsets as 3-element subsets, so $\binom{5}{2} = 10 = \binom{5}{3}$.

4. Probability

- A. Sets and counting methods are also important to **discrete probability**. Discrete probability involves **experiments** with finitely many possible **outcomes**. An **event** is a set of an experiment's possible outcomes.

Example:

Rolling a 6-sided die with faces numbered 1 to 6 is an experiment with 6 possible outcomes. Let's call these outcomes 1, 2, 3, 4, 5, and 6, each number being the one facing up after the roll. Notice that no two outcomes can occur together.

One event in this experiment is that the outcome is 4. This event is written as $\{4\}$.

Another event in the experiment is that the outcome is an odd number. This event has the three outcomes 1, 3, and 5. It is written as $\{1, 3, 5\}$.

B. The probability of an event E is written as $P(E)$ and is a number between 0 and 1, inclusive. If E is the empty set of no possible outcomes, then E is **impossible**, and $P(E) = 0$. If E is the set of all possible outcomes of the experiment, then E is **certain**, and $P(E) = 1$. Otherwise, E is possible but not certain, and $0 < P(E) < 1$. If F is a subset of E , then $P(F) \leq P(E)$.

C. If the probabilities of two or more outcomes are equal, those outcomes are **equally likely**. For an experiment with outcomes that are all equally likely, the probability of an event E is

$$P(E) = \frac{\text{the number of outcomes in } E}{\text{the total number of possible outcomes}}.$$

Example:

In the earlier example of a 6-sided die rolled once, suppose all 6 outcomes are equally likely. Then each outcome's probability is $\frac{1}{6}$. The probability that the outcome is an odd number is

$$P(\{1, 3, 5\}) = \frac{|\{1, 3, 5\}|}{6} = \frac{3}{6} = \frac{1}{2}.$$

D. Given two events E and F , these further events are defined:

- i. “not E ” is the set of outcomes not in E ;
- ii. “ E or F ” is the set of outcomes in E or F or both, that is, $E \cup F$;
- iii. “ E and F ” is the set of outcomes in both E and F , that is, $E \cap F$.

The probability that E doesn't occur is $P(\text{not } E) = 1 - P(E)$.

The probability that “ E or F ” occurs is

$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$. This is based on the general addition rule for two sets, given above in Section 3.4.2.G.

Example:

In the example above of a 6-sided die rolled once, let E be the event $\{1, 3, 5\}$ that the outcome is an odd number. Let F be the event $\{2, 3, 5\}$ that the outcome is a prime number. Then

$$P(E \text{ and } F) = P(E \cap F) = P(\{3, 5\}) = \frac{|\{3, 5\}|}{6} = \frac{2}{6} = \frac{1}{3}.$$

So,

$$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}.$$

We get the same result by noticing that the event “ E or F ” is

$$E \cup F = \{1, 2, 3, 5\}, \text{ so } P(E \text{ or } F) = \frac{|\{1, 2, 3, 5\}|}{6} = \frac{4}{6} = \frac{2}{3}.$$

Events E and F are **mutually exclusive** if no outcomes are in $E \cap F$. Then the event “ E and F ” is impossible: $P(E \text{ and } F) = 0$. The special addition rule for the probability of two mutually exclusive events is $P(E \text{ or } F) = P(E) + P(F)$.

- E. Two events A and B are **independent** if neither changes the other's probability. The multiplication rule for independent events E and F is $P(E \text{ and } F) = P(E)P(F)$.

Example:

In the example above of the 6-sided die rolled once, let A be the event $\{2, 4, 6\}$ and B be the event $\{5, 6\}$. Then A 's probability is

$P(A) = \frac{|A|}{6} = \frac{3}{6} = \frac{1}{2}$. The probability of A occurring **if B occurs** is $\frac{|A \cap B|}{|B|} = \frac{|\{6\}|}{|\{5, 6\}|} = \frac{1}{2}$, the same as $P(A)$.

Likewise, B 's probability is $P(B) = \frac{|B|}{6} = \frac{2}{6} = \frac{1}{3}$. The probability of B occurring **if A occurs** is $\frac{|B \cap A|}{|A|} = \frac{|\{6\}|}{|\{2, 4, 6\}|} = \frac{1}{3}$, the same as $P(B)$.

So, neither event changes the other's probability. Thus, A and B are independent. Therefore, by the multiplication rule for independent events, $P(A \text{ and } B) = P(A)P(B) = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{1}{6}$.

Notice the event " A and B " is $A \cap B = \{6\}$, so

$$P(A \text{ and } B) = P(\{6\}) = \frac{1}{6}.$$

The general addition rule and the multiplication rule discussed above imply that if E and F are independent,

$$P(E \text{ or } F) = P(E) + P(F) - P(E)P(F).$$

F. An event A is **dependent** on a nonempty event B if B changes the probability of A .

The probability of A occurring if B occurs is written as $P(A|B)$. So, the statement that A is dependent on B can be written as $P(A|B) \neq P(A)$.

A general multiplication rule for any dependent or independent events A and B is $P(A \text{ and } B) = P(A|B)P(B)$.

Example:

In the example of the 6-sided die rolled once, let A be the event $\{4, 6\}$ and B be the event $\{4, 5, 6\}$.

Then the probability of A is $P(A) = \frac{|A|}{6} = \frac{2}{6} = \frac{1}{3}$. But the

probability that A occurs **if B occurs** is

$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{|\{4, 6\}|}{|\{4, 5, 6\}|} = \frac{2}{3}. \text{ Thus, } P(A|B) \neq P(A),$$

so A is dependent on B .

Likewise, the probability that B occurs is $P(B) = \frac{|B|}{6} = \frac{3}{6} = \frac{1}{2}$.

But the probability that B occurs **if A occurs** is

$$P(B|A) = \frac{|B \cap A|}{|A|} = \frac{|\{4, 6\}|}{|\{4, 6\}|} = 1. \text{ Thus, } P(B|A) \neq P(B), \text{ so}$$

B is dependent on A .

In this example, by the general multiplication rule for events,

$$P(A \text{ and } B) = P(A|B)P(B) = \left(\frac{2}{3}\right)\left(\frac{1}{2}\right) = \frac{1}{3}. \text{ Likewise,}$$

$$P(A \text{ and } B) = P(B|A)P(A) = (1)\left(\frac{1}{3}\right) = \frac{1}{3}.$$

Notice the event “ A and B ” is $A \cap B = \{4, 6\} = A$, so

$$P(A \text{ and } B) = P(\{4, 6\}) = \frac{1}{3} = P(A).$$

G. The rules above can be combined for more complex probability calculations.

Example:

In an experiment with events A , B , and C , suppose $P(A) = 0.23$, $P(B) = 0.40$, and $P(C) = 0.85$. Also suppose events A and B are mutually exclusive, and events B and C are independent. Since A and B are mutually exclusive,

$$P(A \text{ or } B) = P(A) + P(B) = 0.23 + 0.40 = 0.63.$$

Since B and C are independent,

$$P(B \text{ or } C) = P(B) + P(C) - P(B)P(C) = 0.40 + 0.85 - (0.40)(0.85) = 0.91$$

.

$P(A \text{ or } C)$ and $P(A \text{ and } C)$ can't be found from the information given. But we can find that $P(A) + P(C) = 1.08 > 1$. So, $P(A) + P(C)$ can't equal $P(A \text{ or } C)$, which like any probability cannot be greater than 1. This means that A and C can't be mutually exclusive, and that $P(A \text{ and } C) \geq 0.08$.

Since $A \cap B$ is a subset of A , we can also find that

$$P(A \text{ and } C) \leq P(A) = 0.23.$$

And C is a subset of $A \cup C$, so $P(A \text{ or } C) \geq P(C) = 0.85$.

Thus, we've found that $0.85 \leq P(A \text{ or } C) \leq 1$ and that $0.08 \leq P(A \text{ and } C) < 0.23$.

5. Estimation

- A. Calculating exact answers to complex math questions is often too hard or slow. Estimating the answers by simplifying the questions may be easier and faster.

One way to estimate is to **round** the numbers in the original question: replace each number with a nearby number that has fewer digits.

For any integer n and real number m , you can **round m down** to a multiple of 10^n by deleting all of m 's digits to the right of the digit that stands for multiples of 10^n .

To **round m up** to a multiple of 10^n , first add 10^n to m , then round the result down.

To **round m to the nearest 10^n** , first find the digit in m that stands for a multiple of 10^{n-1} . If this digit is 5 or higher, round m up to a multiple of 10^n . Otherwise, round m down to a multiple of 10^n .

Examples:

- i. To round 7651.4 to the nearest hundred (multiple of 10^2), first notice the digit standing for tens (multiples of 10^1) is 5.

Since this digit is 5 or higher, round up:

First add 100 to the original number: $7651.4 + 100 = 7751.4$.

Then drop all the digits to the right of the one standing for multiples of 100 to get 7700.

Notice that 7700 is closer to 7651.4 than 7600 is, so 7700 is the nearest 100.

- ii. To round 0.43248 to the nearest thousandth (multiple of 10^{-3}), first notice the digit standing for ten-thousandths (multiples of 10^{-4}) is 4. Since $4 < 5$, round down: just drop all the digits to the right of the digit standing for thousandths to get 0.432.

B. Rounding can simplify complex calculations and give rough answers. If you keep more digits of the original numbers, the answers are usually more exact, but the calculations take longer.

Example:

You can estimate the value of $\frac{(298.534 + 58.296)}{1.4822 + 0.937 + 0.014679}$ by rounding the numbers in the dividend to the nearest 10 and the numbers in the divisor to the nearest 0.1:

$$\frac{(298.534 + 58.296)}{1.4822 + 0.937 + 0.014679} \approx \frac{300 + 60}{1.5 + 0.9 + 0} = \frac{360}{2.4} = 150$$

- C. Sometimes it's easier to estimate by rounding to a multiple of a number other than 10, like the nearest square or cube of an integer.

Examples:

i. You can estimate the value of $\frac{2447.13}{11.9}$ by noting first that both the dividend and the divisor are near multiples of 12: 2448 and 12. So, $\frac{2447.13}{11.9} \approx \frac{2448}{12} = 204$.

ii. You can estimate the value of $\sqrt{\frac{8.96}{24.82 \times 4.057}}$ by noting first that each decimal number in the expression is near the square of an integer: $8.96 \approx 9 = 3^2$, $24.82 \approx 25 = 5^2$, and $4.057 \approx 4 = 2^2$. So,

$$\sqrt{\frac{8.96}{24.82 \times 4.057}} \approx \sqrt{\frac{3^2}{5^2 \times 2^2}} = \sqrt{\frac{3^2}{10^2}} = \frac{3}{10}.$$

- D. Sometimes finding a **range** of possible values for an expression is more useful than finding a single estimated value. The range's **upper bound** is the smallest number found to be greater than (or no less than) the expression's value. The range's **lower bound** is the largest number found to be less than (or no greater than) the expression's value.

Example:

In the equation $x = \frac{2.32^2 - 2.536}{2.68^2 + 2.79}$, each decimal is greater than 2 and less than 3. So, $\frac{2^2 - 3}{3^2 + 3} < x < \frac{3^2 - 2}{2^2 + 2}$. Simplifying these fractions, we find that x is in the range $\frac{1}{12} < x < \frac{7}{6}$. The range's lower bound is $\frac{1}{12}$, and the upper bound is $\frac{7}{6}$.

6. Sequences and Series

- A. A **sequence** is an ordered list of elements, starting with a first element. Any sequence may be defined as the range of a function $a(n)$ with a domain that is the positive integers. Its value a_i for a specific positive integer i is its ***i*th term**.

An **arithmetic sequence** has the form $a_n = b + cn$, where b and c are constants. The first term of an arithmetic sequence is $a_1 = b + c$. For each term a_i , the next term $a_{i+1} = a_i + c$.

A **geometric sequence** has the form $a_n = bc^n$, where b and c are constants. The first term of an arithmetic sequence is $a_1 = bc$. For each term a_i , the next term $a_{i+1} = a_i c$.

Sometimes a sequence is defined only by giving the value of the first term or terms, then giving an equation for finding each subsequent term's value based on the value of one or more earlier terms.

Examples:

- i. The function $a(n) = n^2 + \left(\frac{n}{5}\right)$ with a domain that is the positive integers is an infinite sequence $a_1, a_2, a_3, \dots$. Its third term is its value at $n = 3$, which is $a_3 = 3^2 + \frac{3}{5} = 9.6$.
- ii. $a(n) = n^2 + \left(\frac{n}{5}\right)b_n = (-1)^n(n!)$ The function $a(n) = 2 + 3n$ with a domain that is the positive integers is an arithmetic sequence. Its first term $a_1 = 5$. For any positive integer i , $a_i + 1 = a_i + 3$. Thus, $a_2 = a_1 + 3 = 8$.
- iii. The function $b(n) = 2(3^n)$ with a domain that is the positive integers is a geometric sequence. Its first term $b_1 = 6$. For any positive integer i , $b_i + 1 = 3b_i$. Thus, $b_2 = 3 \times 6 = 18$.
- iv. We can define a sequence $c(n)$ just by stating that its first term $c_1 = 1$, its second term $c_2 = 1$, and each subsequent term $c_n + 2 = c_n + c_n + 1$. Thus, $c_3 = c_1 + c_2 = 1 + 1 = 2$.

B. A **series** is the sum of a sequence's terms.

For an infinite sequence $a(n)$, the **infinite series** $\sum_{n=1}^{\infty} a(n)$ adds up the sequence's infinitely many terms, $a_1 + a_2 + a_3 + \dots$. Some infinite series add up to finite sums, but others don't.

Example:

The infinite series based on the function $a(n) = n^2 + \left(\frac{n}{5}\right)$ is

$\sum_{i=1}^{\infty} n^2 + \left(\frac{n}{5}\right)$. It adds up the infinitely many terms

$\left(1^2 + \frac{1}{5}\right) + \left(2^2 + \frac{2}{5}\right) + \left(3^2 + \frac{3}{5}\right) + \dots$. This series has no finite sum.

The partial sum of the first three terms $a(n) = n^2 + \left(\frac{n}{5}\right)$ is

$$\sum_{i=1}^3 a_i = \left(1^2 + \frac{1}{5}\right) + \left(2^2 + \frac{2}{5}\right) + \left(3^2 + \frac{3}{5}\right) = 1.2 + 4.4 + 9.6 = 15.2.$$

3.5 Reference Sheets

Arithmetic and Decimals

ABSOLUTE VALUE:

$|x|$ is x if $x \geq 0$ and $-x$ if $x < 0$.

For any x and y , $|x + y| \leq |x| + |y|$,
and $|xy| = |x||y|$.

$$\sqrt{x^2} = |x|.$$

EVEN AND ODD NUMBERS:

Even $\times$ Even = Even	Even $\times$ Odd = Even
Odd $\times$ Odd = Odd	Even + Even = Even
Even + Odd = Odd	Odd + Odd = Even

ADDITION AND SUBTRACTION:

$$x + 0 = x = x - 0$$

$$x - x = 0$$

$$x + y = y + x$$

$$x - y = -(y - x) = -y + x$$

$$(x + y) + z = x + (y + z)$$

If x and y are both positive, then
 $x + y$ is positive.

If x and y are both negative, then
 $x + y$ is negative.

QUOTIENTS AND REMAINDERS:

The quotient q and the remainder r of
dividing positive integer x by positive
integer y are unique positive integers
such that

$$y = xq + r \text{ and } 0 < r < x.$$

The remainder r is 0 if and only if y is
divisible by x . Then x is a factor of y .

MULTIPLICATION AND DIVISION:

$$x \times 1 = x = \frac{x}{1}$$

$$x \times 0 = 0$$

$$\text{If } x \neq 0, \text{ then } \frac{x}{x} = 1.$$

$$\frac{x}{0} \text{ is undefined.}$$

$$xy = yx$$

$$\text{If } x \neq 0 \text{ and } y \neq 0, \text{ then } \frac{x}{y} = \frac{1}{\left(\frac{y}{x}\right)}$$

.

$$(xy)z = x(yz)$$

$$xy + xz = x(y + z)$$

If $y \neq 0$, then

$$\left(\frac{x}{y}\right) + \left(\frac{z}{y}\right) = \frac{(x + z)}{y}.$$

DECIMALS WITH MANY FINITE DIGITS:

Add or subtract decimals by lining up their decimal points:

17.6512	653.2700
+ 653.2700	-17.6512
670.9212	635.6188

To multiply decimal A by decimal B :

First, ignore the decimal points and multiply A and B as if they were integers.

Next, if decimal A has n digits to the right of its decimal point and decimal B has m digits to the right of its decimal point, place the decimal point in $A \times B$ so it has $m + n$ digits to its right.

To divide decimal A by decimal B , first move the decimal points of A and B equally many digits to the right until B is an integer. Then divide as you would integers.

If x and y are both positive, then xy is positive.

If x and y are both negative, then xy is positive.

If x is positive and y is negative, then xy is negative.

If $xy = 0$, then $x = 0$ or $y = 0$, or both.

SCIENTIFIC NOTATION:

To convert a number in scientific notation $A \times 10^n$ into regular decimal notation, move A 's decimal point n places to the right if n is positive, or $|n|$ places to the left if n is negative.

To convert a decimal to scientific notation, move the decimal point n spaces so that exactly one nonzero digit is to its left. Multiply the result by 10^n if you moved the decimal point to the left or by 10^{-n} if you moved it to the right.

ARITHMETIC SHORTCUTS:

For any two nonnegative integers x and y , the final digit of $x + y$ is the final digit of the sum of x 's final digit and y 's final digit. And for any two integers x and y , the final digit of xy is the final digit of the product of x 's final digit and y 's final digit.

An integer is divisible by	if and only if
2	its final digit is 0, 2, 4, 6, or 8.
3	the sum of its digits is divisible by 3.
4	its final two digits make a number divisible by 4.
5	its final digit is 5 or 0.
9	the sum of its digits is divisible by 9.
10	its final digit is 0.

Exponents

SQUARES, CUBES, AND SQUARE ROOTS:

EXPONENTIATION:

Every positive number has two real square roots, one positive and the other negative. The table below shows the positive square roots rounded to the nearest hundredth.

n	n^2	n^3	$\sqrt{n}$
1	1	1	1
2	4	8	1.41
3	9	27	1.73
4	16	64	2
5	25	125	2.24
6	36	216	2.45
7	49	343	2.65
8	64	512	2.83
9	81	729	3
10	100	1,000	3.16

Formula	Example
$x^1 = x$	$2^1 = 2$
If $x \neq 0$, then $x^0 = 1$.	$2^0 = 1$
If $x \neq 0$, then $x^{-1} = \frac{1}{x}$.	$2^{-1} = \frac{1}{2}$
If $x > 1$ and $y > 1$, then $x^y > x$.	$2^3 = 8 > 2$
If $0 < x < 1$ and $y > 1$, then $x^y < x$.	$0.2^3 = 0.008 < 0.2$
$(x^y)^z = x^{yz} = (x^z)^y$	$(2^3)^4 = 2^{12} = (2^4)^3$
$x^{y+z} = x^y x^z$	$2^7 = 2^3 2^4$
If $x \neq 0$, then $x^{y-z} = \frac{x^y}{x^z}$.	$2^{5-3} = \frac{2^5}{2^3}$
$(xz)^y = x^y z^y$	$6^4 = 2^4 3^4$
If $z \neq 0$, then $\left(\frac{x}{z}\right)^y = \frac{x^y}{z^y}$.	$\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \frac{9}{16}$

If $z \neq 0$, then

$$x^{\frac{y}{z}} = (x^y)^{\frac{1}{z}} = (x^{\frac{1}{z}})^y$$

.

$$4^{\frac{2}{3}} = (4^2)^{\frac{1}{3}} = (4^{\frac{1}{3}})^2$$

Algebraic Expressions and Linear Equations

TRANSLATING WORDS INTO MATH OPERATIONS:

$x + y$	$x - y$	xy	$\frac{x}{y}$	x^y
x added to y x increased by y x more than y x plus y the sum of x and y the total of x and y	x decreased by y difference of x and y y fewer than x y less than x x minus y x reduced by y y subtracted from x	x multiplied by y the product of x and y x times y If y equals two: double x twice x If y equals three: triple x	x divided by y x over y the quotient of x and y the ratio of x to y If y equals two: half of x x halved	x to the power of y x to the y super th power If y equals two: x squared If y equals three: x cubed

MANIPULATING ALGEBRAIC EXPRESSIONS:

Technique	Example
Factor to combine like terms	 three times x times y minus nine times y equals three times y times left parenthesis x minus three right parenthesis
Divide out common factors	 equation sequence part 1 left parenthesis three times x times y minus nine times y right parenthesis divided by left parenthesis x minus three right parenthesis equals part 2 three times y times left parenthesis x minus three right parenthesis divided by left parenthesis x minus three right parenthesis equals part 3 three times y of one equals part 4 three times y
Multiply two expressions by multiplying each term of one expression by each term of the other	 multiline equation row 1 left parenthesis three times x minus four right parenthesis times left parenthesis nine times y plus x right parenthesis equals three times x times left parenthesis nine times y plus x right parenthesis minus four times left parenthesis nine times y plus x right parenthesis row 2 equals three times x of nine times y plus three times x of x minus four times left parenthesis nine times y right parenthesis minus four times left parenthesis x right parenthesis row 3 equals 27 times x times y plus three times x squared minus 36 times y minus four times x
Substitute constants for variables	If  x equals three and  y equals negative two, then

 equation sequence part 1 three times
x times y minus x squared plus y
equals part 2 three times left
parenthesis three right parenthesis
times left parenthesis negative two
right parenthesis minus left
parenthesis three right parenthesis
squared plus left parenthesis negative
two right parenthesis equals part 3
negative 18 minus nine minus two
equals part 4 negative 29 full stop

SOLVING LINEAR EQUATIONS:

Technique	Example
Isolate a variable on one side of an equation by doing the same operations on both sides of the equation.	Solve the equation left parenthesis five times x minus six right parenthesis divided by three equals four like this: (1) Multiply both sides by 3 to get five times x minus six equals 12. (2) Add 6 to both sides to get five times x equals 18. (3) Divide both sides by 5 to get x equals 18 divided by five.
To solve two equations with two variables x and y: (1) Express x in terms of y using one of the equations. (2) Substitute that expression for x to make the second equation have only the variable y. (3) Solve the second equation for y. (4) Substitute the solution for y into the first equation to solve for x.	Solve the equations A: x minus y equals two and B: $3x + 2y = 11$: (1) From A, x equals two plus y. (2) In B, substitute two plus y for x to get three times left parenthesis two plus y right parenthesis plus two times y equals 11 . (3) Solve B for y: sum with 3 summands six plus three times y plus two times y equals 11 multiline equation row 1 six plus five times y equals 11 row 2 five times y equals five row 3 y equals one

	(4) Since $y = 1$, from A we find  equation sequence part 1 x equals part 2 two plus one equals part 3 three .
Alternative technique: (1) Multiply both sides of one equation or both equations so that the coefficients on  y have the same absolute value in both equations. (2) Add or subtract the two equations to remove  y and solve for x. (3) Substitute the solution for  x into the first equation to find the value of  y.	Solve the equations A:  x minus y equals two and B: $3x + 2y = 11$: (1) Multiply both sides of A by 2 to get  two times x minus two times y equals four . (2) Add the equation in (1) to equation B:  multiline equation row 1 sum with 3 summands two times x minus two times y plus three times x plus two times y equals four plus 11 row 2 five times x equals 15 row 3 x equals three (3) Since  x equals three, from A we find $3 - y = 2$, so  y equals one.

Factoring, Quadratic Equations, and Inequalities

SOLVING EQUATIONS BY FACTORING:

Techniques	Example
(1) Start with a polynomial equation. (2) Add or subtract from both sides until 0 is on one side of the equation. (3) Write the nonzero side as a product of factors. (4) Set each factor to 0 to find simple equations giving the solutions to the original equation.	multiline equation row 1 $x^3 - 2x^2 + x = -5(x - 1)$ sum with 3 summands $x^3 - 2x^2 + x + 5(x - 1) = 0$ left parenthesis i right parenthesis $(x - 1)(x^2 - x + 5) = 0$ left parenthesis ii right parenthesis $(x - 1)(x^2 - x + 5) = 0$ left parenthesis iii right parenthesis $(x - 1)(x^2 - x + 5) = 0$  $x + 5 = 0$ or $x - 1 = 0$  So, $x = -5$ or  $x = 1$.

FORMULAS FOR FACTORING:

THE QUADRATIC FORMULA:

left parenthesis a plus one divided by a right parenthesis squared equals sum with 3 summands a squared plus one divided by a squared plus two

$$a^2 - b^2 = (a - b)(a + b)$$

sum with 3 summands a squared plus two times a times b plus b squared equals left parenthesis a plus b right parenthesis squared

a squared minus two times a times b plus b squared equals left parenthesis a minus b right parenthesis squared

equation sequence part 1 a squared plus b squared equals part 2 left parenthesis a plus b right parenthesis squared minus two times a times b equals part 3 left parenthesis a minus b right parenthesis squared plus two times a times b

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

left parenthesis a plus b right parenthesis cubed equals sum with 4 summands a cubed plus three times a squared times b plus three times a times b squared plus b cubed

For any quadratic equation

sum with 3 summands a times x squared plus b times x plus c equals zero

with a not equals zero, the roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ and}$$

x equals negative b minus Square root of b squared minus four times a times c divided by two times a

These roots are two distinct real numbers if

b squared minus four times a times c greater than or equals zero

.

If

b squared minus four times a times c equals zero

, the equation has only one root: $-\frac{b}{2a}$.

If

b squared minus four times a times c less than zero

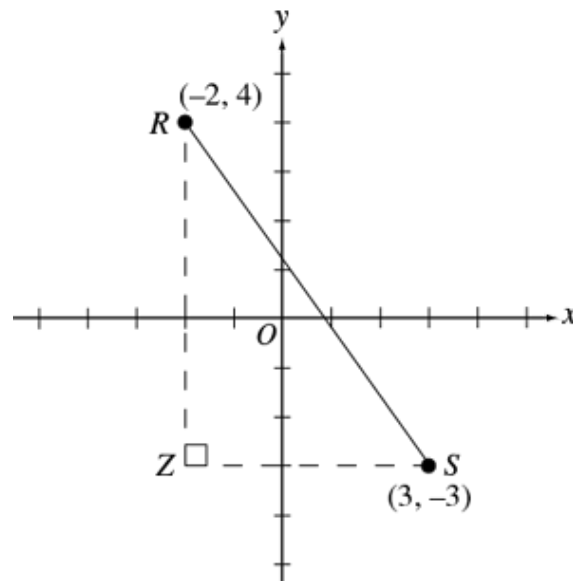
, the equation has no real roots.

SOLVING INEQUALITIES:

Explanation	Example
As in solving an equation, the same number can be added to or subtracted from both sides of the inequality, or both sides can be multiplied or divided by a positive number, without changing the order of the inequality. But multiplying or dividing an inequality by a negative number reverses the order of the inequality. Thus, $6 > 2$, but $(-1)(6) < (-1)(2)$.	To solve the inequality  five times x minus one  divided by negative two less than three for x, isolate : (1)  five times x minus one greater than negative s (multiplying both sides by -2, reversing the order of the inequality) (2) $5x > -5$ (add 1 to both sides) (3)  x greater than negative one (divide both sides by 5)

LINES IN THE COORDINATE PLANE:

An equation y equals m times x plus b defines a line with slope m , which has a y -intercept that is b .



For a line through two points

(x_1, y_1) and

(x_2, y_2) with

$x_1 \neq x_2$, the slope is $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the

known point (x_1, y_1) and the slope m , any other point

(x, y) on the line satisfies the

equation $m = \frac{y - y_1}{x - x_1}$.

Above, the line's slope is

$(-3 - 4) \div (3 - (-2))$ equals $-7 \div 5$

. To find an equation of the line, we can use the point $(3, -3)$:

multiline equation row 1 $y - 3 = \frac{-7}{5}(x - 3)$ row 2 $y + 3 = \frac{-7}{5}(x + 21)$ row 3 $y = \frac{-7}{5}x + \frac{6}{5}$

So, the y -intercept is $\frac{6}{5}$.

Find the x -intercept like this:

$$\begin{aligned} 0 &= \left(-\frac{7}{5}\right)x + \frac{6}{5} \\ \left(\frac{7}{5}\right)x &= \frac{6}{5} \\ x &= \frac{6}{7} \end{aligned}$$

The graph shows both these intercepts.

Rates, Ratios, and Percents

FRACTIONS:

Equivalent or Equal Fractions:

Two fractions stand for the same number if dividing each fraction's numerator and denominator by their greatest common divisor makes the fractions identical.

Adding, Subtracting, Multiplying, and Dividing Fractions:

 a divided by b plus c divided by d equals a times d divided by b times d plus b times c divided by b times d

$$; \frac{a}{b} + \frac{c}{d} = \frac{ad}{bd} + \frac{bc}{bd}$$

 a divided by b multiplication c divided by d equals a times c divided by b times d

$$; \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

MIXED NUMBERS:

A mixed number of the form

 a times b divided by c equals the fraction $\frac{ac + b}{c}$.

RATE:

distance = rate $\times$ time

PROFIT:

Profit = Revenues – Expenses, or

Profit = Selling price – Cost.

PERCENTS:

 x percent equals x divided by 100.

$x\%$ of  y equals $\frac{xy}{100}$.

To convert a percent to a decimal, drop the percent sign, then move the decimal point two digits left.

To convert a decimal to a percent, add a percent sign, then move the decimal point two digits right.

PERCENT INCREASE OR DECREASE:

The percent increase from  x to y is  100 times left parenthesis y minus x divided by x right parenthesis percent.

The percent decrease from x to  y is $100 \left(\frac{x - y}{x} \right) \%$.

DISCOUNTS:

A price discounted by  n percent becomes $(100 - n)$ percent of the original price.

A price discounted by  n percent and then by m percent becomes  left parenthesis 100 minus n right parenthesis times left parenthesis 100 minus m right parenthesis percent of the original price.

	WORK:
INTEREST: Simple annual interest = (principal) $\times$ (interest rate) $\times$ (time) Compound interest over n periods = (principal) $\times (1 + \text{interest per period})^n$ - principal	$\frac{1}{r} + \frac{1}{s} = \frac{1}{h}$, where r is how long one individual takes to do an amount of work, s is how long a second individual takes to do that much work, and h is how long they take to do that much work when both are working at the same time.

MIXTURES:

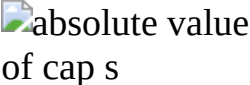
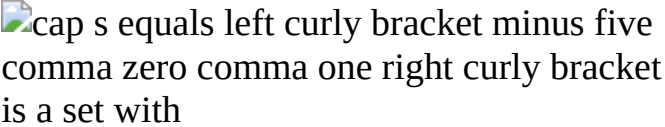
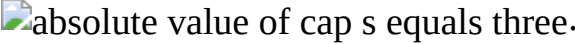
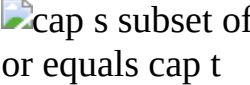
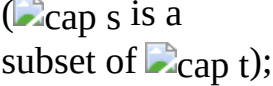
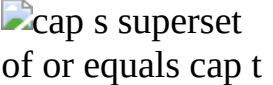
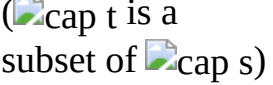
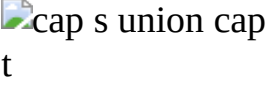
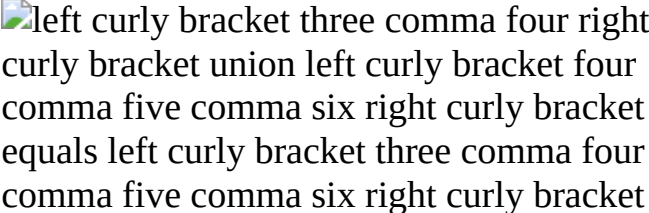
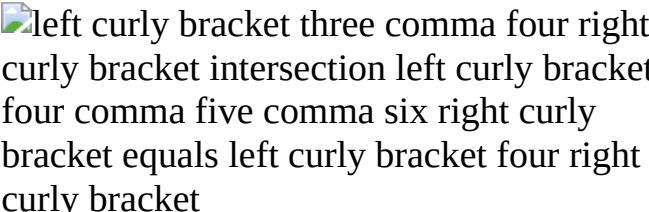
	Number of units of a substance or mixture	Amount of an ingredient per unit of the substance or mixture	Total amount of that ingredient in the substance or mixture
Substance A	X	M	$X \times M$
Substance B	Y	N	$Y \times N$
Mixture of A and B	$X + Y$	$\frac{(X \times M) + (Y \times N)}{X + Y}$	$(X \times M) + (Y \times N)$

Statistics, Sets, and Counting Methods

STATISTICS:

Concept	Definition for a data set of n values ordered from least to greatest	Example with data set {4, 4, 5, 7, 10}
Mean	The sum of the n values, divided by n	$\frac{(4 + 4 + 5 + 7 + 10)}{5} = \frac{30}{5} = 6$
Median	The middle value if  n is odd; The mean of the two middle values if  n is even.	5 is the middle value in {4, 4, 5, 7, 10}.
Mode	The value that appears most often in the data set.	4 is the only value that appears more than once in {4, 4, 5, 7, 10}.
Range	The largest number in the data set minus the smallest	$10 - 4 = 6$
Standard Deviation	Calculate like this: (1) Find the arithmetic mean (2) Find the differences between each of the  n values and the mean (3) Square each of the differences (4) Find the average of the squared differences, and (5) Take the nonnegative square root of this average	(1) The mean is 6. (2) -2, -2, -1, 1, 4 (3) 4, 4, 1, 1, 16 (4)  26 divided by five equals 5.2 (5)  Square root of 5.2

SETS:

Concept	Notation for finite sets  and 	Example
Number of elements		 
Subset	   	$\{-5, 0, 1\}$ is a subset of $\{-5, 0, 1, 4, 10\}$.
Union		
Intersection		

The general addition rule for two sets

 absolute value of $S \cup T$ equals absolute value of S plus absolute value of T minus absolute value of $S \cap T$

 multiline equation line 1 absolute value of $S \cup T$ equals line 2 absolute value of S plus absolute value of T minus absolute value of $S \cap T$ line 3 equation sequence part 1 absolute value of $S \cup T$ plus absolute value of $S \cap T$ minus absolute value of $S \cap T$ equals part 2 two plus three minus one equals part 3 four full stop

COUNTING METHODS:

Concept and Equations	Examples
Multiplication Principle: The number of possible choices of 1 element apiece from the data sets  cap a sub one comma cap a sub two comma ellipsis comma cap a sub n is  absolute value of cap a sub one multiplication absolute value of cap a sub two multiplication ellipsis multiplication absolute value of cap a sub n .	The number of possible choices of 1 element apiece from the data sets  cap s equals left curly bracket negative five comma zero comma one right curly bracket ,  cap t equals left curly bracket three comma four right curly bracket , and  cap u equals left curly bracket three comma four comma five comma six right curly bracket is  equation sequence part 1 absolute value of cap s multiplication absolute value of cap t multiplication absolute value of cap u equals part 2 three multiplication two multiplication four equals part 3 24 .
Factorial:  multiline equation line 1 n factorial equals n multiplication left parenthesis n minus one right parenthesis multiplication ellipsis multiplication one line 2 zero factorial equals one factorial equals one line 3 n factorial equals left parenthesis n minus one right parenthesis factorial left parenthesis n right parenthesis	$4! = 4 \times 3 \times 2 \times 1 = 24$ $4! = 3! \times 4$
Permutations: A data set of  n objects has  n! permutations	The data set of letters A, B, and C has $3! = 6$ permutations: ABC, ACB, BAC, BCA, CAB, and CBA.

Combinations:

The number of possible choices of  k objects from a data set of  n objects is

 vector element 1 n element 2 k equals n factorial divided by k factorial left parenthesis n minus k right parenthesis factorial

.

The number of 2-element subsets of data set $\{A, B, C, D, E\}$ is

 equation sequence part 1 vector element 1 five element 2 two equals part 2 five factorial divided by two factorial three factorial equals part 3 120 divided by left parenthesis two right parenthesis times left parenthesis six right parenthesis equals part 4 10

.

The 10 subsets are: $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, $\{A, E\}$, $\{B, C\}$, $\{B, D\}$, $\{B, E\}$, $\{C, D\}$, $\{C, E\}$, and $\{D, E\}$.

Probability and Sequences

PROBABILITY:

Concept	Definition, Notation, and Equations	Example: Rolling a die with 6 numbered sides once
Event	A set of outcomes of an experiment	The event of the outcome being an odd number is the set {1, 3, 5}.
Probability	The probability P of an event E is a number between 0 and 1, inclusive. If each outcome is equally likely, P of E equals $\frac{\text{number of possible outcomes in } E}{\text{total number of possible outcomes}}$.	If the 6 outcomes are equally likely, the probability of each outcome is $\frac{1}{6}$. The probability that the outcome is an odd number is equation sequence part 1 P times left parenthesis left curly bracket one comma three comma five right curly bracket right parenthesis equals part 2 absolute value of left curly bracket one comma three comma five right curly bracket divided by six equals part 3 three divided by six equals part 4 one divided by two .
Conditional Probability	The probability that E occurs if F occurs is P of E vertical line F equals absolute value of E intersection F divided by absolute value of F.	equation sequence part 1 P of left curly bracket one comma three comma five right curly bracket vertical line left curly bracket one comma two right curly bracket equals part 2 absolute value of left curly bracket one right curly bracket divided by absolute value of left curly bracket one comma two right curly bracket equals part 3 one divided by two

Not  bold-italic cap e	The set of outcomes not in event  cap e:  cap p of not cap e equals one minus cap p of cap e .	 equation sequence part 1 cap p of not of three equals part 2 six minus one divided by six equals part 3 five divided by six
 bold-italic cap e and  bold-italic cap f	The set of outcomes in both  cap e and  cap f, that is,  cap e intersection cap f ;  equation sequence part 1 cap p of cap e and cap f equals part 2 cap p times left parenthesis cap e intersection cap f right parenthesis equals part 3 cap p of cap e vertical line cap f cap p of cap f .	For  cap e equals left curly bracket one comma three comma five right curly bracket and  cap f equals left curly bracket two comma three comma five right curly bracket :  multiline equation line 1 equation sequence part 1 cap p of cap e and cap f equals part 2 cap p times left parenthesis cap e intersection cap f right parenthesis equals part 3 cap p of left curly bracket three comma five right curly bracket equals part 4 line 2 equation sequence part 1 absolute value of left curly bracket three comma five right curly bracket divided by six equals part 2 two divided by six equals part 3 one divided by three
 bold-italic cap e or  bold-italic cap f	The set of outcomes in  cap e or  cap f or both, that is,  cap e union cap f;	For  cap e equals left curly bracket one comma three comma five right curly bracket and

	 cap p of cap e or cap f equals cap p of cap e plus cap p of cap f minus cap p of cap e and cap f .	 cap f equals left curly bracket two comma three comma five right curly bracket :  multiline equation line 1 cap p of cap e or cap f equals cap p of cap e plus cap p of cap f minus cap p of cap e and cap f line 2 equation sequence part 1 equals part 2 three divided by six plus three divided by six minus two divided by six equals part 3 four divided by six equals part 4 two divided by three full stop
Dependent and Independent Events	 cap e is dependent on  cap f if  cap p of cap e vertical line cap f not equals cap p of cap e .  cap e and  cap f are independent if neither is dependent on the other. If  cap e and  cap f are independent,  cap p of cap e and cap f equals cap p of cap e times cap p of cap f .	For  cap e equals left curly bracket two comma four comma six right curly bracket and  cap f equals left curly bracket five comma six right curly bracket :  equation sequence part 1 cap p of cap e vertical line cap f equals part 2 cap p of cap e equals part 3 one divided by two , and  equation sequence part 1 cap p of cap f vertical line cap e equals part 2 cap p of cap f equals part 3 one divided by three , so  cap e and  cap f are independent. Thus

	 equation sequence part 1 cap p of cap e times and times cap f equals part 2 cap p of cap e times cap p of cap f equals part 3 left parenthesis one divided by two right parenthesis times left parenthesis one divided by three right parenthesis equals part 4 one divided by six .
--	---

SEQUENCE:

An ordered list of elements, starting with a first element.

Example: The function

 a of n equals n squared plus left parenthesis n divided by five right parenthesis

with the domain of all positive integers

 n equals one comma two comma three, ... is an infinite sequence  a sub n.

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4.0 Verbal Review

4.0 Verbal Review

To prepare for the GMAT™ exam's Verbal Reasoning section and Data Insights section, and to succeed in graduate business programs, you need basic skills in analyzing and evaluating texts and the ideas they express. This chapter explains concepts to help you develop these skills. This is only a brief overview. So, if you find unfamiliar terms or concepts, consult outside resources to learn more.

The sections below can help you develop skills you need for the Verbal Reasoning and Data Insights sections of the GMAT exam.

[Section 4.1](#), “Analyzing Passages,” includes:

1. Arguments
2. Explanations and Plans
3. Narratives and Descriptions

[Section 4.2](#), “Inductive Reasoning,” includes:

1. Inductive Arguments
2. Generalizations and Predictions
3. Causal Reasoning
4. Analogies

[Section 4.3](#), “Deductive Reasoning,” includes:

1. Deductive Arguments
2. Logical Operators
3. Reasoning with Logical Operators
4. Necessity, Probability, and Possibility

5. Quantifiers

6. Reasoning with Quantifiers

4.1 Analyzing Passages

1. Arguments

A. An **argument** gives one or more ideas as reasons to accept one or more other ideas. Often some of these ideas are implied but not stated.

Example:

The sidewalk is dry, so it must not have rained last night.

This argument gives the observation that the sidewalk is dry as a reason to accept that it didn't rain last night. The argument implies but doesn't say that rain typically leaves sidewalks wet.

B. A **premise** is an idea that an argument gives as a reason to accept another idea. An argument can have any number of premises.

The words and phrases below often mark premises:

<i>after all</i>	<i>for one thing</i>	<i>moreover</i>
<i>because</i>	<i>furthermore</i>	<i>seeing that</i>
<i>for</i>	<i>given that</i>	<i>since</i>
<i>for the reason that</i>	<i>in light of the fact that</i>	<i>whereas</i>

Example:

Our mayor shouldn't support the proposal to expand the freeway **because** the expansion's benefits wouldn't justify the costs.

Furthermore, most voters oppose the expansion.

This is an argument with two stated premises. The word **because** marks the first premise, that the expansion's benefits wouldn't justify the costs. The word **furthermore** marks the second premise, that most voters oppose the expansion. These premises are given as reasons the mayor shouldn't support the proposal.

- C. A **conclusion** is an idea an argument supports with one or more premises. An **intermediate conclusion** is a conclusion the argument uses to support another conclusion. A **main conclusion** is a conclusion the argument doesn't use to support any other conclusion.

The words and phrases below often mark conclusions:

<i>clearly</i>	<i>it follows that</i>	<i>suggests that</i>
<i>entails that</i>	<i>proves</i>	<i>surely</i>
<i>hence</i>	<i>shows that</i>	<i>therefore</i>
<i>implies that</i>	<i>so</i>	<i>thus</i>

Example:

Julia just hiked fifteen kilometers, **so** she must have burned a lot of calories. **Surely**, she's hungry now.

This argument has a premise, an intermediate conclusion, and a main conclusion. The word **so** marks the intermediate conclusion: that Julia must have burned a lot of calories. The word **surely** marks the main conclusion: that Julia is hungry now. The premise that Julia just hiked fifteen kilometers supports the intermediate conclusion, which in turn supports the main conclusion.

Conclusions may be stated before, between, or after premises. Sometimes no marker words show which statements are premises and which are conclusions. To find premises and conclusions without marker words, consider which statements the author gives as reasons to accept which other statements. The reasons given are the premises. The ideas the author tries to persuade readers to accept are the conclusions.

Example:

For healthy eating, Healthful Brand Donuts are the best donuts you can buy. Unlike any other donuts on the market, Healthful Brand Donuts have plenty of fiber and natural ingredients.

In this argument, the author tries to persuade the reader that Healthful Brand Donuts are the best donuts to buy for healthy eating. So, the first sentence is the conclusion. The statement about Healthful Brand Donuts' ingredients is a premise because it's given as a reason to accept the conclusion. Since the author's intent is clear, no marker words are needed.

- D. A **valid** argument is one whose conclusions follow from its premises. A valid argument can have false premises and conclusions. In a valid argument with false premises, the conclusion **would** follow if the premises **were** true.

A **sound** argument is a valid argument with true premises. Since a sound argument's premises are true, and its conclusions follow from those premises, its conclusions must also be true.

Examples:

(i) Everyone who tries fried eggplant is guaranteed to love the taste. So, if you try it, you'll love the taste too.

In example (i), the premise is false: not everyone who tries fried eggplant is guaranteed to love the taste. So, example (i) is not a sound argument. But it is a valid argument because if everyone who tried fried eggplant **were** guaranteed to love the taste, it would follow that you, too, would love the taste if **you** tried it.

(ii) Some people who try fried eggplant dislike the taste. So, if you try it, you'll probably dislike the taste too.

In example (ii), the premise is true: some people who try fried eggplant do dislike the taste. However, example (ii) is an invalid argument, so it's not sound. **Some** people dislike the taste of fried eggplant, but that does not mean **you personally** will **probably** dislike the taste.

E. An **assumption** is an idea taken for granted. An assumption may be a premise in an argument, a claim about a cause or effect in a causal explanation, a condition a plan relies on, or any other type of idea taken for granted. A conclusion is never an assumption—an argument doesn't take a conclusion for granted, but rather gives reasons to accept it.

A passage may also have **implicit assumptions** the author considers too obvious to state. These unstated ideas fill logical gaps between the passage's statements.

An argument, plan, or explanation with implausible assumptions is weak and vulnerable to criticism.

F. A **necessary assumption** of an argument is an idea that must be true for the argument's stated premises to be good enough reasons to accept

its conclusions. That is, a necessary assumption is one the argument needs in order to work.

Example:

Mario has booked a flight scheduled to arrive at 5:00 p.m.—which should let him get here by around 6:30 p.m. So, by 7:00 p.m. we'll be going out to dinner with Mario.

In this argument, one necessary assumption is that the flight Mario booked will arrive not much later than scheduled. A second necessary assumption is that Mario caught his flight. Unless these and all the argument's other necessary assumptions are true, the argument's stated premises aren't good enough reasons to accept the conclusion.

G. A **sufficient assumption** of an argument is an idea whose truth would make the argument's main conclusion follow from the stated premises. That is, adding a sufficient assumption to an argument makes the argument valid.

Example:

The study of poetry is entirely without value since poetry has no practical use.

In this argument, one sufficient assumption is that studying anything with no practical use is entirely without value. This assumption, together with the argument's stated premise, is enough to make the conclusion follow. If both the premise and the assumption are true, the conclusion must also be true. But this sufficient assumption is not a necessary assumption.

H. Arguments are often classified based on what kinds of conclusions they have.

i. A **prescriptive** argument has a conclusion about what should or shouldn't be done. Prescriptive arguments may advocate for or

against policies, procedures, strategies, goals, laws, or ethical norms.

Example:

Our company's staff is too small to handle our upcoming project. So, to make sure the project succeeds, the company **should** hire more employees.

Another example of a prescriptive argument is shown above in Section 4.1.1.B. That argument is prescriptive because it concludes that the mayor **should not** support the proposed freeway expansion.

- ii. An **evaluative** argument concludes that something is good or bad, desirable or undesirable, without advocating any particular policy or actions.

Example:

This early novel is clearly one of the greatest of all time. Not only did it pioneer brilliantly innovative narrative techniques, but it did so with exceptional grace, subtlety, and sophistication.

- iii. An **interpretive** argument has a conclusion about something's underlying significance. An interpretive argument may be about the meaning, importance, or implications of observations, a theory, an artistic or literary work, or a historical event.

Example:

Many famous authors have commented emphatically on this early novel, either praising or condemning it. This suggests the novel has had an enormous influence on later fiction.

- iv. A **causal** argument concludes that one or more factors did or did not contribute to one or more effects. A causal argument may be

about the causes, reasons, or motivations for an event, condition, decision, or outcome. For example, a causal argument may support or oppose an account of the influences behind a literary or artistic style or movement.

Example:

Our houseplant started to thrive only when we moved it to a sunny window. So, probably the reason it was sickly before then was that it wasn't getting enough sunlight.

Another causal argument can be found in the example in Section 4.1.1.C above, which concludes that Julia must be hungry now.

- v. A basic ***factual*** argument has a factual conclusion that doesn't fit in any other category explained above.

Example:

All dogs are mammals. Rover is a dog. Therefore, Rover is a mammal.

2. Explanations and Plans

- A. A ***causal explanation*** claims that one or more factors contribute to one or more effects. A causal explanation might not be an argument. It might have no premises or conclusions. But a causal explanation can be a premise or conclusion in an argument.

The words and phrases below often mark a causal explanation:

<i>as a result</i>	<i>due to</i>	<i>results in</i>
<i>because</i>	<i>leads to</i>	<i>that's why</i>
<i>causes</i>	<i>produces</i>	<i>thereby</i>
<i>contributes to</i>	<i>responsible for</i>	<i>thus</i>

Some of these words can also mark premises or conclusions in arguments. To tell what the words mark, you may have to judge whether the author is giving reasons to accept a conclusion or only saying what causes an effect. If the author is only saying what causes an effect, without trying to persuade the reader that the effect is real, then the passage is a causal explanation but not an argument.

Just as an argument may have premises, intermediate conclusions, and main conclusions, a causal explanation may claim that one or more factors cause one or more intermediate effects that, in turn, cause further effects.

Example:

Julia just hiked fifteen kilometers, **thereby** burning a lot of calories. **That's why** she's hungry now.

This is a causal explanation claiming that a factor (Julia's fifteen-kilometer hike) caused an intermediate effect (Julia burning a lot of calories) that, in turn, caused another effect (Julia being hungry now). The word **thereby** marks the intermediate effect, and the phrase **that's why** marks the final effect. This explanation doesn't try to convince the reader that Julia's hungry. It just explains what made her hungry. So, it's not an argument.

- B. An **observation** is a claim that something was observed or is otherwise directly known. In the example of a causal explanation above, the claims that Julia just hiked fifteen kilometers and that she's hungry are observations. But if her burning of calories was not directly known or

observed, the claim that she burned a lot of calories is not an observation.

- C. A **hypothesis** is a tentative idea neither known nor assumed to be true. A hypothesis can be an argument's conclusion. Causal explanations are often hypotheses. A passage may discuss **alternative hypotheses**, such as competing explanations for the same observation. Sometimes a passage gives pros and cons of alternative hypotheses without arguing for any particular hypothesis as a conclusion.

Example:

A bush in our yard just died. The invasive insects we've seen around the yard lately might be the cause. Or the bush might not have gotten enough water. It's been a dry summer.

This example presents two alternative hypotheses. The first hypothesis gives the observation that invasive insects have been in the yard as a possible causal explanation for the bush's observed death. This hypothesis assumes the insects can hurt bushes like the one in the yard. The second hypothesis provides the observation that it's been a dry summer as an alternative causal explanation for the bush's observed death. This hypothesis assumes dry weather can result in bushes getting too little water. The passage presents observations to tentatively support each hypothesis. But it doesn't argue for either hypothesis as a conclusion.

- D. A **plan** describes an imagined set of actions meant to work together to achieve one or more goals. A plan is not itself an argument. Its actions aren't proposed as reasons to accept a goal, but rather as ways to reach the goal. However, a prescriptive argument may recommend or oppose a plan. A plan may also be among an argument's premises.

Just as an argument may have premises, intermediate conclusions, and main conclusions, a plan may suggest actions to reach intermediate goals that, in turn, are meant to help achieve main goals. And like an argument, a plan may have assumptions, including necessary and sufficient assumptions. A necessary assumption of a plan is one that must be true for the plan to achieve its goals. And a sufficient

assumption of a plan is one whose truth guarantees the plan would achieve its goals if followed.

A plan is not a causal explanation. The actions a plan suggests haven't been done yet, so they can't have caused anything. However, any plan does assume possible future causal links between its proposed actions and its goals.

Example:

To repaint our house, we'll need to buy gallons of paint. To do that, we could go to the hardware store.

In this plan, going to the hardware store is an action imagined to help reach the intermediate goal of getting gallons of paint. The intermediate goal is imagined to help reach the main goal of repainting the house. This plan isn't itself an argument. But, combined with the premise that we **should** repaint the house, the plan could be part of a prescriptive argument that we **should** go to the hardware store.

3. Narratives and Descriptions

- A. A **narrative** describes a sequence of related events. A narrative is not an argument, an explanation, or a plan. But it may contain one or more arguments, causal explanations, or plans—or be contained in them.

The words and phrases below often show narrative sequence:

<i>after</i>	<i>earlier</i>	<i>then</i>
<i>afterwards</i>	<i>later</i>	<i>thereafter</i>
<i>before</i>	<i>previously</i>	<i>until</i>
<i>beforehand</i>	<i>since</i>	<i>while</i>
<i>during</i>	<i>subsequently</i>	<i>when</i>

Example:

While Julia was hiking fifteen kilometers, she burned a lot of calories. **Afterwards**, she felt hungry.

This narrative describes a sequence of three events. The word **while** shows that Julia's hike and her burning of calories happened at the same time. The word **afterwards** shows that her hunger arose soon after the first two events. Although you can reasonably assume these events were causally linked, the narrative doesn't say they were. So, it's not an explicit causal explanation. And since the narrative doesn't report the events as reasons to accept a particular conclusion, it's not an argument either.

- B. Not all passages are arguments, causal explanations, plans, or narratives. Some passages report on views, findings, innovations, places, societies, artistic works, devices, organisms, etc., without arguing for a conclusion, explaining what caused what, suggesting actions to reach a goal, or narrating what happened.
- C. Likewise, not all statements in passages are premises, conclusions, observations to be explained, hypotheses, or reports of events. Statements in passages can also:
- Give background information to help the reader understand the rest of the passage

- Describe details of something that the passage is discussing
- Express the author's attitude toward material in the passage
- Provide examples to illustrate general statements
- Summarize ideas that the passage is arguing against

4.2 Inductive Reasoning

1. Inductive Arguments

- A. In an ***inductive argument***, the premises are meant to support a conclusion but not to fully prove it. For example, the premises may just be meant to give evidence that the conclusion is **probably** true, leaving a chance that the conclusion is false despite that evidence.
- B. An inductive argument may be ***strengthened*** by adding reasons that directly support the conclusion or that help the argument's premises better support the conclusion. Conversely, an inductive argument may be ***weakened*** by adding reasons that directly cast doubt on the argument's conclusion, or that make the argument's premises less effective at supporting the conclusion. Below, we discuss how various types of inductive arguments are evaluated, strengthened, and weakened.

2. Generalizations and Predictions

- A. An argument by ***generalization*** often uses premises about a sample of a population to support a conclusion about the whole population.

Example:

Six of the eight apartments available for lease in this building are studio apartments. So, probably about $\frac{3}{4}$ of all the apartments in the building are studio apartments.

In this example, the whole set of apartments in the building is a population. The apartments available for lease are a **sample** of that population. Since six of the eight apartments available for lease are studio apartments, $\frac{3}{4}$ of the apartments in the sample are studio apartments. The argument generalizes from this by assuming the whole population is probably like the sample. It concludes that as in the sample, probably about $\frac{3}{4}$ of the apartments in the population are studio apartments.

- B. A similar type of argument by generalization uses premises about a whole population to support a conclusion about part of that population.

Example:

About $\frac{3}{4}$ of all the apartments in the building are studio apartments. So, probably about $\frac{3}{4}$ of the apartments on the building's second floor are studio apartments.

This example uses a premise about the proportion of studio apartments in the population of all the apartments in the building to support the conclusion that there's a similar proportion of studio apartments in just a part of the population—the apartments on the second floor.

- C. A **predictive** argument by generalization uses a premise about the sample observed so far in a population to support a conclusion about another part of the population.

Example:

Of the eight apartments I've visited in this building so far, six have been studio apartments. So, probably about six out of the next eight apartments I visit in the building will also be studio apartments.

In this example, the apartments the author has visited so far are a sample of the total population of apartments in the building. The observation about the proportion of studio apartments in the sample supports a prediction that roughly the same proportion of studio apartments will be found in another part of the population—the next eight apartments the author will visit in the building.

- D. The strength of an argument by generalization partly depends on how similar the sample is to the overall population, or to the unobserved part of the population a prediction is about. A sample chosen in a way likely to make it relevantly different than the population is a **biased sample**. An argument using a biased sample is flawed.

Example:

In a telephone survey of our city's residents, about four out of every five respondents said they usually answer the phone when it rings. So, about four out of every five residents of our city usually answer the phone when it rings.

In this example, the sample is the respondents to the telephone survey. People who usually answer the phone when it rings are more likely than other people to respond to telephone surveys. Because the sample was selected through a telephone survey, probably a greater proportion of the sample than of all city residents usually answer the phone when it rings. So, the argument is flawed because the sample is biased.

- E. The strength of an argument generalizing from a sample also partly depends on the sample's size. The smaller the sample, the weaker the argument. This is because a smaller sample is statistically likely to differ more from the population in its average traits. An argument by

generalization that uses too small a sample to justify its conclusion is flawed by ***hasty generalization***.

Example:

A coin came up heads five of the eight times Beth flipped it. This suggests the coin she flipped is weighted to make it come up heads more often than tails.

In this argument, the sample is the eight flips of the coin, and the population is all the potential flips of the same coin. The sample is probably not biased because Beth's flips of the coin are probably no more likely than anyone else's flips of the same coin to come up heads or tails. However, the sample is too small to justify the conclusion that the coin is weighted to favor heads. A fair, unweighted coin flipped eight times usually comes up heads more or fewer than exactly four times—just by chance. So, this argument is flawed by hasty generalization. If Beth and other people flipped the coin thousands or millions of times, and still saw it come up heads in five out of every eight flips, that would strengthen the argument. But no matter how many times the coin was flipped to confirm this pattern, a tiny chance would be left that the coin was not weighted to favor heads and that the results had been purely random.

- F. An argument by generalization is weaker when its conclusion is more precise, and stronger when its conclusion is vaguer, given the same premises. That's because a sample usually doesn't precisely match the population it's extracted from. A less precise conclusion allows a larger range of potential mismatches between sample and population. So, it's more likely to be true, given the same evidence. An argument whose conclusion is too precise for its premises to justify is flawed by the ***fallacy of specificity***.

Example:

Biologists carefully caught, weighed, and released fifty frogs out of the hundreds in a local lake. These fifty frogs weighed an average of 32.86 grams apiece. So, the frogs in the lake must also weigh an average of 32.86 grams apiece.

In this example, the sample might be biased because frogs of certain types might have been easier for the biologists to catch. But even if the biologists avoided any sampling bias, the average weight of the fifty sampled frogs probably wouldn't exactly match the average weight of the hundreds of frogs in the lake. The conclusion is too precise, so the argument suffers from the fallacy of specificity. A stronger argument might use the same evidence to conclude less precisely that the frogs in the lake weigh on average between 25 and 40 grams apiece. This less precise conclusion would still be true even if the average weight of the sampled frogs didn't exactly match the average weight of all the frogs in the lake. Since the less precise conclusion is more likely to be true given the same evidence, that evidence justifies it better. That means the argument for the less precise conclusion is stronger.

3. Causal Reasoning

- A. Causal arguments use premises about correlations or causal links to support conclusions about causes and effects. Causal reasoning is hard because causal links can't be directly observed. And there's no scientific or philosophical consensus about what causality is. But saying that one type of situation causally contributes to another usually implies that after a situation of the first type, situations of the second type are more likely. It also implies that situations of the first type help to explain situations of the second type.

Example:

Bushes of the species in our yard tend to die after several weeks without water. So, they must need water at least every few weeks to survive.

In this example, the premise is that after situations of one type (bushes of a certain species getting no water for several weeks), situations of another type become more likely (the bushes dying). The conclusion implies that the first type of situation causes the second: getting no water for several weeks causes bushes of that species to die.

- B. A causal argument may use a general correlation to support the conclusion that a situation of one type caused a situation of another type.

Example:

A bush in our yard just died. There's been no rain this summer, and no one has been watering the yard. Bushes of the species in our yard tend to die after several weeks without water. So, probably, the bush died because it didn't get enough water.

In this example, the first premises say that a **specific** situation of one type (a bush of a specific species dying) followed a specific situation of another type (the bush going without water for weeks). The final premise says that **in general**, situations of the first type (bushes of that species dying) follow situations of the second type (bushes of that species getting no water for weeks). These premises together support the conclusion that the lack of water caused the bush to die.

- C. Causal arguments can be weakened by observations that suggest alternative causal explanations. A way to check which of two competing explanations is stronger is to look at situations with the possible cause from one explanation but not the possible cause from the other.

Example:

Explanation 1: Bushes of this species tend to die after several weeks without water. Maybe the lack of water kills the bushes.

Explanation 2: Bushes of this species grow only in a region where long dry spells are always very hot. Maybe the heat alone kills these bushes during weeks without water.

To check which of these two hypothetical explanations is stronger, we can run two experiments. Each experiment creates a situation with one of the two proposed causes but not the other.

Experiment 1: Water some of the bushes often during weeks of extreme heat and see how well they survive.

Experiment 2: Keep some of the bushes dry in cooler weather and see how well they survive.

Finding that the bushes survive well in Experiment 1 but not in Experiment 2 would support Explanation 1 and cast doubt on Explanation 2.

Finding that the bushes survive well in Experiment 2 but not in Experiment 1 would support Explanation 2 and cast doubt on Explanation 1.

Finding that the bushes always die in both experiments would support both explanations, suggesting either heat or drought alone can kill the bushes.

Finding that the bushes survive well in both experiments would cast doubt on both explanations. It would suggest that something other than drought or heat is killing the bushes—or that drought and heat must occur together to kill the bushes.

D. Experiments to test causal hypotheses shouldn't add or remove possible causal factors other than those tested for.

Examples:

(i) To run Experiment 1 above, a scientist planted some of the bushes in a tropical rainforest where very hot days are usually rainy.

(ii) To run Experiment 2 above, a scientist planted some of the bushes under an awning where rain couldn't reach them in cooler weather.

These versions of the two experiments are problematic because they add other possible causal factors. In example (i), the rainforest's soil type, insects, or humidity might make it harder or easier for the bushes to survive, regardless of the heat and rainfall. In example (ii), putting the bushes under an awning would likely help keep them shaded. That, too, might make it easier or harder for the bushes to survive, regardless of the heat and rainfall. These experimental design flaws cast doubt on any argument that cites these versions of the experiments as evidence to support Explanation 1 or Explanation 2.

Testing a hypothesis through experiments usually means reasoning by generalization: a conclusion about a whole population is reached by observing a sample in the experiment. Causal arguments based on experiments can have the flaws discussed above in Section 4.2.2, "Generalizations and Predictions." A causal argument is weak if it generalizes from a sample that's too small or chosen in a biased way. In small or biased samples, correlations between factors that aren't causally linked often appear just by chance or because of outside factors.

E. Even when two factors clearly are causally linked, it can be hard to tell which causes which, or whether a third, underlying factor causes them both.

Example:

One type of earthworm is far more often found in soil under healthy bushes of a certain species than in soil under sickly bushes of that same species.

Even if the earthworms' presence is causally linked to the bushes' health, the causal link could be that:

- (i) the earthworms improve the bushes' health, or
- (ii) healthier bushes attract the earthworms, or
- (iii) certain soil conditions both improve the bushes' health and attract the earthworms.

More than one of these causal links, and others, may hold at once. A way to untangle the causal links is to find out: (i) how healthy the bushes are in the same soil conditions without earthworms,

(ii) how attracted the earthworms are to those soil conditions without the bushes, and

(iii) whether the earthworms tend to appear around healthier bushes even in different soil conditions.

- F. Even reliable correlations sometimes arise just by chance. One way to check whether a correlation between two types of situations is just a coincidence is to test whether stopping situations of the first type also stops situations of the other type. Even when no test is possible, you can consider whether there's any plausible way either type of situation could cause the other.

Example:

For years, Juan has arrived at work at a hair salon every weekday at exactly 8:00 a.m. Five hundred miles to the north, over the same years, Ashley has arrived at work at a car dealership every weekday at exactly 8:01 a.m. So, Juan's daily arrival at his work must make Ashley arrive at her work a minute later.

In this example, even though for years Ashley has always arrived at work a minute after Juan has, the argument is absurdly weak. That's because Juan's arrival at his work has no apparent way to affect when Ashley arrives at her work. But we could still test the hypothesis in the argument's conclusion. For example, we could persuade Juan to vary his arrival times, then see whether Ashley's arrival times change to match. With no plausible link between the two workers' arrival times, we'd need a lot of evidence like this to reasonably overcome the suspicion that the correlation is pure coincidence. However, finding out that Juan and Ashley know each other and have reasons to coordinate their work schedules could give us a plausible causal link between their arrival times. That would greatly strengthen the argument that their arrival times are indeed causally linked.

4. Analogies

- A. An argument by ***analogy*** starts by saying two or more things are alike in certain ways. The argument then gives a claim about one of those two things as a reason to accept a similar claim about the other.

Example:

Laotian cuisine and Thai cuisine use many of the same ingredients and cooking techniques. Ahmed enjoys Thai cuisine. So, if he tried Laotian cuisine, he'd probably enjoy it.

This example starts by saying how Laotian cuisine and Thai cuisine are alike: they use similar ingredients and cooking techniques. The argument then makes a claim about Thai cuisine: that Ahmed enjoys it. By analogy, these premises support a similar claim about Laotian cuisine: Ahmed would enjoy it if he tried it.

- B. For an argument by analogy to work well, the noted similarities must be **relevant** to whether the two things are also similar in the way the conclusion claims. The argument in the example above meets this standard. A cuisine's ingredients and cooking techniques usually affect how much a specific person would enjoy it. Noting that Laotian cuisine and Thai cuisine have similar ingredients and cooking techniques is relevant to whether Ahmed is likely to similarly enjoy the two cuisines.

An argument by analogy is weaker if its premises only note similarities that are less relevant to its conclusion.

Example:

Laotian cuisine and Latvian cuisine both come from nations whose English names start with the letter *L*. Ahmed enjoys Latvian cuisine. So, if he tried Laotian cuisine, he'd probably enjoy it too.

This example notes that Laotian cuisine and Latvian cuisine are similar with respect to the English names of the nations they're from. Since the spelling of a nation's name almost never affects how much anyone enjoys that nation's cuisine, this similarity is irrelevant to whether Ahmed would similarly enjoy the two cuisines. The analogy is absurd, so the argument is flawed. To save the argument, we'd need a good reason why the noted similarity is relevant after all—for example, evidence that Ahmed is an unusual person whose enjoyment of different cuisines depends on English spellings.

- C. A reasonable argument by analogy can be strengthened by noting other relevant similarities between the things compared, or weakened by noting relevant dissimilarities.

Example:

Beth and Alan are children living on the same block in the Hazelfern School District. Beth attends Tubman Primary School. Therefore, Alan probably does as well.

Noting that Beth and Alan are both in the same grade would strengthen this moderately reasonable argument, because that similarity increases the odds that they attend the same school. However, noting that Beth is eight years older than Alan would weaken the argument, because that dissimilarity suggests Alan may be too young to attend the school Beth attends.

4.3 Deductive Reasoning

1. Deductive Arguments

- A. The premises of a **deductive argument** are given to fully prove its conclusion. A valid deductive argument with only true premises **must** have a true conclusion. An argument presented as deductive is flawed if its premises can all be true while its conclusion is false. However, a flawed deductive argument might still work well as an inductive argument if the author doesn't wrongly present the premises as **proving** the conclusion. Deductive arguments often use **logical operators** or **quantifiers** or both, as explained below.

2. Logical Operators

- A. A **logical operator** shows how the truth or falsehood of one or more statements affects the truth or falsehood of a larger statement made from those statements and the operator. The basic logical operators are **negations**, **logical conjunctions**, **disjunctions**, and **implications**.
- B. A statement's **negation** is true just when the statement is false. Words and phrases like **not**, **it is false that**, and **it is not the case that** often mark negation.

Statements are often vague, ambiguous, context-sensitive, or subjective. They may be true in one sense and false in another, they may be only partly true, or their truth may be indefinite. If a statement is true only in one way or to a limited degree, its negation is false in the same way and to the same degree.

Example:

“The cat is on the mat” can have the negation “The cat is not on the mat.” Either of these statements is true if the other is false—but only when both are about the same cat and the same mat, in the same sense and the same context. If you make the first statement while the cat is sleeping on the mat, but then the cat wakes up and leaves before you make the second statement, the context has changed. Then your second statement isn’t the negation of your first. If the cat is only partly on the mat when both statements are made, then the second statement is **partly** false just as much as the first is partly true, and in just the same way.

- C. A **logical conjunction** of two statements is true just when both are true. The words and phrases below can mark a logical conjunction of statements **A** and **B**:

<i>A and B</i>	<i>A even though B</i>	<i>not only A but also B</i>
<i>Although A, B</i>	<i>A. Furthermore, B.</i>	<i>A, whereas B</i>
<i>A but B</i>	<i>A, however, B</i>	

The conjunction markers **and**, **furthermore**, and **not only . . . but also** usually imply that **A** and **B** are relevant to each other or mentioned for similar reasons—for example, that both are premises supporting the same conclusion. On the other hand, the conjunction markers **although**, **but**, **even though**, **however**, and **whereas** suggest tension between **A** and **B**—for example, that it’s surprising **A** and **B** are both true, or that **A** and **B** support conflicting conclusions, or that **A** and **B** differ in some other unexpected way.

Examples:

(i) Raul has worked for this company a long time, **and** he's searching for another job.

(ii) **Although** Raul has worked for this company a long time, he's searching for another job.

Both these examples say that Raul has worked for the company a long time, and that he's searching for another job. But in example (ii), **although** suggests Raul's search for another job is **surprising**, given that he's worked for the company a long time. In contrast, the **and** in example (i) suggests Raul's search for another job is **unsurprising** now that he's worked for the company a long time.

D. A **disjunction** of two statements is true only when one of them is true. The words and phrases **A or B**, **either A or B**, and **A unless B** often mark a disjunction of **A** and **B**.

There are two kinds of disjunction. An **inclusive disjunction** of two statements is true when **at least** one of them is true, and **also** when both are. An **exclusive disjunction** of two statements is true just when **exactly** one of them is true—**not** when both are. English disjunctions often aren't clearly inclusive or clearly exclusive. But **A or B or both** clearly means inclusive disjunction. And **A or B but not both** clearly means exclusive disjunction.

Examples:

(i) It will **either** rain **or** snow tomorrow.

(ii) It will rain tomorrow **unless** it snows.

These examples both say that at least one of the statements “It will rain tomorrow” and “It will snow tomorrow” is true. But neither example clearly says whether or not it might **both** rain **and** snow tomorrow. To clarify, we can say: (iii) Tomorrow, it will rain or snow—or both. (***inclusive disjunction***)

or

(iv) Tomorrow, it will either rain or snow, but not both. (***exclusive disjunction***)

E. A ***conditional*** says that for one statement to be true, another must be true. In other words, a conditional means the first statement entails the second. The words and phrases below all mark the same conditional link between statements **A** and **B**:

<i>A would mean that B</i>	<i>B if A</i>	<i>A only if B</i>
<i>If A, then B</i>	<i>Not A unless B</i>	<i>B provided that A</i>

Conditionals of these forms do not mean that **A** is true, nor that **B** is. They do not give **A** as a reason to accept **B**. So, in a conditional, **A** isn’t a premise and **B** isn’t a conclusion. That is, a conditional is not an argument. However, the conditional ***if A, then B*** does mean that correctly assuming **A** as a premise lets you correctly reach **B** as a conclusion.

Examples:

(i) It will snow tonight **only if** the temperature falls below 5 degrees Celsius.

(ii) It **won't** snow tonight **unless** the temperature falls below 5 degrees Celsius.

(iii) **If** it snows tonight, it'll mean the temperature has fallen below 5 degrees Celsius.

These three conditionals all mean the same thing. Each says that snow tonight would require a temperature below 5 degrees Celsius. They do not say that it **will** snow tonight, nor that the temperature **will** be below 5 degrees Celsius. But they do suggest, for example, that seeing it snow tonight would tell you the temperature must be below 5 degrees Celsius.

Although conditionals often make or suggest causal claims, their meaning isn't always causal. The examples above do not mean that snow tonight would **cause** the temperature to fall below 5 degrees Celsius, nor vice versa.

None of these examples imply that it **must** snow if the temperature falls below 5 degrees Celsius tonight. A conditional **if A, then B** does not imply that **if B, then A**.

F. Two **logically equivalent** statements **A and B** are always both true or both false under the same conditions. Each implies the other. That is, **if A, then B**, and **if B, then A**. These two conditionals can be combined as **A if and only if B**.

3. Reasoning with Logical Operators

A. Here's a list of some types of logically equivalent statements made with logical operators. In this list, **not** means negation, **and** means logical conjunction, **or** means inclusive disjunction, and **if . . . then** means conditional implication.

Logical Equivalences with Logical Operators		
<i>A and B</i>	is logically equivalent to	<i>B and A</i>
<i>not (A and B)</i>	is logically equivalent to	<i>not-A or not-B</i>
<i>A or B</i>	is logically equivalent to	<i>B or A</i>
<i>not (A or B)</i>	is logically equivalent to	<i>not-A and not-B</i> (in other words, <i>neither A nor B</i>)
<i>if A, then B</i>	is logically equivalent to	<i>if not-B, then not-A</i>
<i>if A, then (B and C)</i>	is logically equivalent to	<i>(if A, then B) and (if A, then C)</i>
<i>if A, then (B or C)</i>	is logically equivalent to	<i>(if A, then B) or (if A, then C)</i>
<i>if (A or B), then C</i>	is logically equivalent to	<i>(if A, then C) and (if B, then C)</i>

- B. Of any two logically equivalent statements, either can be a premise supporting the other as a conclusion in a valid deductive argument. For any line in the list above, a valid deductive argument has a premise of the form on one side and a logically equivalent conclusion of the form on the other side.

Examples:

The second line in the list above says that for any statements **A** and **B**, the statement **not (A and B)** is logically equivalent to **not-A or not-B**. This gives us two valid deductive arguments: (i) **not (A and B), therefore not-A or not-B**

and

(ii) **not-A or not-B, therefore not (A and B)**

For example, the statement **Ashley and Tim don't both live in this neighborhood** is logically equivalent to **Either Ashley doesn't live in this neighborhood or Tim doesn't**. This lets us make two valid deductive arguments: (iii) Ashley and Tim don't both live in this neighborhood. Therefore, either Ashley doesn't live in this neighborhood or Tim doesn't.

and

(iv) Either Ashley doesn't live in this neighborhood or Tim doesn't. Therefore, Ashley and Tim don't both live in this neighborhood.

C. Here's a list of some other valid deductive argument forms with logical operators, and of invalid forms often confused with them.

Valid and Invalid Inferences with Logical Operators	
Valid: <i>A and B, therefore A</i>	Invalid: <i>A, therefore A and B</i>
Valid: <i>A, therefore A or B</i>	Invalid: <i>A or B, therefore A</i>
Valid: <i>not-A and not-B, therefore not (A and B)</i>	Invalid: <i>not (A and B), therefore not-A and not-B</i>
Valid: <i>not (A or B), therefore not-A or not-B</i>	Invalid: <i>not-A or not-B, therefore not (A or B)</i>
Valid: <i>if A, then B; and A; therefore B</i>	Invalid: <i>if A, then B; and B; therefore A</i>
Valid: <i>if A, then B; and not-B; therefore not-A</i>	Invalid: <i>if A, then B; and not-A; therefore not-B</i>

Examples:

The third line in the table above says that ***not-A and not-B, therefore not (A and B)*** is valid, but ***not (A and B), therefore not-A and not-B*** is invalid. A simple example of a **valid** argument is: (i) Ashley doesn't live in this neighborhood, and Tim doesn't either. Therefore, Ashley and Tim don't both live in this neighborhood.

But swapping argument (i)'s premise with its conclusion makes this **invalid** argument:

(ii) Ashley and Tim don't both live in this neighborhood. Therefore, Ashley doesn't live in this neighborhood and Tim doesn't either.

As another example, the table's fifth line says that ***if A, then B; and A; therefore B*** is valid, but ***if A, then B; and B; therefore A*** is invalid. So, another **valid** argument is: (iii) If Ashley lives in this neighborhood, so does Tim. Ashley does live in this neighborhood. Therefore, Tim also lives in this neighborhood.

However, swapping argument (iii)'s second premise with its conclusion makes this **invalid** argument:

(iv) If Ashley lives in this neighborhood, so does Tim. Tim does live in this neighborhood. Therefore, Ashley lives in this neighborhood.

4. Necessity, Probability, and Possibility

A. Some words and phrases mark how likely a statement is to be true. For example, they may mean that:

- The statement is ***necessarily*** true; that is, there's a 100 percent chance the statement is true; or
- The statement is ***probably*** true; that is, there's a good chance the statement is true;

- The statement is ***possibly*** true; that is, the odds are greater than 0 percent that the statement is true.

Saying a claim is possibly true or probably true usually implies it's not necessarily true.

B. The table below shows three categories of words and phrases that can stand for degrees of probability:

Words Standing for Necessity, Probability, and Possibility		
Necessity	Probability	Possibility
<i>certainly</i>	<i>probably</i>	<i>can</i>
<i>clearly</i>	<i>likely</i>	<i>could</i>
<i>definitely</i>	<i>more likely than not</i>	<i>may</i>
<i>must</i>		<i>maybe</i>
<i>necessarily</i>		<i>might</i>
<i>surely</i>		<i>perhaps</i>
		<i>possibly</i>

The words ***probably*** and ***likely*** sometimes mean high probability, like a 95 percent chance. Other times they mean a medium chance, even one below 50 percent. Don't give these terms any exact meanings when you find them on the GMAT exam.

C. The table below lists some valid deductive argument forms with necessity, probability, and possibility, as well as invalid forms often confused with them.

Valid and Invalid Inferences with Necessity, Probability, and Possibility	
Valid: <i>Probably A, therefore possibly A</i>	Invalid: <i>Possibly A, therefore probably A</i>
Valid: <i>Possibly (A and B), therefore possibly A and possibly B</i>	Invalid: <i>Possibly A and possibly B, therefore possibly (A and B)</i>
Valid: <i>Probably (A and B), therefore probably A and probably B</i>	Invalid: <i>Probably A and probably B, therefore probably (A and B)</i>
Valid: <i>Probably A or probably B, therefore probably (A or B)</i>	Invalid: <i>Probably (A or B), therefore probably A or probably B</i>
Valid: <i>Necessarily A or necessarily B, therefore necessarily (A or B)</i>	Invalid: <i>Necessarily (A or B), therefore necessarily A or necessarily B</i>

Examples:

The table's second line says that ***possibly (A and B), therefore possibly A and possibly B*** is valid, while ***possibly A and possibly B, therefore possibly (A and B)*** is invalid. A simple **valid** argument is:

(i) Possibly Tim and Ashley both live in this house. So, possibly Tim lives in this house, and possibly Ashley does.

Swapping argument (i)'s premise with its conclusion gives us this **invalid** argument:

(ii) Possibly Tim lives in this house, and possibly Ashley does. So, possibly both Tim and Ashley live in this house.

To see that argument (ii) is invalid, suppose you know only one person lives in the house, but you don't know whether that person is Tim, Ashley, or someone else. Then argument (ii)'s premise would be true, but its conclusion would be false.

5. Quantifiers

A. A **quantifier** is a word or phrase for a proportion, number, or amount. Some basic quantifiers are ***all***, ***most***, ***some***, and ***none***.

- i. A quantifier like ***all*** means 100 percent of the individuals in a category, or the whole of an amount.
- ii. A quantifier like ***most*** usually means more than half the individuals in a category, or more than half of a whole. ***Most*** usually implies ***not all***, but not always. Writing ***most but not all***, or else ***most or all***, can clarify the meaning.
- iii. A quantifier like ***some*** often means one or more individuals in a category, or part of a whole. ***Some*** usually but not always implies ***not all***. However, ***only some*** clearly does imply ***not all***, while ***at least some*** clearly doesn't. ***Some*** with a plural usually means ***more than one***. In contrast, ***some*** with a singular usually means ***exactly one***. For example, "some dogs" usually means "more than one dog," while "some dog" usually means "exactly one dog." But "some dog or dogs" means "at least one dog."

- iv. A quantifier like **no** or **none of** means something is being denied about all the individuals in a category, or about all of some whole.
- v. Other common quantifiers have more nuanced meanings. For example, **a few** vaguely means a small number more than two. The upper limit of what counts as **a few** depends on context. For example, “a few Europeans” might mean thousands of people (still a tiny part of Europe’s population). But “a few residents in our building” might mean only three or four people if the building has only fifteen residents.

B. The table below classifies some quantifier words by their meanings.

Basic Quantifier Words			
“All” and similar quantifier words	“Most” and similar quantifier words	“Some” and similar quantifier words	“No” and similar quantifier words
<i>all</i>	<i>generally</i>	<i>a number</i>	<i>never</i>
<i>always</i>	<i>a majority</i>	<i>a portion</i>	<i>no</i>
<i>any</i>	<i>most</i>	<i>any</i>	<i>none</i>
<i>both</i>	<i>more than half</i>	<i>at least one</i>	<i>not any</i>
<i>each</i>	<i>usually</i>	<i>occasionally</i>	<i>not one</i>
<i>every</i>		<i>one or more</i>	<i>nowhere</i>
<i>everywhere</i>		<i>some</i>	
<i>whenever</i>		<i>sometimes</i>	
<i>wherever</i>		<i>somewhere</i>	

Notice the table shows **any** both as a quantifier like **all** and as a quantifier like **some**. That’s because **any** can have either meaning. For example, **Any of the students would prefer chocolate ice cream** means **Each of the students would prefer chocolate ice cream**. However, **I**

don't know if any of the students would prefer chocolate ice cream means *I don't know if even one of the students would prefer chocolate ice cream.*

- C. A quantifier used with a category usually implies the category isn't empty. But this doesn't always hold in hypothetical statements, in conditionals, or with the quantifier **any**.

Examples:

(i) **All** life forms native to planets other than Earth **are** carbon-based.

(ii) **Any** life forms native to planets other than Earth **would be** carbon-based.

In (i), the words **all** and **are** show the author is claiming there really are life forms native to planets other than Earth. But in (ii), the words **any** and **would be** show the author is carefully avoiding that claim.

- D. Statements with two or more quantifiers sometimes look alike but differ in meaning because of word order and phrasing.

Example:

(i) Some beverage must be the favorite of every student in the class.

(ii) Each student in the class must have some favorite beverage.

Statement (i) suggests that every student in the class must have **the same** favorite beverage. In contrast, statement (ii) can be true even if each student has a **different** favorite beverage.

6. Reasoning with Quantifiers

- A. Here's a list of some logically equivalent statement forms with quantifiers. In this list, **some** means **one or more**. The forms in the list use plurals, but similar equivalences can hold without plurals. For

example, **No water is fire** is logically equivalent to **No fire is water**, even though those two statements don't have plurals like the forms in the list do.

Logical Equivalences with Quantifiers		
<i>All As are Bs</i>	is logically equivalent to	<i>No As are not Bs.</i>
<i>Some As are Bs</i>	is logically equivalent to	<i>Some Bs are As.</i>
<i>No As are Bs</i>	is logically equivalent to	<i>No Bs are As.</i>
<i>Some As are not Bs</i>	is logically equivalent to	<i>Not all As are Bs.</i>

However, **All As are Bs** is **not** equivalent to **All Bs are As**. And **Some As are not Bs** is **not** equivalent to **Some Bs are not As**.

Examples:

(i) The true statement **All ostriches are birds** is not equivalent to the false statement **All birds are ostriches**.

(ii) The true statement **Some birds are not ostriches** is not equivalent to the false statement **Some ostriches are not birds**.

B. As explained above, either of two logically equivalent statements can be a premise supporting the other as a conclusion in a valid deductive argument. This works for equivalences with quantifiers just like it does for equivalences with logical operators.

C. A **syllogism** is a type of simple deductive argument whose two premises have one quantifier apiece, and whose conclusion also has one quantifier.

Here's a list of some valid syllogism forms along with invalid forms sometimes confused with them. As above, in this list, **some** means **one or more**.

Valid and Invalid Syllogisms

Valid: *All As are Bs. All Bs are Cs. So, all As are Cs.*

Invalid: *All As are Bs. All Bs are Cs. So, all Cs are As.*

Invalid: *All As are Bs. All Cs are Bs. So, all As are Cs.*

Invalid: *All Bs are As. All Bs are Cs. So, all As are Cs.*

Valid: *Some As are Bs. All Bs are Cs. So, some As are Cs.*

Invalid: *All As are Bs. Some Bs are Cs. So, some As are Cs.*

Invalid: *Some As are Bs. Some Bs are Cs. So, some As are Cs.*

Valid: *All As are Bs. No Bs are Cs. So, no As are Cs.*

Invalid: *No As are Bs. All Bs are Cs. So, no As are Cs.*

Invalid: *No As are Bs. No Bs are Cs. So, all As are Cs.*

Invalid: *No As are Bs. All Bs are Cs. So, some As are not Cs.*

Examples:

The list's first line says that ***All As are Bs. All Bs are Cs. So, all As are Cs*** is valid. And here's a **valid** syllogism of that form: (i) All the trees in the local park were planted by the town arborist. All the trees the arborist planted have been labeled by her. So, all the trees in the park must have been labeled by the arborist.

A similar-looking but **invalid** syllogism has the form ***All As are Bs. All Bs are Cs. So, all Cs are As*** from the list's second line: (ii) All the trees in the local park were planted by the town arborist. All the trees the arborist planted have been labeled by her. So, all the trees the arborist has labeled must be in the park.

To see that argument (ii) is invalid, notice that even if both premises are true, the arborist might also have labeled trees outside the park—maybe even trees she didn't plant.

Another **invalid** syllogism has the form ***All As are Bs. All Cs are Bs. So, all As are Cs*** from the list's third line: (iii) All the trees in the local park were planted by the town arborist. All the trees the arborist has labeled are trees she planted. So, all the trees in the park must have been labeled by the arborist.

To see that argument (iii) is invalid, notice that even if both premises are true, the arborist might not have labeled every tree she planted, nor even every tree she planted in the park.

The form in the list's fourth line (***All Bs are As. All Bs are Cs. So, all As are Cs***) gives us yet another **invalid** syllogism: (iv) All the trees the town arborist has planted are in the local park. All the trees the arborist planted have been labeled by her. So, all the trees in the park must have been labeled by the arborist.

To see that argument (iv) is invalid, notice that even if both premises are true, the park might have many trees that the arborist neither planted nor labeled.

D. Some quantifier words in the table “Basic Quantifier Words” in Section 4.3.5.B above refer to time or place. For example, ***whenever***

means **every time**, **usually** means **most times**, and **never** means **at no time**. Understanding these meanings can help you rewrite deductive arguments using these words into standard syllogisms to check their validity.

Example:

Max never goes running when the sidewalks are icy. The sidewalks are usually icy on January mornings, so Max must not go running on most January mornings.

We can rewrite this argument in the valid syllogism form **No As are Bs. Most Cs are Bs. So, most Cs are not As**: No occasions when Max goes running are occasions when the sidewalks are icy. Most January mornings are occasions when the sidewalks are icy. So, most January mornings are not occasions when Max goes running.

Since arguments in this syllogism form are valid, we can tell that the argument in this example is valid.

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5.0 Data Insights Review

5.0 Data Insights Review

For the GMAT™ Data Insights section, and for graduate business programs, you need basic skills in analyzing data. This chapter is about the kinds of data in tables and charts. It's only a brief overview. If you find unfamiliar terms, consult outside resources to learn more. Before reading this chapter, read this book's Section 3.4.1, "Statistics," and Section 4.2.2, "Generalizations and Predictions," to review basic statistics and inductive generalization.

[Section 5.1](#), "Data Sets and Types," includes:

1. Data Sets
2. Qualitative Data
3. Quantitative Data

[Section 5.2](#), "Data Displays," includes:

1. Tables
2. Qualitative Charts
3. Quantitative Charts

[Section 5.3](#), "Data Patterns" includes:

1. Distributions
2. Trends and Correlations

5.1 Data Sets and Types

1. Data Sets

- A. A **data set** is an organized collection of data about a specific topic. A data set can be shown with one or more tables, charts, or both.

Example:

A data set might have data about all the employees in a company's human resources department, such as their first names, last names, home addresses, phone numbers, positions, salaries, hiring dates, and full-time or part-time statuses.

- B. A **case** is an individual or thing that a data set holds data about. Often a data set has the same types of data about many cases together. Then a table or chart can show those types of data for all the cases.

Example:

In the example above, each employee in the human resources department is a case.

- C. A **variable** is any specific type of data a data set holds about its cases. For each case, the variable has a **value**.

Example:

In the example above, each type of data held about the employees is a variable: *first name* is one variable, *home address* is another variable, *phone number* is a third variable, and so on. If Zara is an employee in the human resources department, one value of the variable *first name* is "Zara."

In a simple table, the top row often names the variables. Each column shows one variable's values for all the cases. In a chart, labels usually say which variables are shown.

- D. The value of a ***dependent variable*** depends on the values of one or more other variables in the data set. An ***independent variable*** is not dependent.

Example:

In a data set of revenues, expenses, and profits, *revenue* and *expense* are independent variables. But *profit* is a dependent variable because profit is calculated as revenue minus expense. For each case, the value of the variable *profit* depends on the values of the variables *revenue* and *expense*.

- E. A ***data point*** gives the value of one variable in one case. A cell in a table usually stands for a data point.

Example:

In the example above, one data point might be that Zara's position is assistant manager. That is, the data point gives the value "assistant manager" for the variable *position* in Zara's case.

- F. A ***record*** is a set of the data points for one case. A row in a table usually shows one record. In a chart, a record might be shown as a point, a line, a bar, or a small shape with a specific position or length.

Example:

In the example above, one record might list Zara's first name, last name, home address, phone number, position, salary, hiring date, and full-time or part-time status.

2. Qualitative Data

- A. ***Qualitative data*** is any type of data that doesn't use numbers to stand for a quantity. Statements, words, names, letters, symbols, algebraic expressions, colors, images, sounds, computer files, and web links can

all be qualitative data. Even data that looks numeric is qualitative if the numbers don't stand for quantities.

Example:

Phone numbers are qualitative data. That's because they don't stand for quantities and aren't used in math—for example, they're generally not summed, multiplied, or averaged.

- B. **Nominal data** is any type of qualitative data that's not ordered in any relevant way. The statistical measures of mean, median, and range don't apply to nominal data, because those measures need an ordering that nominal data lacks. But even in a set of nominal data, some values may appear more often than others. So, the statistical measure of mode does apply to nominal data, because the mode is simply the value that appears most often. To review these statistical terms, refer to Section 3.4.1, "Statistics."

Example:

In the example above, the first names of the human resources department's employees are nominal data if their alphabetical order doesn't matter in the data set. Suppose three of the employees share the first name "Amy," but no more than two of the employees share any other first name. Then "Amy" is the mode of the first names of the department's employees, because it's the first name that appears most often in the data set.

- C. **Ordinal data** is qualitative data ordered in a way that matters in the data set. Because ordinal data is qualitative, its values can't be added, subtracted, multiplied, or divided. So, the statistical measures of mean and range don't apply to ordinal data, because they require those arithmetic operations. However, the statistical measure of median does apply to ordinal data, because finding a median only requires putting the values in an order. The statistical measure of mode also applies, just as it does for nominal data.

Example:

In a data set for a weekly schedule of appointments, the weekdays Monday, Tuesday, Wednesday, Thursday, and Friday are ordinal data. These days are in an order that matters to the schedule, but they're not numbers and don't measure quantities. Suppose the data set lists seven appointments: two on Monday, two on Tuesday, and three on Thursday. The fourth appointment is the median in the schedule, because three appointments are before it and three are after it. The fourth appointment is on Tuesday, so "Tuesday" is the median value of the variable *weekday* for these appointments. "Thursday" is the mode, because more of the appointments are on Thursday than on any other day.

- D. **Binary data** takes only two values, like "true" and "false." Binary data is ordinal if the order of the two values matters, but nominal otherwise. Tables may show binary values with two words like "yes" and "no," or with two letters like "T" and "F," or with a check mark or "X" standing for one of the two values and a blank space standing for the other.

Example:

In the example above of the data set of employees in a human resources department, their employment status is a binary variable with two values: "full time" and "part time." A table might show the employment status data in a column simply titled "Full Time," with a check mark for each full-time employee and a blank for each part-time employee.

- E. **Partly ordered data** has an order among some cases but not others. The statistical measure of median does not apply to a set of partly ordered values, though it might apply to a subset whose values are fully ordered.

Example:

Suppose a family tree shows how people over several generations are related as parents, children, and siblings, but doesn't show when each person was born. This tree lets us partly order the family members by the variable *age*. For example, suppose the family tree shows that Haruto's children are Honoka and Akari, and that Honoka's child is Minato. Since we know that all parents are older than their children, we can tell that Haruto is older than Honoka and Akari, and that Honoka is older than Minato. That also means Haruto is older than Minato. But we can't tell whether Akari is older or younger than her sister Honoka. We can't even tell whether Akari is older or younger than her nephew Minato. So, the family tree only partly orders the family members by age.

3. Quantitative Data

- A. **Quantitative data** is data about quantities measured in numbers. Quantitative values can be added, multiplied, averaged, and so on. The statistical measures of mean, median, mode, range, and standard deviation all apply to quantitative data.

Example:

In the example above of the data set of the employees in the human resources department, the salaries are quantitative data. They're amounts of money shown as numbers.

- B. Quantitative data is **continuous** if it measures something that can be infinitely divided.

Examples:

Temperatures are continuous data. That's because for any two temperatures, some third temperature is between them—warmer than one and cooler than the other. Likewise, altitudes are continuous data. For any two altitudes, some third altitude is higher than one and lower than the other.

A set of continuous quantitative values rarely has a mode. Because infinitely many of these values are possible, usually no two of them in a data set are exactly alike.

C. Quantitative data that isn't continuous is **discrete**.

Examples:

The numbers of students taking different university courses are discrete data, because they're whole numbers. A course normally can't have a fractional number of students.

As another example, prices in a currency are discrete data, because they can't be divided beyond the currency's smallest denomination. Suppose one price in euros is €3.00, and another is €3.01. No price in euros is larger than the first and smaller than the second, because the currency has no denomination below one euro cent (1/100 of a euro). That means you can't have a price of €3.005, for example. The prices in euros aren't continuous, so they're discrete.

Counted numbers of people, objects, or events are generally discrete data.

D. **Interval data** uses a measurement scale whose number zero doesn't stand for a complete absence of the factor measured. So, for interval data, a measurement above zero doesn't show how much greater than nothing the measured quantity is. Because of this, the ratio of two measurements in interval data isn't the ratio of the two measured quantities.

Examples:

Dates given in years are interval data. In different societies' calendars, the year 0 stands for different years. The year 0 in the Gregorian calendar was roughly the year 3760 in the Hebrew calendar, -641 in the Islamic calendar, and -78 in the Indian National calendar. In none of these calendars does 0 stand for the very first year ever. This means, for example, that the ratio of the numbers in the two years 500 CE and 1500 CE in the Gregorian calendar isn't the ratio of two amounts: the year 500 CE isn't $\frac{1}{3}$ the amount of the year 1500 CE.

As another example, temperatures in Celsius or Fahrenheit are also interval data. The temperature 0 degrees Celsius is 32 degrees Fahrenheit. Neither temperature scale uses 0 degrees to mean absolute zero, the complete absence of heat. This means, for example, that the ratio of the numbers in the temperatures 30°F and 60°F isn't the ratio of two amounts of heat. Thus, 60°F isn't twice as hot as 30°F.

In a measurement scale for interval data, each unit stands for the same amount. That is, any two measurements that differ by the same number of units stand for two quantities that differ by the same amount.

Examples:

In the example above of dates given in years, the year 1500 CE was 1000 years after 500 CE, because $1500 - 500 = 1000$. Likewise, the year 1600 CE was 1000 years after 600 CE, because $1600 - 600 = 1000$. Although you can't divide one year by another to find a real ratio, you can subtract one year from another to find a real length of time—in this case, 1000 years.

Likewise, 60°F is the same amount warmer than 40°F as it is cooler than 80°F. A 20°F difference in two temperatures always stands for the same real difference in heat between those temperatures.

E. **Ratio data** uses a measurement scale whose number zero stands for the absence of the measured factor. In ratio data, as in interval data, the difference between two measurements stands for the actual difference between the measured amounts. However, in ratio data, unlike interval data, the ratio of two measurements also stands for the actual ratio of the measured amounts.

Examples:

Measured weights are ratio data, whether they're in kilograms or pounds. That's because 0 kilograms stands for a complete absence of weight, as does 0 pounds. So, the ratio of 10 kilograms to 5 kilograms is a ratio of two real weights. Because the ratio of 10 to 5 is 2 to 1, 10 kilograms is twice as heavy as 5 kilograms, and 10 pounds is twice as heavy as 5 pounds.

As another example, temperatures in Kelvin are ratio data. That's because 0 K stands for absolute zero, the complete absence of heat. Thus, 200 K is really twice as hot as 100 K. As explained above, this isn't true for temperatures in Celsius or Fahrenheit, which are interval data.

F. **Logarithmic data** use a measurement scale whose higher units stand for amounts exponentially farther apart. For logarithmic data, as for ratio data, the number zero stands for a complete absence. But in logarithmic data, when two measurements a certain number of units apart are higher, the real difference between the measured amounts is greater.

Example:

Noise measured in decibels is logarithmic data. Although 0 decibels means complete silence, 30 decibels is 10 times as loud as 20 decibels, not just 1.5 times as loud. And the real difference in loudness between 40 decibels and 30 decibels is 10 times the real difference in loudness between 30 decibels and 20 decibels, even though the two differences are each 10 decibels.

Because higher units on a logarithmic scale stand for greater amounts of difference, you can't just sum and divide logarithmic data to find a statistical mean. Nor can you subtract one logarithmic measurement from another to find a statistical range. Finding the mean and range requires more complex calculations, which you won't have to do on the GMAT.

5.2 Data Displays

1. Tables

- A. A **table** shows data in rows and columns. In a simple table, the top row shows the names of the variables and is called the **header**. Below the header, usually each row shows one record, each column shows one variable, and each cell shows one data point. Sometimes another row above the header has the table's title or description. Sometimes a column or a few rows within the table are used only to group the records by category. Likewise, a row or a few columns within the table are sometimes used only to group the variables by category. Sometimes a row or column shows totals, averages, or other operations on the values in other rows or columns.

Example:

The table below shows revenues, expenses, and profits for two branches of Village Shoppe for one year. The top row is the header listing the independent variables *revenue* and *expense* and the dependent variable *profit*. In each row, the profit is just the revenue minus the expense.

The table has only four rows of records. These are the third row showing the Mapleton branch's January–June finances, the fourth row showing the Mapleton branch's July–December finances, the sixth row showing the Elmville branch's January–June finances, and the seventh row showing the Elmville branch's July–December finances. The second row, “Mapleton branch” serves to group the third and fourth rows together. Likewise, the fifth row, “Elmville branch” serves to group the sixth and seventh rows together. The bottom row, “Annual grand totals” doesn't show a record but rather sums the values in the four records for each of the three variables.

Village Shoppe	Revenue	Expense	Profit
Mapleton branch			
January–June	125000	40000	85000
July–December	90000	35000	55000
Elmville branch			
January–June	85000	30000	55000
July–December	115000	25000	90000
Annual grand totals	415000	130000	285000

- B. Sometimes a table's title or description explains the data shown. To understand the data, always study any title or description.

Example:

In the table below, the title says the populations are in **thousands**. So, to find the total population aged 44 and under, add 63,376 thousand and 86,738 thousand. This gives 150,114 thousand, which is 150,114,000. If you only read the numbers without noticing the title, you'll get the wrong result for the total population aged 44 and under.

Population by Age Group (in thousands)	
Age	Population
17 years and under	63,376
18–44 years	86,738
45–64 years	43,845
65 years and over	24,051

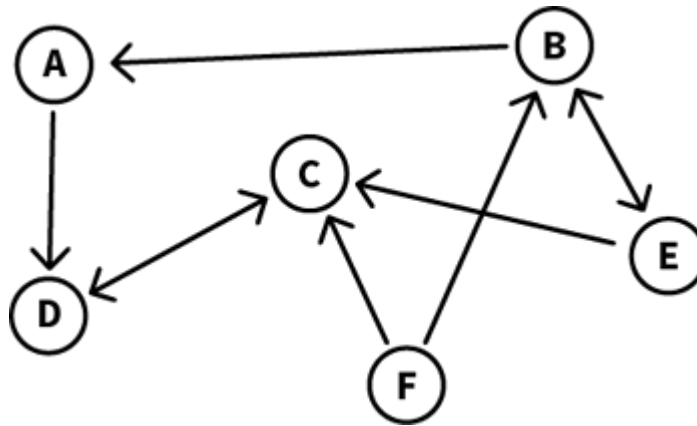
2. Qualitative Charts

- A. Many different types of charts show qualitative data. This book's Section 3.4.2, "Sets," discusses one type of qualitative chart, the **Venn diagram**. Here we describe a few other common types. But the GMAT may also challenge you with unfamiliar types of charts this book doesn't discuss. To understand any type of chart, study its labels. The labels will tell you how to read the chart's various points, lines, shapes, symbols, and colors. Several labels may be together, sometimes inside a rectangle on the chart, to make a **legend**.

- B. A **network diagram** has lines connecting small circles or other shapes. Each small shape is a **node** standing for an individual, and each line stands for a relationship between two individuals. In some network diagrams, the lines are one-way or two-way arrows standing for one-way or two-way relationships.

Example:

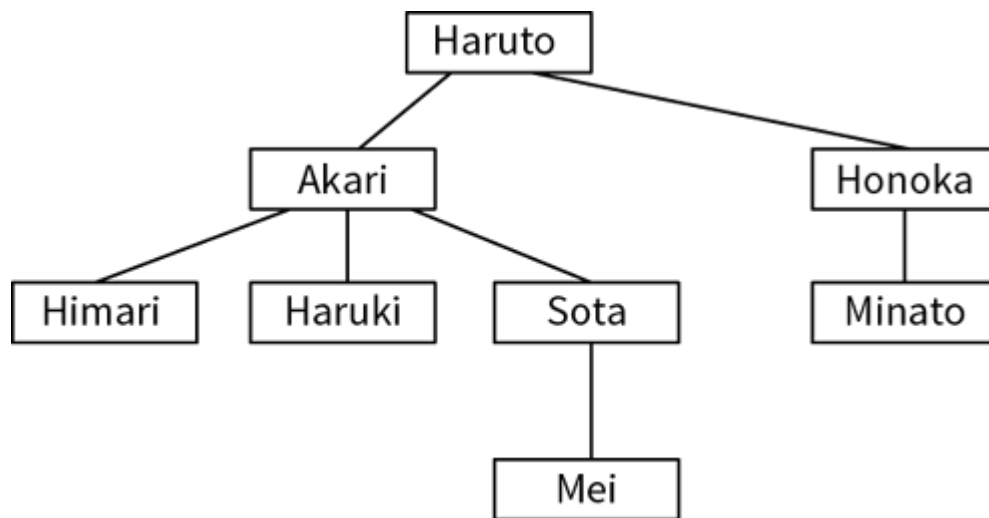
In the network diagram below, the lettered nodes stand for six pen pals: Alice, Ben, Cathy, Dave, Ellen, and Frank. The arrows show who got a letter from whom in the past month. Each arrow points from the pen pal who sent the letter to the one who got it. A two-way arrow means both the pen pals got letters from each other. This diagram tells us many facts about the pen pals and their relationships over the past month. For example, it shows who got the most letters (Cathy received three) and who got the fewest (Frank received none).



- C. A **tree diagram** is a type of network diagram that shows partly ordered data like organizational structures, ancestral relationships, or conditional probabilities. In a tree diagram, each relationship is one way.

Example:

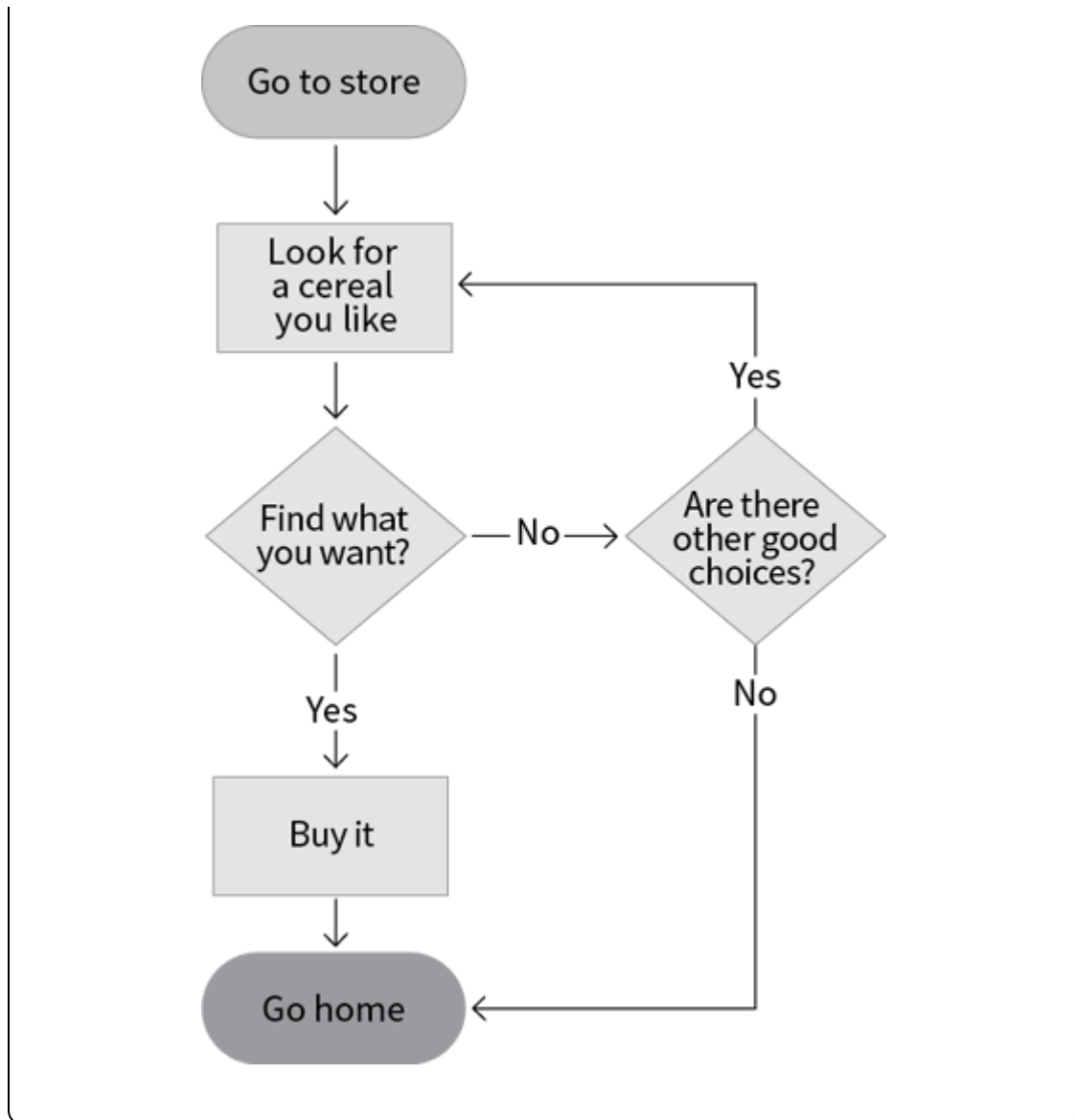
Expanding on the example in Section 5.1.2.E above, the tree diagram below shows how Haruto and his descendants are related. Each line connects a parent above to his or her child below. The diagram shows that Haruto has two children, four grandchildren, and one great-grandchild. From the diagram we can tell that Akari is older than her grandchild Mei, and that Himari and Minato are cousins. But we can't tell whether Akari, Mei, or Himari is older than Minato.



- D. In a **flowchart**, each node stands for a step in a process. Arrows direct you from each step to the next. An arrow pointing back to an earlier step tells you to repeat that step. A flowchart usually has at least three types of nodes. **Process nodes** stand for actions to take and are usually rectangles. **Decision nodes** show questions to answer and are usually diamond shaped. At least two labeled arrows lead from each decision node to show how choosing the next step depends on how you answer the question. **Terminal nodes** show the start or end of the process and are usually oval.

Example:

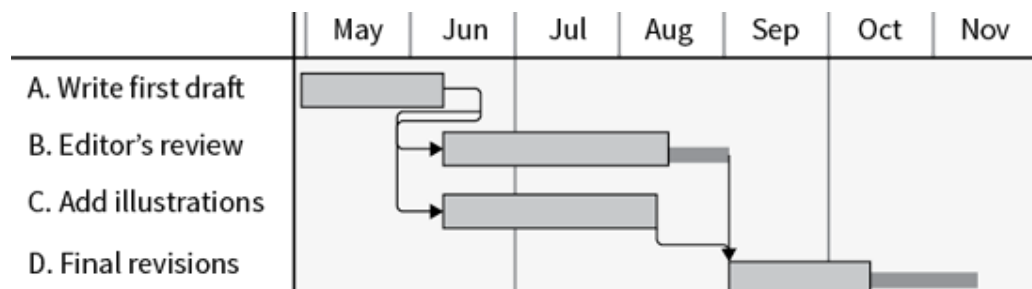
The flowchart below shows a simple process for getting cereal from a store. The top oval shows the first step, going to the store. The arrow below it then takes us to a process node saying to look for a cereal we like. For the third step, we reach a decision node asking whether we've found the cereal we want. If we have, we follow the decision node's "Yes" arrow to another process node telling us to buy the cereal we found, and then we move on to the bottom terminal node telling us to go home. On the other hand, if we haven't found the cereal we want, we follow the decision node's "No" arrow to a second decision node asking whether the store offers other good cereal choices. If it doesn't, we follow another "No" arrow telling us to give up and go home. But if the store does offer other good cereal choices, we follow a "Yes" arrow back to the "Look for a cereal you like" step to peruse the choices again. We repeat the loop until we either find a cereal we want to buy or else decide the store has no more good cereal choices. Since the store won't have infinitely many cereals, we eventually end at the bottom terminal node and go home.



E. A **Gantt chart** is used to plan and schedule projects. The horizontal axis has a time scale showing the period planned for a project. The time units usually are qualitative ordinal units, such as months or calendar dates. The vertical axis is divided into labeled rows, each typically standing for one task in the project. In each row, a bar stretches from the planned start time to the planned end time for that row's task. Sometimes a second bar stretches out past the first to show how a task might take longer than expected. An arrow pointing from the end of one task's bar to the beginning of another's bar means the second task can't begin until the first is finished.

Example:

The Gantt chart below shows a schedule for writing a children's book. The top bar shows that Task A, writing the first draft, should be done over about five weeks in May and early June. Task B, the editor's review, and Task C, adding illustrations, start when Task A ends. The lines pointing from the end of Task A to the beginnings of Tasks B and C mean that Tasks B and C can't start until Task A is finished. So, if Task A isn't finished on time, Tasks B and C will both have to be delayed. Tasks B and C are both scheduled to be finished by mid-August. But the thinner gray bar at the end of Task B's main bar means the plan allows the editor's review to take until the beginning of September if necessary—perhaps in case the editor needs extra time to review the illustrations. The arrows from Tasks B and C to Task D, the final revisions, mean that the final revisions can't start until Tasks B and C are finished. Although Task D is supposed to be done by mid-October, a final thinner gray bar at the end of Task D's main bar shows that the plan allows Task D to take until mid-November in case of delays. Notice that although this chart uses qualitative data, it can still support quantitative estimates of how long different parts of the project may take under various conditions.



3. Quantitative Charts

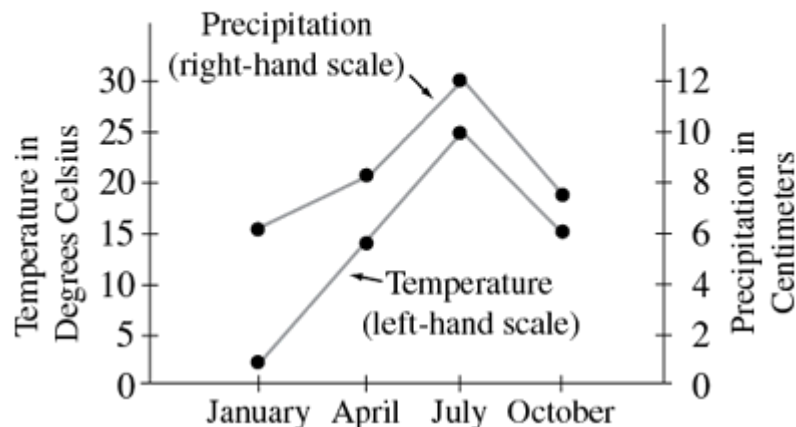
- A. Here we discuss a few common types of charts that normally show quantitative data. The GMAT may also use rarer types this book doesn't discuss. To understand any type of quantitative chart, study the

description and any labels to find out what cases and variables are shown, and how. Also notice any axes. These show the measurement scales. Some quantitative charts have no axes. Others have one, two, or, rarely, three or more. For each axis, notice whether it starts below, at, or above 0. Note the numbers on the axes, as well as the units the numbers stand for. You must read the axes correctly to understand the data.

Example:

In the chart below, labels say that the scale on the left is for temperature data and the scale on the right is for precipitation data. The bottom axis shows four months of the year, spaced three months apart. The chart's title tells us each data point gives the **average** temperature or precipitation in City X during a given month. This implies that the temperatures and precipitations shown are averages over many years. Suppose we're asked to find the average temperature and precipitation in City X in April. To do this, we don't have to calculate averages of any of the values shown. Those values are **already** averages. So, to find the average temperature for April, we simply read the April temperature data point by noting it's slightly lower than the 15 on the temperature scale at the left. Likewise, to find the average precipitation for April, we read the April precipitation data point by noticing it's about as high as the 8 on the precipitation scale on the right. This means the chart says that in April, the average temperature is around 14° Celsius and the average precipitation around 8 centimeters. Since the question is only about April, the data shown for January, July, and October are irrelevant.

**AVERAGE TEMPERATURE AND
PRECIPITATION IN CITY X**

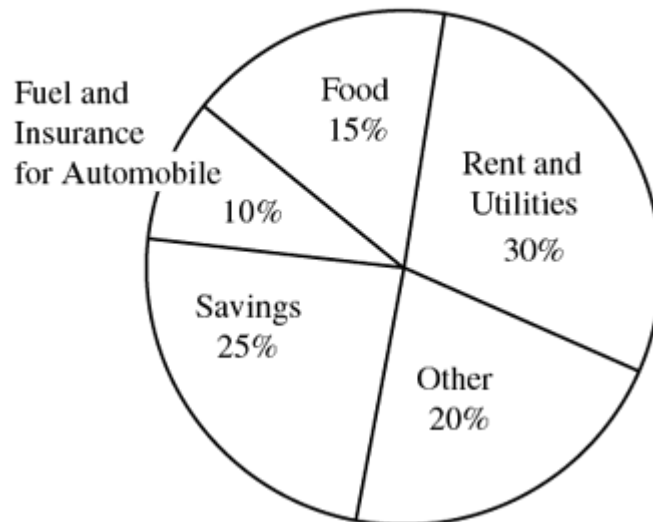


B. A **pie chart** has a circle divided into sections like pie slices. The sections make up the whole circle without overlapping. They stand for parts that together make up some whole amount. Usually, each section is sized in proportion to the part it stands for and labeled with that part's fraction, or percent, of the whole amount. You can use these fractions or percents in calculations. Refer to [Section 3.3](#), "Rates, Ratios, and Percents," to review how.

Example:

In the pie chart below, the sections are sized in proportion to their percent amounts. These percents add up to 100%. Suppose we're told that Al's weekly net salary is \$350 and asked how many of the categories shown each individually took at least \$80 of that \$350. To answer, first we find that $\frac{\$80}{\$350}$ is about 23%. This means each category that took at least \$80 of Al's salary took at least 23% of his salary. So, the graph shows exactly two categories that each took at least \$80 of Al's salary: **Savings** took 25% of his salary, and **Rent and Utilities** took 30%.

DISTRIBUTION OF AL'S WEEKLY NET SALARY



C. A **bar chart** has vertical or horizontal bars standing for cases. A simple bar chart has only one quantitative variable. The bars' different heights

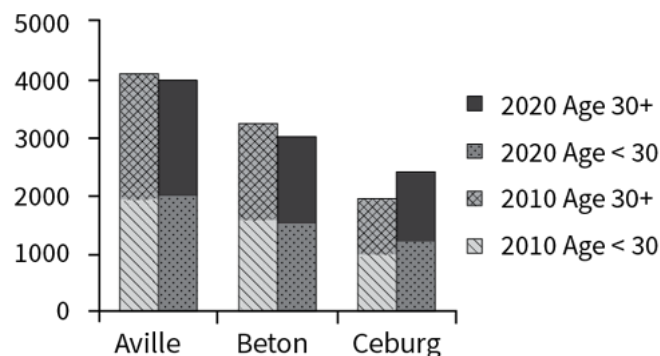
or lengths show this variable's values for different cases.

A **grouped** bar chart may have more than one quantitative variable. Its bars are grouped together. Each group shows the values either of different variables for one case or of related cases for one variable.

In a **stacked** bar chart, segments are stacked into bars. Each segment inside a bar stands for part of an amount, and the whole bar stands for the whole amount.

Example:

The bar chart below is both grouped and stacked. Each pair of grouped bars shows population figures for one of three towns. In each pair, the bar on the left shows the town's population in 2010, and the bar on the right shows the town's population in 2020. Inside each bar, the lower segment shows how many people in the town were under age 30, and the upper segment shows how many were age 30 or older. For example, the chart's fifth bar shows that in 2010, Ceburg's population was around 2,000, including about 1,100 people under 30 and 900 people 30 or older. And the chart's sixth bar shows that by 2020, Ceburg's population had grown to around 2,400, including about 1,200 people under 30 and 1,200 people 30 or older. By reading the amounts shown, we can also estimate various other amounts, like the three towns' combined number of residents 30 or older in 2020.

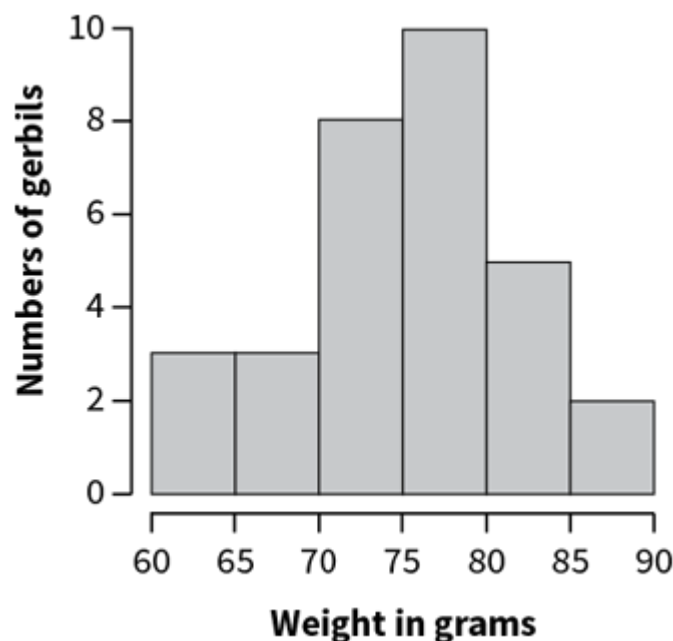


D. A **histogram** looks like a bar chart but works differently. In a histogram, each bar stands for a range of values that the same variable

can take. These ranges don't overlap. Together they usually include every value the variable can take, or at least every value it does take in some population. The bars are in order from the one farthest left, which stands for the lowest range of values, to the one farthest right, which stands for the highest range. Each bar's height shows the number or proportion of times the variable's value is in the range the bar stands for. A bar chart shows clearly how the values are distributed.

Example:

The histogram below shows the weights of 31 gerbils. Each bar's height shows how many gerbils were in a specific weight range. For example, the bar farthest left says 3 gerbils each weighed from 60 to 65 grams. The histogram doesn't show any individual gerbil's weight. However, it does give us some statistical information. For example, by adding the numbers of gerbils in the different weight ranges, we can tell that the 16th-heaviest of the 31 gerbils weighed between 75 and 80 grams. This means the gerbils' median weight was in that range. The histogram also shows that the gerbils mainly weighed between 70 and 85 grams apiece, though several weighed less or more.



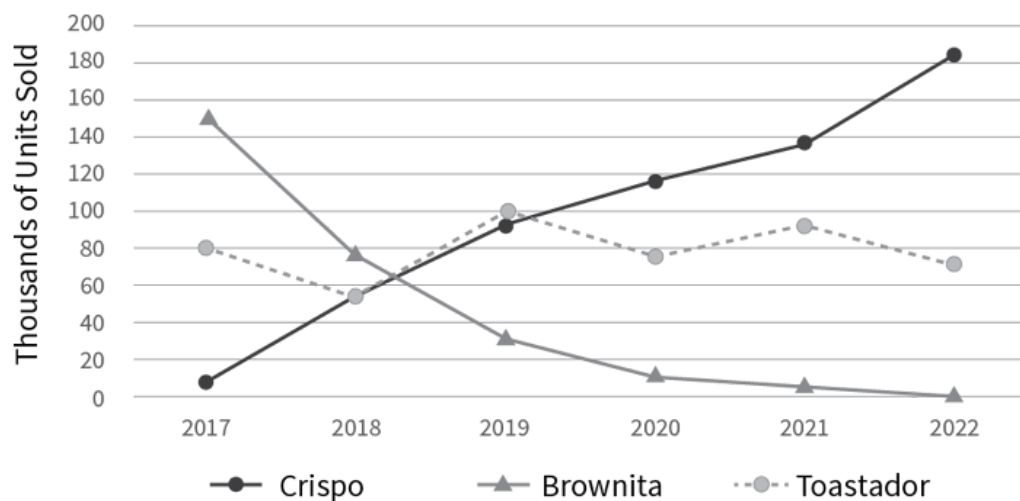
E. A **line chart** often shows how the values of one or more quantitative variables change over time. Typically, the horizontal axis has the time scale, and the vertical axis has one or more scales for the variable or variables. One or more lines connect the data points. Different lines may stand either for different variables or for a single variable applied to different cases. Line charts make it easy to see trends and correlations.

Some line charts show probability distributions instead of changes over time, as we'll see in Section 5.3.1 below.

Examples:

In Section 5.2.3.A above, the chart of average monthly temperatures and precipitations is a line chart whose two variables have two separate scales, one on the left and one on the right.

The line chart below shows how many toasters of three different brands were sold each year from 2017 to 2022. Each sloping line shows how sales changed for one of the brands during those years. The legend on the bottom says which line stands for which brand. All three lines use the scale on the left, whose numbers stand for thousands of units sold annually. The chart shows that over the six years, annual sales of Crispo toasters increased dramatically, annual sales of Brownita toasters declined to almost zero, and annual sales of Toastador toasters fluctuated.

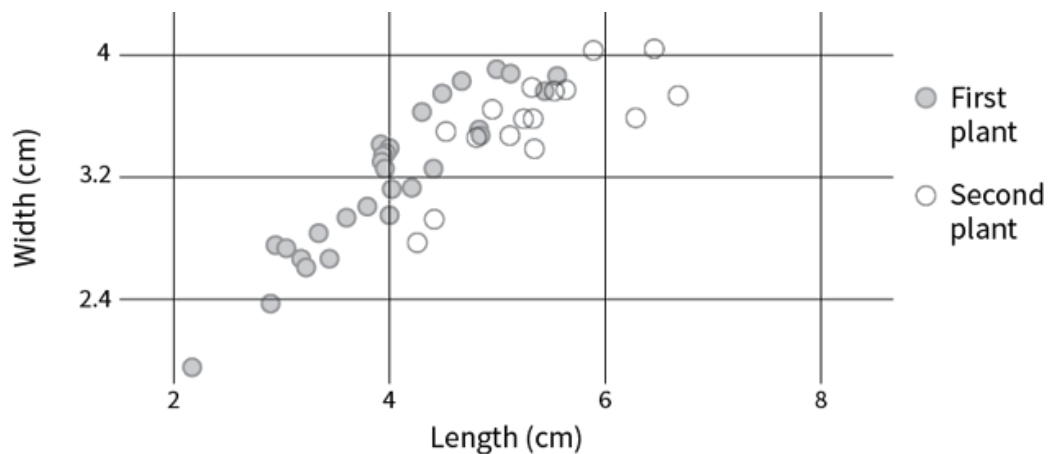


- F. A **scatterplot** has at least two quantitative variables, one on each axis. For each case in the data, a dot's position shows the variables' values. No lines connect the scattered dots, but a line through the scatterplot may show an overall trend in the data. Sometimes the dots have a few different shapes or colors to show they stand for cases in different categories. And in scatterplots called **bubble charts**, the dots have different sizes standing for values of a third variable.

Scatterplots are useful for showing correlations. They also show how much individual cases fit an overall correlation or deviate from it.

Example:

The scatterplot below shows measured widths and lengths of leaves on two plants. Each dot stands for one leaf. The dot's position stands for the leaf's width and length in centimeters, as the scatterplot's two axes show. The legend says how one set of dots stands for leaves on the first plant, and another set for leaves on the second. The scatterplot shows that, in general, the longer leaves tend to be wider, and vice versa. It also shows how leaves on the second plant tend to be somewhat longer and wider than those on the first, with some exceptions.



5.3 Data Patterns

1. Distributions

- A. A data ***distribution*** is a pattern in how often different values appear in data. How data in a sample is distributed can tell you how likely different values are to appear in the same population outside the sample. In this book, Section 4.2.2, “Generalizations and Predictions,” discusses generalizations and predictions based on samples.

- B. A distribution is **uniform** if the values each occur about equally often. It's less uniform when the values differ more in how often each occurs. The more uniform a distribution is for a sample, the better it supports the conclusion that the distribution for other cases in the population is likewise uniform.

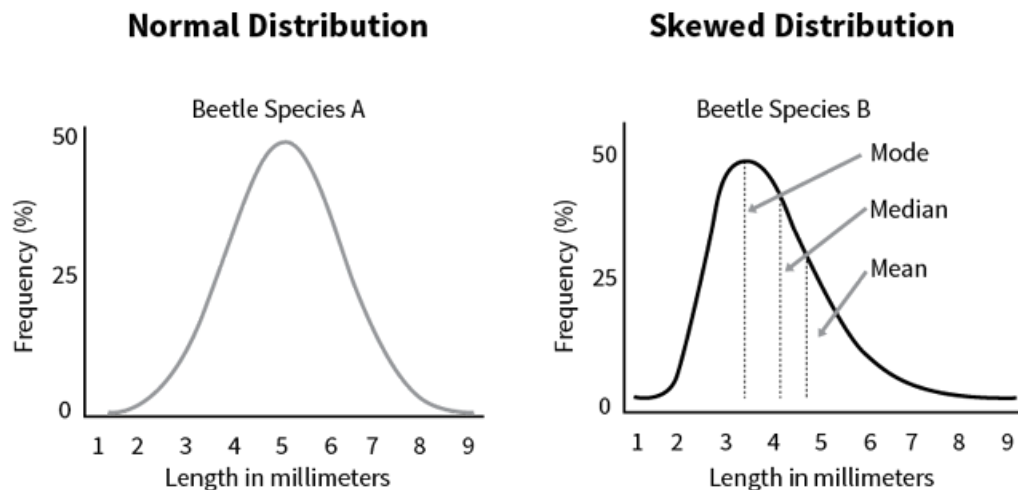
Example:

Suppose the six faces on a die are numbered 1 to 6. And suppose that when rolled sixty times, the die comes up 1 eleven times, 2 nine times, 3 ten times, 4 eleven times, 5 ten times, and 6 nine times. This isn't a **perfectly** uniform distribution, because the six values 1 to 6 didn't occur exactly ten times apiece. But because each value did occur between nine and eleven times, the distribution is **fairly** uniform. This suggests that each of the six values is about equally likely to occur whenever the die is rolled, and that their distribution for future rolls will stay fairly uniform.

- C. A variable's values are often distributed unevenly. Sometimes one central value occurs most often, with other values occurring less often the farther they are from the central value. When this type of distribution is **normal**, each value below the central value occurs just as often as the value equally far above the central value. For a perfectly normal distribution, the central value is the mean, the median, and the mode. When plotted on a chart, a normal distribution is bell shaped, with a central hump tapering off equally into tails on both sides. But a distribution with a larger tail on one side of the hump than the other is not normal but **skewed**. For a skewed distribution, the mean is often farther out on the larger tail than the median is.

Example:

The two charts below show distributions of lengths for two beetle species. The chart on the left shows that Species A beetles have roughly a normal distribution of lengths. The central hump is symmetrical, with equal tails on both sides. From the chart on the left, we can tell that the mode, the median, and the mean of the lengths for Species A are all around 5 millimeters. In contrast, the chart on the right shows that Species B beetles have a skewed distribution of lengths. The tail on the right side of the hump is larger than the tail on the left side. This means more beetles of Species B have lengths above the mode than below it. As a result, the median length for Species B is above the mode, and the mean is above the median.



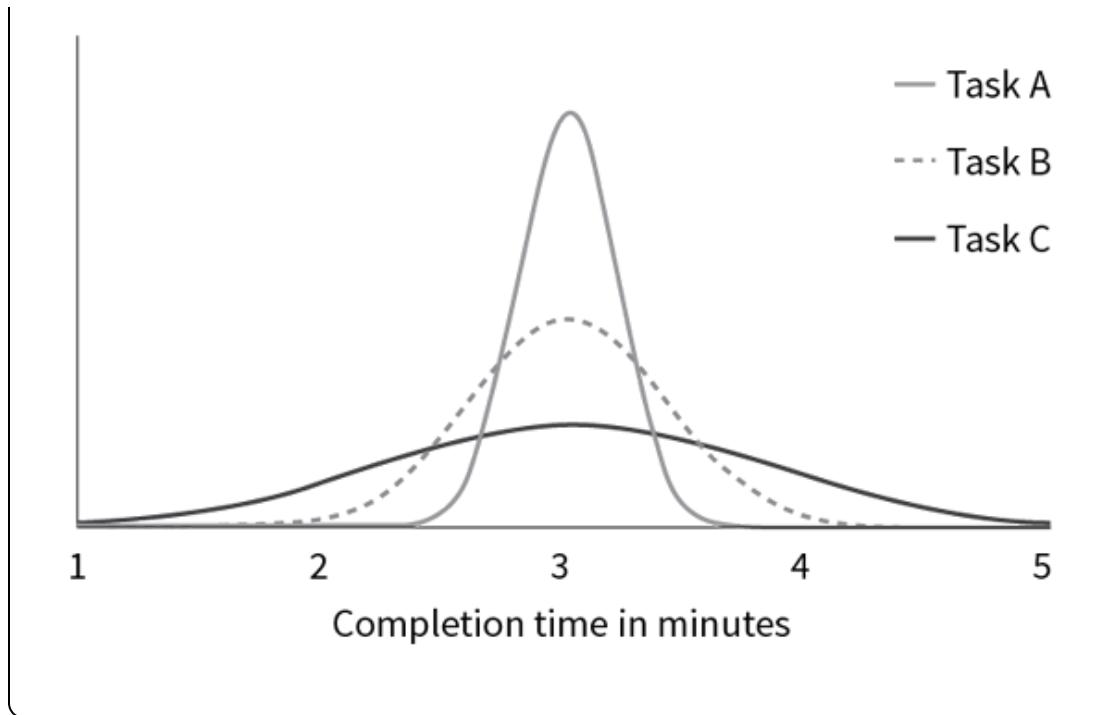
- D. The more tightly clustered a distribution is around a central value, the higher and narrower the hump is, and the smaller the tails are. A more tightly clustered distribution also has a smaller standard deviation, as discussed in Section 3.4.1, “Statistics.” The more tightly clustered a distribution is for an observed sample, the more likely a new case from the population outside that sample is to have a value near the distribution’s central value.

Example:

For each of three tasks, the chart below shows how frequently it takes workers different lengths of time in minutes to complete that task. Even though the bottom axis shows times from 1 to 5 minutes, it doesn't stand for a single period starting at 1 minute and ending at 5 minutes. Instead, it stands for a range of lengths of task completion times.

Notice how the chart uses smooth curves to show the frequencies of different completion times. That suggests it shows trends idealized from the observed data points, which could be shown more precisely as separate dots or bars.

For each task, the distribution of completion times is uniform, with a mode, median, and mean of 3 minutes. But the completion times are most tightly clustered around 3 minutes for Task A, and least tightly clustered for Task C. That means the standard deviation of the completion times is lowest for Task A and highest for Task C. It also means the probability is higher that an individual worker will take close to 3 minutes to complete Task A than to complete Task C. A worker's completion time for Task C is less predictable and likely to be farther from 3 minutes. So, the chart gives **stronger evidence** for the conclusion that finishing Task A will take a worker between 2 and 4 minutes than for the conclusion that finishing Task C will take that worker between 2 and 4 minutes.



E. Data distributions take many other shapes too. Some have two or more humps, and others have random variations in frequency among adjacent values. In general, the less the values in a sample cluster around one central hump, the larger the standard deviation is, and the less predictable the values are for cases in the population outside the observed sample.

2. Trends and Correlations

A. Charts often show trends over time. They may show that a variable's values increase, decrease, fluctuate, or change in cycles. They may also show values of different plotted variables changing in the same ways or opposite ways.

Usually, an observed trend over a period is evidence that the same trend extends at least slightly before and after that period. This evidence is stronger the longer the observed trend has lasted and the more varied the conditions it's lasted through. Generalizing from a longer observed period with more varied conditions is like generalizing from a larger sample, as discussed in Section 4.2.2, "Generalizations and Predictions." But the odds increase of other

factors disrupting the observed trend at times farther from the observed period, and in conditions that differ more from those observed.

Example:

The line chart in Section 5.2.3.E above shows that annual sales of Crispo toasters rose from fewer than 10,000 in 2017 to over 180,000 in 2022. If this trend continues another year, even more than 180,000 Crispo toasters will be sold in 2023. But many factors might disrupt Crispo's surging popularity. For example, another company might start making better or less expensive toasters, drawing consumers away from Crispo toasters. Or broader social, economic, or technological changes might reduce demand for toasters altogether. The more years that pass outside the observed period of 2017 through 2022, the more likely such disruptions become. So, the observed trend gives stronger evidence that annual Crispo toaster sales will be over 180,000 in 2023 than it gives that they'll still be over 180,000 in 2050.

- B. Two quantitative or ordinal variables are ***positively correlated*** if they both tend to be higher in the same cases. They're ***negatively correlated*** if one tends to be higher in cases where the other is lower.

Examples:

If warmer days tend to be rainier in a certain region, then temperature and precipitation are positively correlated there.

But if warmer days tend to be drier in a different region, then temperature and precipitation are negatively correlated there.

- C. On a line chart, the lines standing for positively correlated variables tend to slope up or down together. The more consistently the slopes match, the stronger the positive correlation. But when two variables are negatively correlated, the line standing for one tends to slope up where the other slopes down, and vice versa. The more consistently the lines slope in opposite directions, the stronger the negative correlation.

Examples:

The chart in Section 5.2.3.A shows a positive correlation between average monthly temperature and precipitation in City X. The temperature and precipitation lines both slope up together from January through July, and then slope down together.

The chart in Section 5.2.3.E shows a negative correlation between annual sales of Crispo toasters and annual sales of Brownita toasters. Throughout this chart, the line standing for Crispo toaster sales slopes up and the line standing for Brownita toaster sales slopes down. But the chart shows no clear positive or negative correlation of Crispo or Brownita toaster sales with Toastador toaster sales. The line standing for Toastador toaster sales fluctuates, sometimes sloping up and sometimes down. It shows no consistent trend relative to the other two lines. So, the chart doesn't clearly support a prediction that in future years Toastador sales will increase or decrease as Crispo or Brownita sales do the same or the opposite.

- D. When a scatterplot shows a positive correlation, the dots tend to cluster around a line that slopes up. When it shows a negative correlation, they tend to cluster around a line that slopes down. The stronger the correlation, the more tightly the dots cluster around the sloped line. If the dots spread farther away from the sloped line, or cluster around a line with a less consistent slope, the correlation is weaker or nonexistent.

Example:

The scatterplot in Section 5.2.3.F shows leaf width as positively correlated with leaf length for both plants. The dots mostly cluster around a line sloping up, which means wider leaves tend to be longer and vice versa. Thus, the scatterplot supports a prediction that if another leaf on one of the plants is measured, it too will probably be wider if it's longer. But the correlation in the scatterplot isn't perfect. For example, the dot farthest to the right is lower than each of about a dozen dots to its left. That dot farthest to the right stands for a leaf that's both longer and thinner than any of the leaves those other dozen dots stand for. This inconsistency in the correlation somewhat weakens the scatterplot's support for the prediction that another leaf measured on one of the plants will be wider if it's longer.

- E. Values of a nominal variable may also be associated with higher or lower values of an ordinal or quantitative variable. Different values of two nominal variables can also be associated with each other.

Examples:

Species is a nominal variable. Individuals of some species tend to weigh more than individuals of other species, so different values of the nominal variable *species* are associated with higher or lower values of the quantitative variable *weight*.

As another example, university students majoring in certain subjects like physics are more likely to take certain courses like statistics than university students majoring in other subjects like theater are, even though a student's *major* and *course enrollment* are both nominal variables.

An association involving a nominal variable like *species* isn't a positive or negative correlation, because nominal variables aren't ordered; one species isn't greater or less than another. Still, these nominal associations can be shown in qualitative charts and tables, and

they can support predictions. Knowing an animal's species gives you some evidence about roughly how much it likely weighs. And knowing a student's major gives you some evidence about what courses that student is more likely to take.

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7.0 GMAT™ Official Guide Data Insights Review Question Index

7.0 GMAT™ Official Guide Data Insights Review Question Index

The Data Insights Review Question Index is organized by the section, difficulty level, and then by mathematical concept. The question number, page number, and answer explanation page number are listed so that questions within the book can be quickly located.

Data Insights — Chapter 6 – Page 122

Data Sufficiency – Page 133

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Applied Problems	8	700305	134	150
Easy	Applied Problems	9	700290	134	150
Easy	Applied Problems	10	700109	134	150
Easy	Applied Problems	14	700107	134	152
Easy	Applied Problems	22	700404	135	156
Easy	Applied Problems	24	700129	135	157
Easy	Applied Problems	27	700140	136	158
Easy	Applied Problems	29	700142	136	159
Easy	Equations	12	700402	134	151
Easy	Evaluate	35	700325	136	161
Easy	First-Degree Equations	4	700298	133	148
Easy	First-Degree Equations	16	700108	134	153
Easy	First-Degree Equations	19	700125	135	154
Easy	Inequalities	26	700135	135	158

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Inference	32	700322	136	160
Easy	Inference	33	700323	136	160
Easy	Inference	34	700324	136	161
Easy	Inference	36	700326	137	162
Easy	Inference	37	700327	137	163
Easy	Inference	38	700328	137	163
Easy	Percents	13	700403	134	152
Easy	Percents	20	700126	135	155
Easy	Percents	30	700143	136	159
Easy	Probability	2	700299	133	147
Easy	Rate Problems	11	700295	134	151
Easy	Ratio and Proportion	3	700401	133	148
Easy	Series and Sequences	25	700132	135	157
Easy	Simultaneous Equations	18	700124	135	154

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Simultaneous Equations	23	700405	135	156
Easy	Statistics	1	700291	133	147
Easy	Statistics	5	700293	133	148
Easy	Statistics	6	700292	133	149
Easy	Statistics	7	700288	134	149
Easy	Statistics	15	700289	134	152
Easy	Statistics	17	700286	135	153
Easy	Statistics	21	700294	135	155
Easy	Statistics	28	700141	136	158
Easy	Statistics	31	700147	136	160
Medium	Applied Problems	39	700406	137	164
Medium	Applied Problems	43	700321	137	166
Medium	Applied Problems	45	700100	138	167
Medium	Applied Problems	47	700111	138	168
Medium	Applied Problems	61	700138	140	175

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Applied Problems	63	700145	140	175
Medium	Applied Problems	65	700148	140	176
Medium	Applied Problems	41	700296	137	165
Medium	Applied Problems	75	700412	141	180
Medium	Estimation	40	700287	137	164
Medium	Evaluate	71	700333	141	178
Medium	First-Degree Equations	56	700131	139	173
Medium	Inference	67	700329	140	177
Medium	Inference	68	700330	140	177
Medium	Inference	69	700331	141	178
Medium	Inference	70	700332	141	178
Medium	Inference	72	700334	141	179
Medium	Inference	73	700335	141	179
Medium	Inference	74	700336	141	179
Medium	Percents	42	700407	137	165

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Percents	48	700409	138	168
Medium	Percents	54	700411	139	171
Medium	Percents	55	700130	139	172
Medium	Percents	60	700137	140	174
Medium	Probability	53	700311	139	171
Medium	Ratio and Proportion	44	700308	138	166
Medium	Simultaneous Equations	50	700410	138	170
Medium	Simultaneous Equations	62	700144	140	175
Medium	Simultaneous Equations	64	700146	140	176
Medium	Statistics	46	700408	138	167
Medium	Statistics	49	700302	138	169
Medium	Statistics	51	700319	138	170
Medium	Statistics	52	700104	139	170

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Statistics	57	700133	139	173
Medium	Statistics	58	700134	139	174
Medium	Statistics	59	700136	139	174
Medium	Statistics	66	700149	140	176
Hard	Applied Problems	77	700414	141	181
Hard	Applied Problems	96	700119	144	189
Hard	Applied Problems	97	700120	144	189
Hard	Applied Problems	99	700122	144	190
Hard	Applied Problems	100	700123	144	191
Hard	Applied Problems	101	700139	144	191
Hard	Applied Problems	108	700343	145	194
Hard	Applied Problems	109	700344	145	195
Hard	Applied Problems	110	700345	145	195
Hard	Applied Problems	111	700346	145	195
Hard	Applied Problems	112	700347	145	196

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Applied Problems	113	700348	145	196
Hard	Applied Problems	87	700285	143	185
Hard	Applied Problems; Rates	94	700318	143	188
Hard	Arithmetic Operations	83	700115	142	183
Hard	Arithmetic Operations; Percents	81	700113	142	183
Hard	Concepts of Sets	86	700314	142	184
Hard	Concepts of Sets; Percents	88	700284	143	185
Hard	First-Degree Equations	98	700121	144	190
Hard	Inference	102	700337	144	191
Hard	Inference	103	700338	144	192
Hard	Inference	104	700339	144	192
Hard	Inference	105	700340	144	193
Hard	Inference	106	700341	145	193

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Inference	107	700342	145	194
Hard	Operations with Fractions	92	700102	143	187
Hard	Percents	78	700297	142	181
Hard	Percents	80	700303	142	182
Hard	Percents	82	700114	142	183
Hard	Percents	84	700112	142	184
Hard	Probability	90	700283	143	186
Hard	Properties of Numbers	85	700300	142	184
Hard	Rate Problems	89	700306	143	186
Hard	Ratio and Proportion	76	700413	141	180
Hard	Sets	91	700310	143	187
Hard	Simultaneous Equations	79	700105	142	182
Hard	Simultaneous Equations	95	700118	143	189

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Work Problems	93	700301	143	188

Two-Part Analysis – Page 198

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Evaluate	119	700257	199	222
Easy	Infer	114	700355	198	219
Easy	Infer	115	700356	198	219
Easy	Infer	116	700155	198	220
Easy	Infer	121	700275	200	224
Easy	Infer	122	700415	200	224
Easy	Recognize	118	700240	199	221
Easy	Recognize	120	700271	200	223
Easy	Strategize	117	700197	199	221
Easy	Strategize	123	700416	201	225
Medium	Apply	127	700184	202	228
Medium	Apply	130	700194	203	230
Medium	Evaluate	124	700350	201	226
Medium	Evaluate	125	700354	201	226

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Evaluate	126	700174	201	227
Medium	Evaluate	128	700191	202	228
Medium	Evaluate	131	700201	203	231
Medium	Evaluate	133	700196	204	232
Medium	Infer	132	700202	204	232
Medium	Infer	134	700198	204	234
Medium	Infer	135	700231	205	234
Medium	Infer	136	700232	205	235
Medium	Infer	140	700256	206	238
Medium	Infer	141	700266	207	238
Medium	Strategize	129	700192	202	229
Medium	Strategize	137	700233	205	235
Medium	Strategize	138	700234	205	236
Medium	Strategize	139	700242	206	237
Hard	Apply	148	700176	209	244

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Apply	153	700205	211	249
Hard	Apply	159	700227	213	253
Hard	Apply	160	700228	213	253
Hard	Apply	161	700239	213	254
Hard	Evaluate	142	700349	207	239
Hard	Evaluate	151	700200	210	247
Hard	Evaluate	152	700203	210	248
Hard	Evaluate	166	700359	215	258
Hard	Infer	143	700351	207	240
Hard	Infer	144	700352	208	241
Hard	Infer	145	700353	208	241
Hard	Infer	146	700157	208	242
Hard	Infer	147	700170	209	243
Hard	Infer	149	700177	209	245
Hard	Infer	150	700186	210	246

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Infer	157	700225	212	251
Hard	Infer	162	700241	214	255
Hard	Infer	165	700364	215	258
Hard	Infer	167	700358	215	259
Hard	Infer	168	700265	216	260
Hard	Recognize	156	700224	212	251
Hard	Recognize	163	700252	214	256
Hard	Recognize	164	700253	214	257
Hard	Recognize	170	700278	216	261
Hard	Strategize	154	700215	211	250
Hard	Strategize	155	700222	211	250
Hard	Strategize	158	700226	212	252
Hard	Strategize	169	700268	216	260

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